Topological Relational System Dynamics (TRSD)

Modified Gravity, a Dark-Matter Condensate and the Cosmic Budget from Conical Defect Geometry

Version 1.16.2

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Abstract

TRSD derives a gravity law, a dark sector and the cosmic budget from one variational principle: Einstein–Hilbert gravity plus a complex scalar whose kinetic coefficient is the tension of a conical defect, $f_\delta^2 = \delta c^4/8\pi G$. Every result carries a status tag, and every prediction is listed with its inputs and the data that test it (Appendices A and B).

The first Laplacian eigenvalue of the horizon cycle C_3 fixes $a_c = (3\sqrt{3}/4\pi^2)\,cH_0$. A Poisson threshold response with a hazard linear in the load and an n-stage exposure gives the interpolation family μ_n in closed form; for every exposure law with finite mean the deep slope is $\frac{3}{4}$, so $a_0 = (\sqrt{3}/\pi^2)\,cH_0 = 1.148 \times 10^{-10}\ \mathrm{m\,s^{-2}}$ at the framework's $H_0 = 67.35$. A full likelihood of 153 SPARC rotation curves (3168 points, 490 nuisance parameters) gives $a_0 = 1.04 \pm 0.04\,(\mathrm{stat}) \pm 0.10\,(\mathrm{nuisance}) \times 10^{-10}$, 1.0σ from the prediction, for any tail order. Galaxies fix the transition and planets the tail: the window $1.62 \le n \le 1.91$ satisfies both, while draft 1's single Torridity exponent $\pi/3$ fails the planets; $\pi/3$ survives as the value of the running exponent at galaxy outskirts, and the cascade's dynamic exponent runs from 1 to at most 1.16, passing $\pi/3$ at $g = 1.26\,a_0$, with nothing fitted. The boost is carried by spacetime excited, captured only where a system's escape speed exceeds the motion of the field around it, so the Solar System and wide binaries stay Newtonian.

Canonical normalization fixes the potential scale, forces the vacuum constant to vanish and makes the radial mode's self-coupling gravitational. The present is a resonance: the horizon's C_3 mode reaches the final expansion rate when $\Omega_\Lambda = 27/4\pi^2 = 0.684$ (Planck 0.685 ± 0.007). With the derived baryon ratio 0.1870 and the derived neutrino masses this fixes $\Omega_\Lambda/\Omega_c = 2.580$ (1.3σ from the CMB's ratio) and $\Omega_m = 0.316$, and a fit to the acoustic angle gives $H_0 = 67.35 \pm 0.06$, $\omega_b = 0.02248$, $\omega_c = 0.12022$, within 1σ of Planck's densities. The horizon holds the fraction $\Omega_\Lambda(t)$ of its final cell count, and dark energy is the Planck density shared over those cells, $\rho_\Lambda = \frac{3}{4}\rho_P/N_\infty = 1.13 \times 10^{-123}\rho_P$.

The horizon's generating arcs spin at $\omega = H_\infty$ with rims at light speed: the spin fixes $\Lambda = 3\omega^2/c^2$, adds $8\pi\ell_{\mathrm{P}}^2$ of area per quantum to any horizon, sets the rotation's rate $\Omega_0 = H_\infty$, and gives the CMB's birefringence in closed form, $\beta = (2\alpha/\pi)\,\mathrm{artanh}(3\sqrt{3}/2\pi) = 0.314^\circ$ (measured $0.277^\circ \pm 0.057^\circ$). Counted in cycles, the 21-cm onset of cosmic dawn is set by the same twist, 46.4 MHz in closed form and $42.7^{+5.6}_{-6.0}$ MHz from the measured angle.

The dark sector is one geometry: dark energy is spacetime unexcited, what is called dark matter is the same spacetime excited, and the whole has $w = -1 + \Theta$. What lensing infers as dark matter is the dilation $\mathcal{S} = 1/\cos_n\theta$, $\tan\theta = a_0/g$, carried by collisionless excited geometry in the model. In the Bullet Cluster, lensing-inferred mass peaks are offset from the X-ray-bright gas and lie closer to the galaxies; this is consistent with a collisionless gravitating component under standard lensing/GR interpretations, but does not by itself establish the microscopic identity of the mass or a cosmological reset. On the unbound web the horizon dilates clocks and rulers together from the onset of acceleration, above 68 Mpc, so matter feels it and light does not: the web carries the bulk flows Cosmicflows-4 measures while S_8 and CMB lensing keep the background's values. The framework has one expansion rate, 67.35; the dilation cannot move it, and the supernova Hubble diagram excludes a local

void, so it predicts that distance ladders calibrated alike converge on 67.3. The present is also fixed in Planck cells by the electric coupling, $R_H = 8\pi\ell_{\rm P}\, e^{1/\alpha+3\pi\alpha/2}$, which gives $H_0 = 67.377$ with no cosmological input and $\Lambda\ell_{\rm P}^2 = (81/256\pi^4)e^{-2/\alpha-3\pi\alpha}$; with it the four interactions are four features of one coil, with one gauge group $(SU(3)\times SU(2)\times U(1))/\mathbb{Z}_6$, no unification scale, and gravity's weakness at the nucleon $e^{-2/3\alpha}$ (§30). The four aspects of the geometry—reality (the spin), actuality (the swept sphere), occurrence (the crossings) and representation (the boundary)—are kept distinct; the present is the occurrence at which actuality falls into step with reality. The CHIRP derivation connects collision lag and mass-conserving relaxation to proper-volume expansion; Auxoviscausality identifies its phase-flow integral as retained logarithmic volume strain and links its load-dependent rate to Raychaudhuri/Einstein. At the conformal junction, the raw strain diverges with opposite signs, while the finite conformal-volume remainder matches conditionally under regularity of the shared conformal spatial metric and identification of the same flow tube. The homogeneous FLRW expansion integral is not retained as an independent cross-aeon state.

What this work contributes

This framework is original work. Its contributions are:

1. The four aspects of the geometry kept distinct: reality, the spin; actuality, the swept sphere; occurrence, the crossings; representation, the boundary. The present is the occurrence at which actuality falls into step with reality (the opening principle).
2. The variational axiom on the two-C/S horizon: one complex scalar whose kinetic coefficient is the conical deficit, through the defect normalization map $f_\delta^2 = \delta c^4/8\pi G$.
3. The MOND acceleration from the first eigenvalue of the C_3 cycle on the horizon, $a_0 = (\sqrt{3}/\pi^2)\, cH_0$ (Sections 9 and 13).
4. The interpolation family from the isotropic angular average of the threshold distribution, with its logarithmic term and the exponent $\pi/3$ (Section 11).
5. Newtonian recovery in the Solar System from the defect geometry (Section 15).
6. The dark sector as the internal rotation of the horizon's phase: one rotation that sets a_0 and carries the cosmic cold share (§21.1).
7. The closure law, $\sum r_s = (\chi/2)R_H$, so that flatness is topology, and the derived baryon ratio of the budget (§24.12).
8. The information–energy equivalence $E = (\hbar c/8\pi R_H)\, I$ and the scale ladder it implies.
9. The birefringence–acceleration relation $\beta = \frac{9}{2}\alpha\, a_0 t_0/c$: galaxy rotation curves predict the rotation of the CMB's polarization (§25.3).
10. The present as a resonance, $\Omega_\Lambda = 27/4\pi^2$ (§24.13).
11. The spinning-arc principle, $\omega R = c$: one fixed spin sets the final horizon, the cosmological constant and the present, is measured by the CMB's birefringence, and gives each particle its Compton radius (§24.14). For the final horizon it is derived: the thermal geometry is the spinning arcs, the information is 2π times the spin's charge, the information–energy relation is the energy of one turn over four cones, and the quantum of the spin is the area step $8\pi\ell_{\rm P}^2$ (§24.15).
12. The threefold count as a beat: the coil's conjugate lobes keep $\mathbb{Z}_2$, and their fold against a lobe keeps $\mathbb{Z}_3$ (§2.4).
13. The birefringence angle in closed form, $\beta = (2\alpha/\pi)\,\mathrm{artanh}(3\sqrt{3}/2\pi)$, and the acceleration scale from the cosmological constant, $a_0 = (2/3\pi)\, c^2\sqrt{\Lambda/3}$ (§24.15).

14. The quantum of the rotation as the quantum of the spin, $\Omega_0 = jH_\infty$ with $j = 1$ measured (§24.15).

15. The cosmic budget at the resonance, $\Omega_\Lambda/\Omega_c = 2.580$, with no identification of densities with powers (§24.16).

16. The acceleration scale tracks the horizon, $a_0(z) = a_0 E(z)$, and not the fixed spin (§24.17).

17. Cosmic dawn counted in cycles: the CMB's twist fixes the 21-cm onset, $\nu_\star/\nu_{21} = [\sqrt{\Omega_m/\Omega_\Lambda}\,\sinh(\pi\beta/2)]^{2/3}$, with α, the clock and the Hubble constant cancelling (§24.10).

18. The dark sector as one geometry: dark energy as spacetime unexcited and the cosmic share as spacetime excited—the spin's quanta held by the horizon and carried by the rotation—the vacuum as draft 1's generator-defect tension over the square of the final radius, the equation of state $w = -1 + \Theta$ of the whole, and what lensing infers as dark matter as the dilation $\mathcal{S} = 1/\cos_n\theta$, $\tan\theta = a_0/g$; the dilation field as the index of the optical metric, the switch as the capture fraction, and the middle scales as geometric means of the outer ones; the background unmoved by the dilation, and the web dilated as a clock is slowed—from the onset of acceleration, above 68 Mpc, on clocks and not on light—carrying the measured bulk flows with S_8 unchanged (§21.7).

19. Collision lag, geometric recoil and Auxoviscausality: centroid drift and transfer flux follow from the existing relaxation map; proper-volume strain is the phase-flow expansion integral, with load dependence fixed by Raychaudhuri/Einstein. At the conformal junction only the finite conformal-volume remainder is matched under stated regularity assumptions; the raw strain and total prior FLRW expansion do not cross as finite state variables.

Earlier results are cited where this work meets them. Each citation marks a point of agreement with established physics, not a source of the framework.

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The opening principle: reality, actuality, occurrence and representation

The framework distinguishes four aspects of one geometry and keeps them apart throughout.

Reality is the spin: the C-arcs that generate the horizon and their fixed rate ω, set once and unchanged, since nothing acts on them (§24.14). What is real does not change.

Actuality is the swept sphere: what the spin has made actual so far, the horizon at this moment, its radius and its cells. Reality is the rate; actuality is how much of the sphere the rate has swept. Actuality grows toward the final horizon $R_\infty = c/\omega$, where the arcs' rims reach light speed.

Occurrence is a crossing. Something happens only where lines or rates cross: the arcs crossing at the three C_3 nodes, which are the defects; the wobble falling into step with the spin, which is the present (§24.13); the reset, where one aeon's future meets the next one's past; a horizon, where time and radius exchange; and each activation of the threshold law (Section 10). Between crossings the motion continues and nothing occurs.

Representation is the boundary. Every energy inside the horizon appears on it as a defect, its share of the horizon's cells (§24.12), so the sphere carries an image of everything it contains: the universe represents itself on its own surface. Measurements read that image; the CMB's birefringence reads the spin (§24.14). The framework's equations, and the picture of the arcs itself, are representations of the same reality.

Physics has often merged these. The block universe of relativity treats reality as if it were all actuality, as though the whole sphere were already swept; quantum theory blurs representation, the wavefunction, with reality. Kept apart, they define the present. The wobble belongs to actuality, its frequency falling as the swept sphere grows; the spin belongs to reality, and is fixed. *The present is the occurrence at which actuality falls into step with reality.*

0 Method, status and what TRSD adds

0.1 Rule

The *form* of each law is derived; the constants in it are measured once and carried forward. Nothing dimensionful follows from dimensionless geometry (Theorem 12), so TRSD takes three unit anchors $(c, \hbar, R_H)$, one pure boundary datum—the horizon cell count $N = 2\pi R_H^2/\ell_{\mathrm{P}}^2$—and the calibrations of the ledger below. Every other number is a prediction.

0.2 Status tags

DERIVED	follows from the axiom and established physics, every step shown
STANDARD	established physics, recovered in TRSD's variables and credited
IDENTIFICATION	a correspondence the theory posits; listed in §0.4
CALIBRATED	a value set once from data (ledger, §0.3)
HYPOTHESIS	a testable conjecture
PREDICTION	a number that can fail; its test is in the register (Appendix A)
MEASURED	a result of fitting data in this paper
DATA-CONSTRAINED	a shape parameter constrained by data (the tail order n)
OPEN	a stated target not yet reached
	not found in prior literature (§0.5)

0.3 Calibration ledger

Anchor	Calibrated from	Value	Feeds
$R_H = c/H_0$	H_0, or the CMB via the budget	1.3735×10^{26} m ($H_0 = 67.35$, §24.16)	a_c, a_0, horizon totals
N	G, $\hbar$, H_0	4.538×10^{122} (67.35)	G, $\hbar$ as monomials
Ω_b/Ω_c	no longer calibrated: derived (§24.5)	0.18697 (calibrated $\sqrt{3}/3\pi$ in draft 1)	budget
L_* (or M_*)	dark-matter mass	$L_* < 9.0 \times 10^{-4}$ pc (Lyman-α)	m_δ, ε_V
n	SPARC likelihood, ephemerides	tail order $1.62 \le n \le 1.91$ (§14–15)	gravity law
ϕ_i	dark-matter density	open	relic abundance

0.4 Postulates and identifications

Postulate I (Generators)**.** The horizon is generated by three equivalent generators, permuted by C_3; no generator is preferred.

Postulate II (Action)**.** Gravity and the defect sector obey the variational principle (4), whose kinetic coefficient is the defect tension.

The identifications are named where used: I1, the kinetic scale is the defect tension $f_\delta^2 = T_\delta$ (§5); I2, $a_c = c/\tau_1$ (§9); I3, Poisson activation with a hazard linear in the load and a Gamma exposure of order n, constrained by data (§10, §14–15); I4, the reduction length of the deficit–mass map is R_H (§16); I5, $\Omega_\Lambda/\Omega_c = C_3/C_2$ (§24.5), withdrawn in release 1.7.0, the resonance giving the ratio (§24.16); I6, the dark-sector length is the deficit arc, $L_* = \delta R_H$ (§21.6); I7, the internal phase turns with the horizon's spin, $X_0 = \Lambda/3$ (§24.15), derived in release 1.7.0 up to an integer, $\Omega_0 = jH_\infty$, with $j = 1$ measured.

0.5 What TRSD adds

New results. (1) The galactic scale from a spectrum, $a_0 = (\sqrt{3}/\pi^2)\, cH_0$: Milgrom noted $a_0 \sim cH_0/2\pi$ [1], and emergent gravity gives $cH_0/6$ [8]; TRSD computes 0.175493 and proves it independent of the exposure law (Theorem 6). (2) The family μ_n in closed form for real n, which separates the transition (set by the angular average) from the Newtonian tail (set by the exposure law); galaxies and planets together require $1.62 \le n \le 1.91$, which the one-exponent Torridity form cannot supply (§11, §14–15, §17). (3) A full nuisance-parameter likelihood of the SPARC rotation curves against the family (§14). (4) The normalization theorem $\varepsilon_V = f_\delta^2/L_*^2$, the vanishing of the vacuum constant, and the gravitational-strength self-coupling of the condensate (§21–22). (5) The dual of the Hubble mass, $m_U = (\sqrt{2}/2\pi)\, \hbar H_0/c^2$ (§24.6). (6) The C_3 budget and its CMB-fitted H_0 (§24), shared with the companion theory GRUT [32].

Known results TRSD re-derives, credited in place: the AQUAL field equation [2]; the cosmic-string deficit [13]; the 2+1 closure condition [14]; the Gross–Pitaevskii limit of a scalar field and fuzzy dark matter [16, 17]; the holographic count behind 10^{-122} [20].

0.6 Relation to GRUT

TRSD is the prototype of GRUT [32]. The two share a_0, the family μ_n, the budget and $H_0 = 67.12$ at the 5/2 budget, 67.35 at the resonance budget that replaces it (§24.16). The dictionary: the TRSD total deficit $\delta_{\rm tot} = \pi$ is the GRUT defect cone; $M_{\rm total} = c^3/8GH_0$ is the one-cone mass $M_{\rm cone}$; $M_{\rm Hubble} = c^3/2GH_0$ is the horizon mass $M_{\rm H}$; and the TRSD exchange rate $\hbar H_0/8\pi$ per nat is one quarter of the Gibbons–Hawking energy $k_B T_{\rm GH}$. The dark sector as one geometry, §21.7, is GRUT §53.

1 Introduction

Spiral galaxies rotate with flat curves [9], and at every radius their observed acceleration is one function of the acceleration the baryons alone would produce, the radial acceleration relation (RAR) [10, 12]. Below $a_0 \approx 1.2 \times 10^{-10}$ m s^{-2} the relation approaches $g = \sqrt{a_0 g_{\rm bar}}$, the MOND regime [1]. MOND and its Lagrangian form AQUAL [2] take a_0 as a free constant and leave its closeness to cH_0 unexplained. Relativistic completions [3, 4] add fields but not the value. Ultralight dark matter [16] suppresses small-scale structure through quantum pressure and forms solitonic cores [18].

TRSD derives the scale and the interpolation function from the horizon's C_3 geometry, and obtains the dark sector from the same scalar field: its phase, turning at the rate the C_3 eigenvalue fixes and held at that rate, behaves as cold dust (§21.1). The paper runs as follows: the coil generator (§2); the axiom and its canonical form (§3–6); the relaxation flow and terminal state (§7); isotropy and the C_3 cycle (§8–9); the threshold response and the interpolation family (§10–13); the SPARC confrontation (§14); Newtonian recovery (§15); mass as deficit (§16); the Torridity family and cascade (§17–18); information and regeneration (§19–20); the dark sector (§21–23); cosmology (§24); the CMB (§25); parameter structure (§26); discussion (§27–28); and the reset (§29). Signature $(-,+,+,+)$; SI units.

2 The Coil Generator

The horizon is modelled by a coil: two conjugate helical lobes, one right-handed and one left-handed, meeting at a waist.

2.1 Parametrization

$$\mathbf{r}_{\pm}(t) = \big(R_0 \sin t \cos \omega t,\ \pm R_0 \sin t \sin \omega t,\ \pm p\, t\big), \qquad t \in [0, \pi], \tag{1}$$

with maximum radius R_0, twist rate ω and pitch p. The radius $R_0 \sin t$ vanishes at $t = 0$ and $t = \pi$, so each lobe pinches at both ends; the lobes are conjugate in twist and in height and share the pinch at $t = 0$. Seen from the side the pair traces a bicone-like envelope, seen at 45° an S (the "C/S" shape of draft 1), and seen along the axis two opposite spirals.

2.2 The tangent bicone

DERIVED Near the waist $R \simeq R_0 t$ and $z = pt$, so the lobes lie on the cone of half-angle

$$\tan \alpha = \frac{R_0}{p}. \tag{2}$$

Away from the waist the envelope $R = R_0 \sin(z/p)$ closes into a spindle; the bicone is its tangent cone at the waist. The pitch $p = R_0$ gives $\alpha = 45°$, the light cone $h = r$; the pitch $p = \sqrt{3}\, R_0$ gives $\alpha = 30°$, the defect cone of §3.3, whose surface unrolls to a half-disk.

2.3 Crossings

Proposition 1 (Crossing count). DERIVED *The axial projections of the two lobes coincide at equal parameter exactly when* $\sin \omega t = 0$. *In* $[0, \pi]$ *there are* $\lfloor \omega \rfloor + 1$ *such crossings, so there are exactly three if and only if* $2 \le \omega < 3$. *The crossing* $t_n = n\pi/\omega$ *lies at azimuth* $n\pi$.

Proof. The projections are $R_0 \sin t\, e^{\pm i\omega t}$. For $0 < t < \pi$ they coincide iff $e^{2i\omega t} = 1$, i.e. $\omega t = n\pi$; the ends $t = 0, \pi$ are the pinches. The values $t_n = n\pi/\omega \le \pi$ are $n = 0, \dots, \lfloor \omega \rfloor$. The azimuth of $\mathbf{r}_+$ at t_n is $\omega t_n = n\pi$. □

The count three is therefore a window of the twist rate, and the three crossings lie on one line through the axis (Figure 1). Their equal $2\pi/3$ spacing is not a property of the crossings; it is Postulate I, which places three equivalent generators at equal azimuth. The draft-1 "Lissajous" argument for the count is withdrawn (Appendix F).

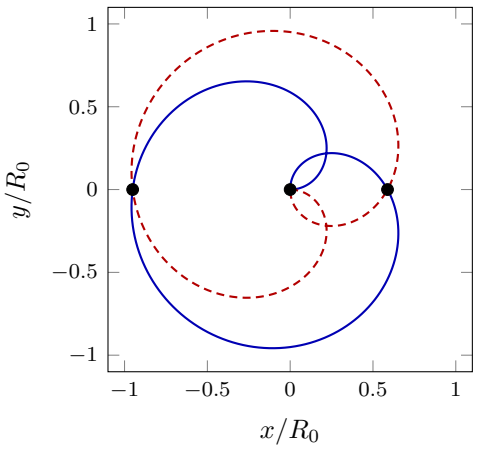


Figure 1: Axial projection of the coil (1) for $\omega = 2.5$. Solid: $\mathbf{r}_+$; dashed: $\mathbf{r}_-$. The three crossings (dots) lie at azimuths $0, \pi, 2\pi$, on one line.

2.4 The threefold beat

DERIVED The lobes (1) turn in opposite senses, $e^{\pm i\omega t}$: they are the conjugate pair, turning once and once backward for each turn, counts $+1$ and -1. Two motions of counts q_1 and q_2, superposed, keep in step under a turn by α only if $e^{i(q_1-q_2)\alpha} = 1$, so they keep the cyclic group $\mathbb{Z}_{|q_1-q_2|}$ and nothing larger. The pair keeps $\mathbb{Z}_2$, which is why its crossings lie π apart (Proposition 1). The fold of the pair—the square of the field, $\phi_+^2 \propto e^{2i\theta}$, the image of the pair under its double cover—turns twice for each turn of a lobe, and against the counter-turning lobe its relative count is three:

$$e^{2i\theta} + e^{-i\theta} = 2\cos\tfrac{3\theta}{2}\, e^{i\theta/2}, \qquad |\phi_+^2 + \phi_-|^2 = \phi^4 + \phi^2 + 2\phi^3\cos 3\theta, \tag{3}$$

the three-petalled rose, with tips at 0, $2\pi/3$ and $4\pi/3$ and all three petals through the centre. The beat keeps exactly $\mathbb{Z}_3$, and its interference term is the $\mathbb{Z}_3$ harmonic of the phase. IDENTIFICATION The threefold wobble of the geometric origin (Section 3) is this beat: it supplies the equal $2\pi/3$ spacing that the crossings alone do not, and it realizes Postulate I without replacing it. The pair against itself carries the $\mathbb{Z}_2$ of the deficit π, the fold against the lobe the $\mathbb{Z}_3$ of the C_3, and together they make $\mathbb{Z}_6$ (GRUT §1.11, §50.10 [32]).

3 Axiom and Variational Principle

Geometric origin. The cosmic horizon is not a static sphere. It is generated by two conjugate C-shaped arcs that spin about perpendicular axes, vertical and horizontal, crossing one another as they sweep the sphere. A threefold wobble in this two-C/S geometry creates three conical defects at equal azimuthal separations, each with deficit angle $\pi/3$; the total deficit is π, and the three defects close into the C_3 cycle. The threefold count is the beat of the fold against the conjugate arc (§2.4): the fold turns twice for each turn of the arc, so against the counter-turning arc it beats three times per turn, and the two keep in step only under thirds of a turn—three nodes at $2\pi/3$. (The figure restored here in release 1.4.0, two equal-frequency perpendicular motions a quarter turn apart, traces a circle and has no nodes; it stays withdrawn, Appendix F.) The spin is physical: its rims move at light speed, and its fixed rate sets the final horizon, the cosmological constant and the present (§24.14). Both are derived from the thermal geometry of the final horizon, in which every line of sight is such an arc, turning about the two points where it meets the horizon, with the observer on its rim (§24.15).

3.1 The action

Postulate II: the action

$$S = \int d^4x\,\sqrt{-g}\,\Big\{\frac{c^4}{16\pi G}R - \frac{\delta c^4}{16\pi G}\,g^{\mu\nu}\partial_\mu\phi_+\partial_\nu\phi_- - \varepsilon_V\Big[\tfrac{1}{2}(\Xi-2)^2 + V_0\Big]\Big\}, \qquad \Xi \equiv \phi_+\phi_-, \tag{4}$$

is stationary. Here δ is the conical deficit, the fields $\phi_\pm$ are dimensionless, the kinetic coefficient $\delta c^4/16\pi G = f_\delta^2/2$ is a tension ($\mathrm{J\,m^{-1}}$), and ε_V is an energy density that sets the scale of the potential. The draft-1 axiom wrote the potential as $2\Xi - \frac{1}{2}\Xi^2 - 4 = -[\frac{1}{2}(\Xi-2)^2 + 2]$ with an implicit unit $\varepsilon_V = 1$; its constant is $V_0 = 2$. Section 21 shows that V_0 must vanish.

3.2 Field structure

STANDARD The fields are the two components of one complex scalar, $\phi_\pm = \phi\, e^{\pm i\theta}$, so $\phi_- = \phi_+^*$ and there is no ghost. Then $\Xi = \phi^2$ and

$$g^{\mu\nu}\partial_\mu\phi_+\partial_\nu\phi_- = \partial_\mu\phi\,\partial^\mu\phi + \phi^2\,\partial_\mu\theta\,\partial^\mu\theta\,: \tag{5}$$

a radial mode ϕ and a phase θ. The potential is a Mexican hat: a maximum at $\Xi = 0$ and a minimum ring at $\Xi = 2$.

3.3 The deficit

DERIVED Three equivalent generators meet at the apex as three equilateral faces, each contributing its interior angle $\pi/3$; the angle around the apex is π, so

$$\delta_{\mathrm{tot}} = 2\pi - \pi = \pi, \qquad \delta_{\mathrm{gen}} = \frac{\pi}{3}. \tag{6}$$

A line defect of deficit δ through the observer leaves the solid angle $4\pi(1-\delta/2\pi)$; for $\delta_{\mathrm{tot}} = \pi$ this is 2π, so the effective horizon area is $A_{\mathrm{eff}} = 2\pi R_H^2$. Four such apexes close a sphere, $4\delta_{\mathrm{tot}} = 4\pi$, the vertex structure of the regular tetrahedron (§16.3).

4 Canonical Normalization

DERIVED The canonical field is $\chi = f_\delta\,\phi$ (units $\sqrt{\mathrm{J\,m^{-1}}}$); the Lagrangian of the scalar sector reads

$$\mathcal{L}_\chi = -\tfrac{1}{2}\partial_\mu\chi\,\partial^\mu\chi - \tfrac{1}{2}\chi^2\partial_\mu\theta\,\partial^\mu\theta - \varepsilon_V\Big[\tfrac{1}{2}\Big(\frac{\chi^2}{f_\delta^2} - 2\Big)^2 + V_0\Big]. \tag{7}$$

Expanding about the ring, $\chi = \sqrt{2}\,f_\delta + \eta$, the potential is $\varepsilon_V\big[4\eta^2/f_\delta^2 + 2\sqrt{2}\,\eta^3/f_\delta^3 + \tfrac{1}{2}\eta^4/f_\delta^4 + V_0\big]$, which gives the radial mode

$$\tilde{m}^2 = \frac{8\varepsilon_V}{f_\delta^2}, \qquad g_3 = \frac{12\sqrt{2}\,\varepsilon_V}{f_\delta^3}, \qquad \lambda = \frac{12\,\varepsilon_V}{f_\delta^4}, \tag{8}$$

with $\tilde{m} = m_\delta c/\hbar$, and a massless phase (the Goldstone mode).

Theorem 2 (Normalization). DERIVED *The dark-matter mass takes the form $m_\delta = \sqrt{8}\,\hbar/cL_*$ if and only if the potential scale is the defect tension spread over L_*^2:*

$$\varepsilon_V = \frac{f_\delta^2}{L_*^2}. \tag{9}$$

Proof. By (8), $\tilde{m}^2 = 8/L_*^2 \iff 8\varepsilon_V/f_\delta^2 = 8/L_*^2$. □

The draft-1 "Convention A" is this relation. It does not remove f_δ from the fluctuation couplings: in natural units, with $f_\delta^2 = \delta M_{\mathrm{P}}^2/8\pi$ (§5),

$$\lambda = \frac{3\tilde{m}^2}{2f_\delta^2} = \frac{12\pi}{\delta}\,\frac{m_\delta^2}{M_{\mathrm{P}}^2}, \tag{10}$$

so the physical self-coupling is suppressed by $(m_\delta/M_{\mathrm{P}})^2$; at $\delta = \pi$ it is $\lambda = 12\,m_\delta^2/M_{\mathrm{P}}^2$ (8×10^{-100} for 10^{-22} eV).

5 Defect Normalization Map

STANDARD A straight defect of tension T (energy per length) carries the deficit $\delta = 8\pi GT/c^4$ [13]. IDENTIFICATION (I1) The kinetic scale is that tension:

$$f_\delta^2 \equiv T_\delta = \frac{\delta c^4}{8\pi G}, \tag{11}$$

which reproduces the coefficient $f_\delta^2/2 = \delta c^4/16\pi G$ of (4). For $\delta_{\rm tot} = \pi$, $f_\delta^2 = c^4/8G = F_{\rm P}/8 = 1.5128 \times 10^{43}$ N, where $F_{\rm P} = c^4/G$ is the Planck force, and $f_\delta = 3.8895 \times 10^{21}\ \sqrt{\rm N}$; in natural units $f_\delta = M_{\rm P}/\sqrt{8}$. For $\delta_{\rm gen} = \pi/3$, $f_\delta^2 = F_{\rm P}/24$ and $f_\delta = 2.2456 \times 10^{21}\ \sqrt{\rm N}$.

6 Field Equations

DERIVED Varying ϕ_+ and integrating by parts,

$$\frac{f_\delta^2}{2}\,\Box\phi_- + \varepsilon_V(2-\Xi)\,\phi_- = 0 \quad\Longrightarrow\quad \Box\phi_\pm + \frac{2\varepsilon_V}{f_\delta^2}\,(2-\Xi)\,\phi_\pm = 0, \tag{12}$$

and with (9) the coefficient is $2/L_*^2$. At $\Xi = 0$ the equation reads $\Box\phi + (4/L_*^2)\phi = 0$: a tachyonic mass, the top of the hat, unstable. On the ring, $\phi = \sqrt{2} + \delta\phi$ gives $(2-\Xi)\phi \simeq -4\,\delta\phi$ and

$$\Box\,\delta\phi - \frac{8}{L_*^2}\,\delta\phi = 0, \tag{13}$$

the Klein–Gordon equation with $\tilde m^2 = 8/L_*^2$, while the phase is massless. The stress tensor is

$$T_{\mu\nu} = f_\delta^2\big[\partial_\mu\phi\,\partial_\nu\phi + \phi^2\partial_\mu\theta\,\partial_\nu\theta\big] - g_{\mu\nu}\Big\{\tfrac{f_\delta^2}{2}\big[(\partial\phi)^2 + \phi^2(\partial\theta)^2\big] + \varepsilon_V V(\phi^2)\Big\}, \tag{14}$$

and at the vacuum it reduces to $T_{\mu\nu} = -\varepsilon_V V_0\, g_{\mu\nu}$, an equation of state $w = -1$.

7 Relaxation Flow and the Terminal State

7.1 Relaxation

DERIVED Overdamped (relaxational) dynamics along a flow parameter s follow the gradient of the potential, $d\phi_\pm/ds = -\Gamma\,\partial V/\partial\phi_\mp = \Gamma(2-\Xi)\phi_\pm$, so

$$\frac{d\Xi}{ds} = 2\Gamma\,(2-\Xi)\,\Xi. \tag{15}$$

Theorem 3 (Reconciliation). DERIVED *In the displacement from the vacuum, $u \equiv 2 - \Xi$,*

$$\frac{du}{ds} = 2\Gamma\,u\,(u-2), \tag{16}$$

which is the draft-1 flow $d\Xi/ds = (4/f_\delta^2)(\Xi-2)\Xi$ with $\Xi \to u$ and $2\Gamma = 4/f_\delta^2$. Its stable fixed point $u = 0$ is the vacuum $\Xi = 2$; the unstable one, $u = 2$, is the top of the hat.

Proof. Substitute $\Xi = 2 - u$ in (15): $du/ds = -2\Gamma u(2-u)$. □

The draft-1 flow was derived from an eikonal replacement $\Box \to d/ds$, which does not hold (the eikonal limit yields the null condition and an amplitude transport equation, not a flow of Ξ); read as relaxation in u, the same equation is correct, and its attractor is the vacuum, as in §29.

7.2 Sourced flow

DERIVED An exchange-symmetric source, $d\phi_\pm/ds = \Gamma(2-\Xi)\phi_\pm - \sigma(s)/2\phi_\mp$, adds $-\sigma$ to (15). Near the vacuum $du/ds + 4\Gamma u = \sigma(s)$, with the exact solution

$$u(s) = e^{-4\Gamma(s-s_0)}u(s_0) + \int_{s_0}^{s} e^{-4\Gamma(s-t)}\sigma(t)\,dt, \tag{17}$$

so $\sigma \to 0$ implies $u \to 0$ exponentially, at the rate $4\Gamma = 8/f_\delta^2$ of draft 1. The draft-1 source $K/\phi_\mp\Xi$ is one choice of σ.

7.3 Terminal state

The condensate relaxes to the vacuum, $u \to 0$. Massive bodies lose peculiar momentum as $p \propto 1/a$ and settle into the comoving frame. If the dark-sector length grows with the horizon in the terminal phase, $L_* \propto R_H$ (§29), then $m_\delta \to 0$ and the dark sector joins the massless one on the null cone. HYPOTHESIS The terminal state is $(u, m_\delta) \to (0, 0)$ with every remaining massless degree of freedom at $v = c$.

8 Isotropy of the Vacuum

DERIVED For k unit vectors $\mathbf{u}_j$, the generating function $W_v(\mathbf{x}) = \ln \sum_j e^{\mathbf{x}\cdot\mathbf{u}_j}$ has zero gradient at $\mathbf{x} = 0$ when $\sum_j \mathbf{u}_j = 0$, and Hessian $T_{ab} = k^{-1} \sum_j u_{j,a} u_{j,b}$. Rotational isotropy requires $T_{ab} = C\delta_{ab}$, and the trace gives $3C = 1$:

$$T_{ab} = \tfrac{1}{3}\,\delta_{ab}, \qquad W_v(\mathbf{x}) = W_v(0) + \frac{|\mathbf{x}|^2}{6} + O(|\mathbf{x}|^3). \tag{18}$$

The second moment $\langle u^2 \rangle = \frac{1}{3}$ is the source of the factor 3 in the angular average of §11: a linear response must return itself, which requires the weight $1/\langle u^2 \rangle = 3$.

9 The C_3 Cycle and the Critical Acceleration

DERIVED The minimal closed cycle that carries a cone is the triangle: C_2 is degenerate and C_n, $n \geq 4$, is not minimal. Dividing the horizon circumference $2\pi R_H$ into three arcs of length $\ell = 2\pi R_H/3$, the discrete Laplacian $(\Delta f)_n = \ell^{-2}(2f_n - f_{n+1} - f_{n-1})$ has eigenvalues $\lambda_m = (2/\ell^2)[1 - \cos(2\pi m/3)]$, and

$$\lambda_1 = \frac{3}{\ell^2} = \frac{27}{4\pi^2 R_H^2}. \tag{19}$$

For C_n, $\lambda_1 R_H^2 = (n^2/2\pi^2)(1 - \cos 2\pi/n)$, which rises to 1 as $n \to \infty$; C_3 gives 0.683918. A light-like mode has $\omega_1 = c\sqrt{\lambda_1} = 3\sqrt{3}\,c/2\pi R_H$ and period $\tau_1 = 4\pi^2 R_H/3\sqrt{3}\,c$.

Definition 4 (I2)**.** The critical acceleration brings a body from rest to c in one cycle period, $a_c \equiv c/\tau_1$. (A centripetal reading, $c\,\omega_1$, would be 2π larger.)

Then

$$a_c = \frac{3\sqrt{3}}{4\pi^2}\,cH_0 = 0.131620\,cH_0, \tag{20}$$

that is 8.612×10^{-11} m s^{-2} at the framework's $H_0 = 67.35$ (9.340×10^{-11} at the legacy calibration 73.04 km s^{-1} Mpc^{-1}).

10 Exposure, Survival and Activation

DERIVED (I3, derived) A mode on the cycle is activated by a local acceleration A as a Poisson process whose rate per unit flow parameter is proportional to the load $x \equiv A/a_c$, with the rate constant β of the relaxation operator $d/ds + \beta$. Its exposure is one memoryless stage per arc: on a cycle of n arcs the exposure is Erlang-distributed with density $p_n(s) = (n\beta)^n s^{n-1} e^{-n\beta s}/(n-1)!$ and mean $1/\beta$.

This is derived (GRUT §22.10 [32]): a gapless mode extended around the cycle, carried at acceleration A, sees the Rindler thermal state [60], and its excitation rate $g^2 A/4\pi^2 c$ is linear

in the load; at weak coupling the excitations are independent, so the count is Poisson and the stages memoryless; and with unit coupling per unit length the cycle's length 2π gives $g^2 = 4\pi^2$, one activation per unit of horizon approach.

Lemma 5 (Survival). DERIVED *The probability that a mode is not activated is*

$$S_n(x) = \Big(1 + \frac{x}{n}\Big)^{-n}, \tag{21}$$

independent of β.

Proof. $S_n = \int_0^\infty p_n(s) e^{-\beta x s} ds = (n\beta)^n/(n\beta + \beta x)^n$. □

The single-stage law is the draft-1 threshold distribution, $S_1 = a_c/(A + a_c)$, and β cancels as draft 1 found; a fixed exposure gives $S_\infty = e^{-x}$. The index n—a Gamma exposure for non-integer values—is the order of the Newtonian tail: a coordinate on shapes, constrained by data (§14–15), not a count of anything. The draft-1 alternative route—a Poisson exponential "matched" to a Pareto tail—does not produce $1/(1 + x)$; the mixture over the exposure does, and it replaces that route. The active fraction is $f_n = 1 - S_n$, with density $p_A = -dS/dA$; for $n = 1$, $p_A = a_c/(A + a_c)^2$.

11 Angular Reduction and the Interpolation Family

IDENTIFICATION The response carried along a direction is the load times the active fraction, $j_n(x) = x\, f_n(x)$. DERIVED Averaging over directions with $u = \cos\theta$: the projected load is xu, its radial projection carries a second u, isotropy weights u uniformly on $[0, 1]$ ($d\Omega = 2\pi\, du$), and the weight $3 = 1/\langle u^2\rangle$ (§8) makes a linear response return itself:

$$\sigma_n(x) = 3\int_0^1 j_n(xu)\, u\, du, \qquad \mu_n(x) \equiv \frac{\sigma_n(x)}{x} = 3\int_0^1 u^2 f_n(xu)\, du. \tag{22}$$

For $n = 1$, long division, $u^3/(1 + xu) = u^2/x - u/x^2 + 1/x^3 - 1/[x^3(1 + xu)]$, gives $\int_0^1 u^3/(1 + xu)\, du = 1/3x - 1/2x^2 + 1/x^3 - \ln(1+x)/x^4$ (Appendix D); for $n = 3$ the substitution $w = 1 + xu/3$ closes the integral. Thus

$$\mu_1(x) = 1 - \frac{3}{2x} + \frac{3}{x^2} - \frac{3\ln(1 + x)}{x^3}, \tag{23}$$

$$\mu_2(x) = 1 - \frac{24}{x^3}\Big[\frac{x}{2} + \frac{x/2}{1 + x/2} - 2\ln\Big(1 + \frac{x}{2}\Big)\Big], \tag{24}$$

$$\mu_3(x) = 1 - \frac{81}{x^3}\Big[\ln\Big(1 + \frac{x}{3}\Big) + \frac{2}{1 + x/3} - \frac{1}{2(1 + x/3)^2} - \frac{3}{2}\Big]. \tag{25}$$

Their series are

$$\mu_1 = \tfrac{3}{4}x - \tfrac{3}{5}x^2 + \tfrac{1}{2}x^3 - \cdots, \quad \mu_2 = \tfrac{3}{4}x - \tfrac{9}{20}x^2 + \tfrac{1}{4}x^3 - \cdots, \quad \mu_3 = \tfrac{3}{4}x - \tfrac{2}{5}x^2 + \tfrac{5}{27}x^3 - \cdots. \tag{26}$$

(draft 1 stated the deep limit as $\sigma/x \to \frac{3}{4}$; the correct statement is $\sigma/x^2 \to 3\int_0^1 u^3 du = \frac{3}{4}$, i.e. $\mu \simeq \frac{3}{4}x$.)

Theorem 6 (Universality of $\frac{3}{4}$). DERIVED *Assume (a) activation is a Poisson process whose hazard is linear in the load, (b) the exposure has a finite mean, used as the unit, (c) the response is the load times the active fraction, and (d) directions are averaged as in (22). Then, for every exposure distribution,* $\mu(x) = \frac{3}{4}x + O(x^2)$.

Proof. S is the Laplace transform of the exposure density in units of its mean, so $S(0) = 1$, $S'(0) = -1$, $f(y) = y + O(y^2)$, and $\mu = 3\int_0^1 u^2(xu)\,du + O(x^2) = \frac{3}{4}x + O(x^2)$. □

The theorem covers the whole class (a)–(d), not only the Erlang laws; outside it the slope changes. Hypothesis (a) is itself required by data:

Proposition 7 (Linear hazard). DERIVED *If the hazard grows as x^p, then $\mu \simeq \frac{3}{3+p}x^p$ and the deep regime gives $g \propto g_{\rm bar}^{1/(1+p)}$, so rotation curves are asymptotically flat, with $V^4 \propto M$, only for $p = 1$. The baryonic Tully–Fisher relation [34] fixes $p = 1$. If the exposure mean diverges, $S'(0)$ does too and μ is not linear at small x.*

Proposition 8 (Transition and tail). DERIVED *For real n other than $1, 2, 3$, with $a = x/n$ and $w = 1 + a$,*

$$1 - \mu_n(x) = \frac{3}{a^3}\Big[\frac{w^{3-n} - 1}{3 - n} - \frac{2(w^{2-n} - 1)}{2 - n} + \frac{w^{1-n} - 1}{1 - n}\Big], \tag{27}$$

so the tail is $1 - \mu_n \simeq \frac{3n^n}{3-n}\,x^{-n}$ for $n < 3$, while the transition moves little: $\mu_n(1) = 0.4206$, 0.4515, 0.4557, 0.4601, 0.4623 for $n = 1$, 1.62, 1.75, 1.91, 2. In the Torridity form $z/(1+z^n)^{1/n}$ one exponent sets both the transition width and the tail, $1 - \mu \simeq z^{-n}/n$.

Proof. Substituting $w = 1 + au$ in $3\int_0^1 u^2(1 + au)^{-n}du$ gives (27); its leading term at large a is $3a^{-n}/(3 - n)$. The Torridity tail is the expansion of $(1 + z^{-n})^{-1/n}$. □

12 Free Energy and the Field Equation

DERIVED With $F_{\rm int}(x) = \int_0^x \sigma_1(s)\,ds = \frac{x^2}{2} - \frac{3x}{2} + \frac{3\ln(1+x)}{x} + 3\ln(1+x) - 3$ and $\mathcal{F}(y) = 2F_{\rm int}(\sqrt{y})$, one has $\mathcal{F}'(y) = \mu_1(\sqrt{y})$ (verified symbolically; the same construction holds for every μ_n). STANDARD The Lagrangian $\mathcal{L} = -\frac{a_c^2}{8\pi G}\,\mathcal{F}\big(|\nabla\Phi|^2/a_c^2\big) - \rho\Phi$ gives

$$\nabla\cdot\Big[\mu\Big(\frac{|\nabla\Phi|}{a_c}\Big)\nabla\Phi\Big] = 4\pi G\rho, \tag{28}$$

the AQUAL form [2] with a derived μ.

13 The Deep Regime and the Scale

DERIVED For spherical symmetry Gauss's theorem turns (28) into $\mu(g/a_c)\,g = g_{\rm bar} \equiv GM(r)/r^2$ exactly. In the deep regime $\mu \simeq \frac{3}{4}g/a_c$, so $\frac{3}{4}g^2/a_c = g_{\rm bar}$ and $g = \sqrt{\frac{4}{3}a_c g_{\rm bar}}$:

$$a_0 = \frac{4}{3}\,a_c = \frac{\sqrt{3}}{\pi^2}\,cH_0 \equiv \alpha_{\rm TRSD}\,cH_0, \qquad \alpha_{\rm TRSD} = 0.175493. \tag{29}$$

(draft 1 printed 0.175435; the correct value is 0.175493.) PREDICTION The scale is $a_0 = 1.148 \times 10^{-10}\ \mathrm{m\,s^{-2}}$ at the framework's $H_0 = 67.35$ (1.2454×10^{-10} at the legacy calibration 73.04). By Theorem 6 it does not depend on the exposure law.

14 Confrontation with the SPARC Radial Acceleration Relation

MEASURED The data are the 2696 points of 147 SPARC galaxies used to define the RAR [10, 11], with $\Upsilon_{\rm disk} = 0.5$ and $\Upsilon_{\rm bul} = 0.7$. The predicted acceleration solves $\mu(g/a_c)\,g = g_{\rm bar}$ point by

point; residuals are $\log_{10} g_{\rm obs} - \log_{10} g_{\rm pred}$. Statistical errors come from resampling galaxies (200 bootstrap draws); the systematic error of the scale is ±20%, dominated by Υ [10].

14.1 No free parameter

With a_c fixed by H_0 and nothing fitted:

law	$H_0 = 67.12$: mean	rms	$H_0 = 73.04$: mean	rms
μ_1 (single stage)	−0.007 dex	0.133	−0.019 dex	0.134
μ_2 (two stages)	+0.016 dex	0.134	+0.003 dex	0.133
μ_3 (three arcs)	+0.024 dex	0.136	+0.011 dex	0.134
Torridity, $n = \pi/3$ (§17)	+0.005 dex	0.133	−0.008 dex	0.133
draft 1 screened law (§15)	–	–	+0.173 dex	0.225

The observed scatter about the relation is 0.13 dex [10]. With no parameter adjusted, every member reproduces the relation to within 0.025 dex (6%) in the mean, at the scatter of the best empirical fit (Figure 2).

14.2 Free scale

Fitting the scale alone:

law	fitted a_0 [10^{-10} m s^{-2}]	rms (dex)	implied $H_0 = a_0/\alpha_{\rm TRSD} c$
empirical $g/g_{\rm bar} = [1 - e^{-\sqrt{g_{\rm bar}/g_\dagger}}]^{-1}$	$g_\dagger = 1.161 \pm 0.071$	0.133	–
μ_1	1.105 ± 0.071	0.133	$64.8 \pm 4.1 \pm 13$
μ_2	1.249 ± 0.075	0.133	$73.2 \pm 4.4 \pm 15$
μ_3	1.301 ± 0.079	0.134	$76.3 \pm 4.6 \pm 15$

The empirical fit reproduces the published $g_\dagger = 1.20 \pm 0.02 \pm 0.24$ within its errors, which validates the pipeline. Read through (29), galaxies measure H_0 at the ±6% statistical and ±20% systematic level; the CMB value 67.35 lies 0.6σ (statistical) from the μ_1 value, 1.4σ from the μ_2 value and 2.0σ from the μ_3 value, and within 0.6σ of both once the systematic is included. The draft-1 statement of agreement at 0.056%, and its inversion to 0.04%, are withdrawn: the relation is one test, and the data determine a_0 to 6%, not 0.06%.

14.3 The rotation-curve likelihood

MEASURED The RAR points take the mass-to-light ratios, distances and inclinations as known. The full likelihood fits the velocities of 153 galaxies ($Q < 3$, $i \geq 30°$; 3168 points) with per-galaxy nuisance parameters: $\log \Upsilon_{\rm disk}$ and $\log \Upsilon_{\rm bul}$ with lognormal priors of 0.1 dex about 0.5 and 0.7, and D and i with Gaussian priors from the SPARC table—490 in all, following [33]. Distances rescale radii and the baryonic velocities as $\sqrt{D}$; inclinations rescale the observed velocities and their errors by $\sin i_0 / \sin i$. Each galaxy's χ^2 is minimized over its nuisance parameters and the global scale is scanned. The best χ^2 is 14370 for 3167 degrees of freedom (median per-galaxy $\chi^2_\nu = 2.1$): the quoted velocity errors omit systematics, and errors below are scaled by $S = \sqrt{\chi^2/\nu} = 2.13$. Two forms are compared—the threshold family with real tail order n and the Torridity form of draft 1 (§17)—with the empirical function of [10] as a benchmark:

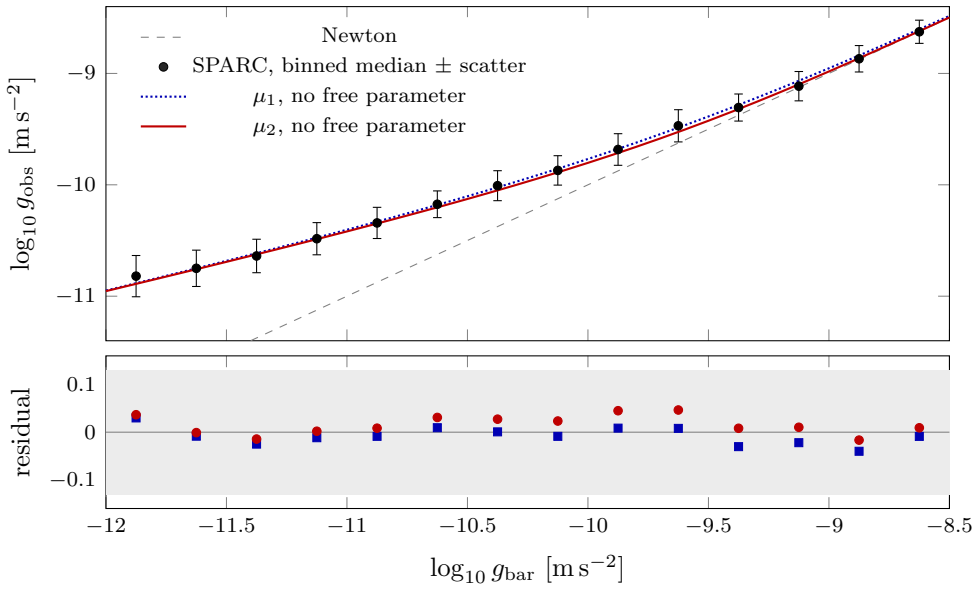


Figure 2: The SPARC radial acceleration relation against TRSD with $H_0 = 67.12$ from the CMB (the withdrawn budget's value; at the framework's 67.35 every curve moves by 0.3%) and no fitted parameter. Top: binned medians of the 2696 points (bars: scatter in each bin). Bottom: median residuals per bin for μ_1 (squares) and μ_2 (circles); shaded, the observed scatter of 0.13 dex.

form	n	a_0 [10^{-10}]	$\Delta\chi^2/S^2$	$\Delta\chi^2/S^2$ at $a_0 = 1.144$
threshold	1	1.063	4.7	3.5
threshold	1.38 (optimum)	1.045	0	5.5
threshold	1.75 (window centre)	1.037	2.2	6.7
threshold	2	1.034	5.1	7.2
threshold	3	1.029	18.6	8.3
Torridity	0.97	1.057	4.4	4.0
Torridity	$\pi/3$	1.081	9.2	2.2
Torridity	$2\pi/3$	1.201	397	2.7
benchmark	–	1.009	−13.6	–

Scale errors are ±0.04 (statistical, scaled) throughout; $\Delta\chi^2/S^2$ is measured from the best threshold member.

Transition and tail. Galaxies constrain the width of the transition, the planets the Newtonian tail (§15). In the threshold family the two are separate (Proposition 8): galaxies give $n = 1.38 \pm 0.20$, the planets $n \geq 1.62$, and the window $1.62 \leq n \leq 1.91$ satisfies both while fitting better than any Torridity exponent ($\Delta\chi^2/S^2 = 4.4$). In the Torridity form one exponent sets both: galaxies give 0.96 ± 0.04, the planets require $n \geq 1.53$ (§17), and no exponent satisfies both.

The benchmark. The empirical function leads the best threshold member by $\Delta\chi^2/S^2 = 13.6$. Its form was chosen on these same data [10], so part of the lead is selection: it is a benchmark, not a competing prediction, and it is derived from nothing.

Scale. The fitted scale depends little on the shape: across the window it runs from 1.041 to 1.037. With the nuisance parameters fixed at their nominal values the same likelihood returns $g_\dagger = 1.206$, reproducing the published 1.20; profiling them lowers every scale by about 16%, while moving the $\Upsilon_{\rm disk}$ prior centre to 0.4 or 0.6 shifts it by only ±1.5%. Taking half the fixed-versus-profiled shift as the systematic, the window gives $a_0 = 1.04 \pm 0.04\,({\rm stat}) \pm 0.10\,({\rm nuisance})$. The prediction 1.148 is 1.0σ high (2.7σ on the statistical error alone), and 2.0σ high at $H_0 = 73.04$; read through (29), galaxies give $H_0 = 60.8 \pm 2.2 \pm 5.9$.

Uncertainty. The agreement is conditional on the nuisance model: distances, inclinations and mass-to-light ratios carry the uncertainty that the scale inherits, and it is not a theory-only error.

14.4 What the data constrain

DATA-CONSTRAINED Not the index. n is a coordinate on shapes, and different forms assign different values to nearly the same shape: the threshold member $n = 1.38$ and the Torridity member $n = 0.96$ describe almost the same transition. What the data constrain is a pair of physical properties—a transition as gradual as the galaxies require and a Newtonian tail as steep as the planets require—and of the two forms only the threshold family supplies both. Numbers below are quoted at the window centre, $n = 1.75$; across the window they change by less than 3%.

15 Newtonian Recovery

DERIVED At high acceleration $1 - \mu_n = 3\int_0^1 u^2 S_n(xu)\,du$, and

$$1-\mu_1 \simeq \frac{3}{2x}, \qquad 1-\mu_2 \simeq \frac{12}{x^2}, \qquad 1-\mu_3 \simeq \frac{81}{x^3}\Big[\ln\frac{x}{3}-\frac{3}{2}\Big]. \tag{30}$$

The single-stage law leaves a constant anomaly $g - g_{\rm bar} \to \frac{3}{2}a_c = 1.28\times10^{-10}$ m s^{-2}. An inward anomaly $A \propto r^k$ turns a perihelion back by $\frac{k+2}{2}\,2\pi A a^2\sqrt{1-e^2}/GM_\odot$ per orbit; for this constant anomaly ($k = 0$) that is -1.7×10^3 mas/cy for Mercury and -2.8×10^3 mas/cy for the Earth, against the ephemeris residuals -2.0 ± 3.0 and 0.19 ± 0.19 mas/cy [7]. A tail x^{-n} has $k = 2n-2$. Requiring both perihelia within 2σ of their residuals gives $n \geq 1.62$ for the threshold family and $n \geq 1.53$ for the Torridity form (§17). At the window centre, $n = 1.75$, the shifts are -0.004 (Mercury) and -0.028 mas/cy (the Earth); at the lower edge of the window the Earth's reaches -0.20 mas/cy, the present sensitivity. (Mercury sits at $x = 4.6\times10^8$; draft 1 gave 2×10^{10}.)

The draft-1 screening is retired. Writing $1-\mu_{\rm scr} = (1-\mu_1)(1+x)^{-\beta}$ gives $\mu_{\rm scr} \simeq (\frac{3}{4}+\beta)\,x$ at small x: the factor changes the deep slope and with it a_0. For $\beta = \pi/2$ the slope is 2.32, the scale falls to $0.43\,a_c$, and the fixed-scale fit to SPARC moves by +0.17 dex with rms 0.225 (§14.1). Every exposure law keeps $\frac{3}{4}$ exactly (Theorem 6); tail orders in the window also fit the rotation curves and pass the planets. DERIVED The solar-system test is the quadrupole induced by the Galactic external field [5], which Cassini constrains [6]; it probes μ near $x \sim 2$. With baryons as the only source the law gives $Q_2 = 3.1\times10^{-26}$ s^{-2} across the window (GRUT §49.10 [32]), twenty times the Cassini/DE440 bound $(1.6\pm1.8)\times10^{-27}$ [41]: a boost that belongs to the baryons' field cannot hold in the Solar System [42]. With the condensate as the source (§15.1) the Sun carries no bound condensate, and the quadrupole is the Galactic tide alone, 8.6×10^{-31} s^{-2}.

15.1 The complete equation

DERIVED The Cassini failure shows that the baryon-only equation is incomplete; the missing source is real matter: the condensate of §21–22 in the condensate reading, the rotation's medium bound around the baryons in the rotation reading (§21.1). The complete equation is

$$\nabla^2\Phi = 4\pi G(\rho_b + \rho_\chi), \qquad \rho_\chi(\mathbf{k}) = \frac{\rho_{\rm ph}(\mathbf{k})}{1+(k/k_J)^4}, \qquad k_J = (16\pi G\bar\rho_\chi)^{1/4}(m_\delta/\hbar)^{1/2}, \tag{31}$$

with $\rho_{\rm ph}$ the phantom density of the system's own baryons (§23), and the bound mass limited by the condensate available. DERIVED The bound density is derived: $1-\mu(x) = 3\int_0^1 u^2 S(xu)\,du$

is the cosine-weighted fraction of condensate elements that survive the threshold, and if a surviving element carries unit gravitational susceptibility the condensate's polarization is $\mathbf{P} = (1-\mu)\,\mathbf{g}/4\pi G$, whose bound density $-\nabla\cdot\mathbf{P}$ is exactly $\rho_{\rm ph}$ (GRUT Theorem 47.6). Real matter obeys the equivalence principle, so a uniform external field does not enter; quantum pressure forbids condensate structure below $\lambda_J = 2\pi/k_J$, which is 40–290 pc for $m_\delta = 10^{-19}$–2×10^{-20} eV. Galaxies, larger than λ_J, keep the results of §14; the Solar System, far smaller, is Newtonian, and its only quadrupole is the Galactic tide, 8.6×10^{-31} s^{-2}, within Cassini's bound at -0.9σ. The field is also bound only where it has been captured: for a Maxwellian field of dispersion σ the captured fraction is $\Gamma = \operatorname{erf} X - (2X/\sqrt{\pi})e^{-X^2}$ with $X = v_{\rm esc}/\sqrt{2}\sigma$, and the Sun, formed inside the Galaxy where the field streams at ~ 165 km s^{-1}, has $\Gamma = 7\times10^{-9}$ at its MOND radius (GRUT §47). The law changes with scale because the matter that carries the boost has a Jeans length and a finite supply: Newtonian below λ_J, the threshold response across galaxies, Keplerian beyond $r_t = (M_\chi/M_b)\sqrt{GM_b/a_0}$. The one identification left is that μ is the gravitational permittivity of the condensate (GRUT §46–47 [32]).

15.2 The relaxation state

DERIVED The complete equation gives the end state of a system's condensate. A bound system starts from its *infall configuration*—on scales above λ_J the condensate is pressureless, so a collapsing system gathers its cosmic share, Ω_c/Ω_b times its baryons, into a collapsed halo—and relaxes toward the *support level*, the polarization equilibrium, with its isothermal halo and flat curve. The *resistance level*, Newton, is reached only where no condensate is bound, as in the Solar System. With f the polarized fraction, $M_\chi = fM_{\rm pol} + (1-f)M_{\rm inf}$. For a representative rich cluster the support law carries 0.28–0.68 of the hydrostatic mass between 250 and 1500 kpc, reduced capture ($\Gamma = 0.5$) only 0.13–0.25, and the infall configuration 0.82–0.84: the unrelaxed state of a bound system is its infall halo, not Newton. A disk has lived about 100 crossing times since assembly and a cluster about four, so if polarization takes $N_{\rm rel}$ crossing times, galaxies reach support and clusters do not for $11 \lesssim N_{\rm rel} \lesssim 34$. A merger before support is a trap, with probability $1 - e^{-t_{\rm rel}/t_{\rm dis}}$: 0.97–1.00 for clusters, 0.10–0.28 for disks. The scatter about the support law is proportional to the spread of $1-f$, so relaxed systems are quiet and systems in transit are not. PREDICTION The mass discrepancy against the support law is ordered by dynamical age since the last major merger; early-assembled fossil groups lie closer to it than clusters of the same mass; disturbed galaxies scatter more about the radial acceleration relation than isolated disks. DERIVED A bound system's clock is fixed at formation, $t_{\rm cross} = 1/10H(t_f)$, and IDENTIFICATION with the polarization renewed once per arc of the horizon's C_3 cycle at that epoch, $t_{\rm rel} = \tau_1(t_f)/3$ and $N_{\rm rel} = 40\pi^2/9\sqrt{3} = 25.3$, inside the window. In the matter era the relaxation state then depends only on $D = t/t_f$, the age of the universe in units of its age at formation: $f = 1 - e^{-(D-1)/x_H}$ with $x_H = 2\pi^2/3\sqrt{3} = 3.80$, the horizon load of §24.5. A cluster assembled at $z_f = 0.5$ has $f = 0.17$ and carries 0.74–0.80 of its hydrostatic mass; a disk formed at $z_f = 6$ has $f = 0.97$ (GRUT §47.6–47.7 [32]).

15.3 Cluster mergers and the post-passage relaxation clock

DERIVED *Collisionless and collisional components.* The phase sector obeys $\nabla_\mu(\lambda\partial^\mu\theta) = 0$ with $X = X_0$ and no baryonic source. With $u_\mu = \partial_\mu\theta/\sqrt{X_0}$, the fixed-norm identity gives $u^\nu\nabla_\nu u_\mu = 0$ and the on-shell phase stress is pressureless, $T^{(\chi)}_{\mu\nu} = 2\lambda\partial_\mu\theta\partial_\nu\theta$. In a merger this component therefore has no gas-shock drag term in the model; the X-ray emitting plasma instead experiences pressure forces and shocks. The collisionless interpretation is a model correspondence, not an observation of the microscopic identity of the lensing mass.

IDENTIFICATION *Conditional relaxation closure.* The framework identifies the C_3 rate as $\omega_{C_3} =$

$(3\sqrt{3}/2\pi)H(z)$ and sets $\omega_{C_3} t_{\rm rel} = 2\pi/3$, giving

$$t_{\rm rel}(z) = \frac{4\pi^2}{9\sqrt{3}\,H(z)}. \tag{32}$$

The exponential transfer fraction follows if the post-passage response is first order, $\dot{f} = (1 - f)/t_{\rm rel}$, with $f(0) = 0$:

$$\boxed{\begin{aligned} &f(\Delta t) = 1 - \exp\left[-\frac{\Delta t}{t_{\rm rel}(z)}\right] = 1 - \exp\left[-\frac{9\sqrt{3}}{4\pi^2} H(z)\Delta t\right], \quad \Delta t \geq 0, \\ &\kappa_{\rm model} = \kappa_b + (1 - f)(\kappa_A + \kappa_B) + f\,\kappa_{\rm pol}, \qquad M_{\chi,i}^{\rm model} = \frac{\Omega_c}{\Omega_b} M_{b,i}^{\rm pre}. \end{aligned}} \tag{33}$$

Here κ_b is the baryonic contribution, κ_A and κ_B are the two pre-passage collisionless-like components, and $\kappa_{\rm pol}$ is the model's later support/polarization template. The mixture form and the pre-merger baryon-share assignment are model closures; neither follows from geodesic flow alone. The transfer law is derived only conditional on these declared closures.

HYPOTHESIS *Correct interpretation of the three phases.* Before first core passage, the subclusters are in an infall configuration. The condition $f = 0$ is the initial condition at the post-passage clock's reset, $\Delta t = 0$; it should not be read as a universal claim that all pre-merger systems have zero relaxation. At core passage, the gas is shock-heated and decelerated while the galaxies and lensing-inferred mass peaks can pass through with smaller drag. At later times f increases according to (33). For the Bullet Cluster, $t_{\rm rel} \simeq 31.4$ Gyr at $z \simeq 0.296$; hence $f(0.15\,{\rm Gyr}) \simeq 0.0048$, $f(0.20\,{\rm Gyr}) \simeq 0.0064$, $f(1\,{\rm Gyr}) \simeq 0.031$, and $f(2\,{\rm Gyr}) \simeq 0.062$. The earlier value 0.0048 corresponds to 0.15 Gyr, not 0.20 Gyr.

STANDARD Observationally, the Bullet Cluster and MACS J0025.4−1222 show spatial separation between lensing-inferred mass and X-ray gas (NASA Bullet Cluster overview; NASA MACS J0025 overview). El Gordo is an ongoing high-redshift merger commonly reconstructed shortly after first core passage, with alternative outgoing/returning solutions (MNRAS dynamical reconstruction); Pandora's Cluster/Abell 2744 is a multi-subcluster collision over an extended interval, not a simple relaxed endpoint (NASA overview). These are distinct systems at different epochs, not a literal time-lapse of one object; merger-phase and projection degeneracies remain (2026 review).

The physical analogy is limited to local dynamics: collisional gas and a collisionless gravitating component respond differently to cluster passage, while a disturbed remnant evolves afterward. It does not establish that the Big Bang was a cluster-like collision, that the conformal scale factor changes sign, or that information crosses an aeon boundary. The proposed cross-scale equality of $t_{\rm rel}H$ is therefore retained as a framework hypothesis/closure, not promoted to an observationally established law. Testing is deferred in this revision, as requested.

DERIVED *Computed, on synthetic maps.* A collision separates an excess on the galaxies from a law computed on the baryons present ($T = 2\ln B = 97.7$, $p_{\rm MC} < 5 \times 10^{-5}$) and a swapped assignment ($T = 99.1$), and does not separate TRSD from particle dark matter ($T = -0.08$, power 0.053). The mesh of the computation is not converged (the peak moves 9.6% from 64 to 96 cells), so offsets below about 30 kpc are unresolved. PREDICTION No galaxy-free peak on the gas above $f(\Delta t) \leq 0.06$ of the excess; a robust one holding $\gtrsim 10\%$ would falsify this (Abell 520 is the test, not a prediction; the dark core of [106, 107] is absent at comparable significance in [108, 109]). Register rows 41–42.

16 Mass as Conical Defect and the Closure Mass

16.1 Curvature of a cone

STANDARD For a slice $ds^2 = dr^2 + r^2 d\theta^2$, $\theta \in [0, 2\pi - \delta)$, the Gauss–Bonnet theorem gives $\int K\, dA = \delta$, and with $R = 2K$, $\int R\sqrt{g}\, d^2x = 2\delta$. (draft 1 used 2δ in one place and δ in another.)

16.2 The deficit–mass map

STANDARD A straight string of mass per length μ has $\delta = 8\pi G\mu/c^2$ [13]; reducing 3+1 gravity along a length L gives the 2+1 coupling $G_3 = G/L$, and a point mass in 2+1 gravity carries $\delta = 8\pi G_3 M/c^2$ [14]. IDENTIFICATION (I4) With $L = R_H$,

$$\delta(M) = \frac{8\pi G M}{c^2 R_H}, \qquad f_\delta^2 = \frac{Mc^2}{R_H} = McH_0. \tag{34}$$

The mass carrying $\delta_{\rm tot} = \pi$ is $M_{\rm total} = c^3/8GH_0$; the mass carrying 4π is $M_{\rm Hubble} = c^3/2GH_0 = 4M_{\rm total}$.

16.3 Closure

Theorem 9 (Closure mass)**.** DERIVED *In the reduced geometry a closed spatial section carrying point masses M_i requires $\sum_i \delta(M_i) = 4\pi$, that is $\sum_i M_i = M_{\rm Hubble}$. Four equal masses $M_{\rm total}$, each carrying $\delta_{\rm tot} = \pi$, realize it at the vertices of a regular tetrahedron.*

Proof. A closed surface that is flat except at cone points has total curvature $2\pi\chi = 4\pi$ (Gauss–Bonnet, or Descartes' theorem), and each point contributes its deficit. A tetrahedral vertex joins three equilateral faces, total angle π, deficit π (§3.3). □

The draft-1 remark that $\delta_{\rm tot} = 4\pi$ is "geometrically impossible" applies to a single line defect, whose solid angle $4\pi(1 - \delta/2\pi)$ vanishes at 2π; as the sum over four apexes on a closed surface, 4π is exactly the closure. The Schwarzschild radius of $M_{\rm Hubble}$ is R_H (§24.3), so closure, the horizon condition and flatness coincide.

17 The Torridity Family

IDENTIFICATION The Torridity function is the n-family

$$\mu_{\rm T}(z) = \frac{z}{(1 + z^n)^{1/n}}, \qquad z = g/a_0, \qquad n = \frac{\delta_{\rm tot}}{3} = \frac{\pi}{3}, \tag{35}$$

with $\mu_{\rm T} \simeq z = \frac{3}{4}x$ in the deep regime, the same scale as (29). MEASURED In the rotation-curve likelihood (§14.3) the Torridity exponent is 0.96 ± 0.04 (on the RAR points with fixed Υ, 0.98 ± 0.03; draft 1's per-galaxy fit, 1.050 ± 0.05), and $\pi/3$ lies at $\Delta\chi^2/S^2 = 4.9$ (2.2σ). The profile has no secondary minimum: draft 1's exponent quantization—$n = k\pi/3$, with $2\pi/3$ for disk galaxies and π for clusters—is refuted, $2\pi/3$ lying at $\Delta\chi^2/S^2 = 397$ (Figure 4). At high acceleration $1 - \mu_{\rm T} \simeq z^{-n}/n$, and the perihelia require $n \geq 1.53$; $\pi/3$ turns them back by 6×10^2 (Mercury) and 1.1×10^3 mas/cy (the Earth), two orders of magnitude beyond the residuals. DERIVED The Torridity form therefore cannot be a complete law for any exponent: the galaxies and the planets ask for different values of its one parameter, which is why draft 1 needed a screening factor. The threshold family does not (Proposition 8).

Proposition 10 (Running exponent). DERIVED *Define the local exponent $m(z)$ of an interpolation function, $z = g/a_0$, as the exponent of the one-parameter form through the same point: $z/(1+z^m)^{1/m} = \mu(z)$. For the threshold family with tail order $n < 3$, $m \to 1$ as $z \to 0$ and $m \to n$ as $z \to \infty$.*

Proof. At small z, $\mu_n = z - cz^2 + O(z^3)$ with $c = 8(n+1)/15n > 0$, while the one-parameter form is $z - z^{1+m}/m + \ldots$; matching gives $m - 1 = \ln(cm)/\ln z \to 0$. At large z, $1 - \mu_n \simeq \frac{3n^n}{3-n}(\frac{4}{3}z)^{-n}$ (Proposition 8), while the one-parameter form gives z^{-m}/m; matching gives $m \to n$. □

There is therefore no universal exponent. It runs with acceleration—from 1 far out to the tail order close in—and each system reads it at its own acceleration:

	$\log_{10} g$	m $(n = 1.62)$	m $(n = 1.75)$	m $(n = 1.91)$
far out	-14	1.014	1.017	1.020
galaxy outskirts	-11.5	1.032	1.038	1.045
galaxy transition	-10.5	1.059	1.070	1.082
inner galaxies	-9.5	1.135	1.161	1.190
Saturn	-4.19	1.507	1.612	1.740
the Earth	-2.23	1.534	1.646	1.782
Mercury	-1.40	1.542	1.656	1.794

Galaxies read the exponent between 1.03 and 1.28; the planets between 1.51 and 1.79. A one-parameter form must use one value at every acceleration, and fails; the running exponent meets both (Figure 3). The requirement the planets place on a one-parameter form, $m \geq 1.534$ at the Earth's acceleration, is met exactly at the lower edge of the window, as it must be: the two planetary thresholds are one condition. The tail order that remains is not the important part—the galactic scale does not depend on it.

draft 1's exponent $\pi/3 = 1.047$ is the running exponent at the accelerations of galaxy outskirts ($m = 1.04$–1.06 at $g = 10^{-11}\ \mathrm{m\,s^{-2}}$ across the window): draft 1 measured it where it holds. The regime dependence draft 1 anticipated with $n = k\pi/3$ is realized by the running exponent; the value $2\pi/3$ anticipated for disks is not what disks read ($m = 1.03$–1.28 across disk accelerations).

The dynamic exponent. DERIVED The tail order in the table above is not a parameter. Each symmetry-preserving level of the cascade (§18) is a closed C_3 structure with $N_a = 2j + 1$ arcs enclosing $N_s = 2j$ regions; its arcs are exponential exposure stages shared equally by its regions, and since the root of $(1 - it)^{-N_a}$ into N_s equal factors is unique, each region is exposed with order $n_j = N_a/N_s = \frac{3}{2}, \frac{5}{4}, \frac{7}{6}, \ldots$ IDENTIFICATION An element is exposed at level j in proportion to the structure still open there, the closure weight $\kappa_{2j-1} = 1/2j(2j+1)$, which over the symmetric ladder sums to $1 - \ln 2$. The interpolation function is then

$$\mu(x) = \sum_{j \geq 1} w_j \, \mu_{n_j}(x), \qquad w_j = \frac{1}{(1 - \ln 2)\, 2j(2j+1)} = 0.543,\ 0.163,\ 0.078, \ldots \tag{36}$$

with every level exposed over the same horizon period, so the deep regime keeps $\mu = \frac{3}{4}x$ exactly. Written in the Torridity form $z/(1 + z^{n(z)})^{1/n(z)}$ its exponent runs with the acceleration:

$\log_{10} g$	-13	-12	-11	-10.5	-10	-9.5	-9	-8
$n(z)$	1.008	1.011	1.018	1.026	1.040	1.065	1.096	1.144

It tends to one in the deep regime, passes $\pi/3$ at $g = 1.26\, a_0$—draft 1's exponent is its value at the transition—and never exceeds about 1.16. The exposure order among the surviving

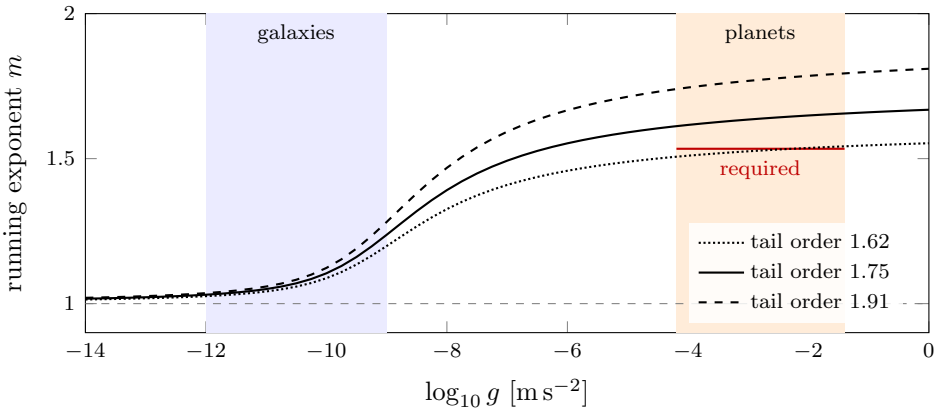


Figure 3: The running exponent of the threshold family: at each acceleration, the exponent of the one-parameter form $z/(1+z^m)^{1/m}$ that passes through the same point. It tends to 1 far out (dashed) and to the tail order close in. Blue: the accelerations of the SPARC galaxies; orange: the planets, where a one-parameter form needs $m \geq 1.534$ (red), which the lower edge of the window meets at the Earth's acceleration.

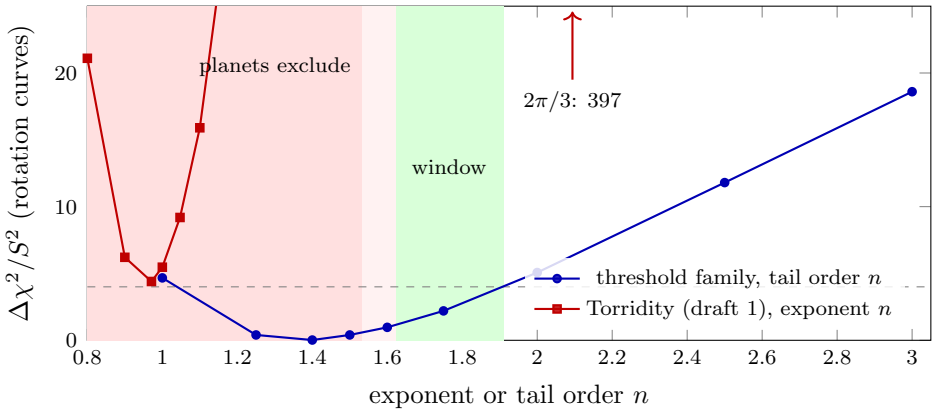


Figure 4: Rotation-curve likelihood against the shape parameter of the two forms, measured from the best threshold member. Red: excluded by the perihelia of Mercury and the Earth (both forms below 1.53; the threshold family below 1.62). Green: the window in which the threshold family meets both. Dashed: 2σ. The Torridity form is excluded by the galaxies wherever the planets allow it; its disk-galaxy exponent $2\pi/3$ lies far off the scale.

elements, which carry the boost, is $\pi^2/24(1-\ln 2) = 1.340$ at low load and falls toward one. With the capture fraction of §15.1 the gravity law is

$$g\left[1-\Gamma\big(1-\mu(g/a_c)\big)\right] = g_{\rm bar}, \tag{37}$$

which gives $g^2 = a_0 g_{\rm bar}$ in the deep regime at full capture, Newton at every acceleration without capture, and a constant enhancement $g_{\rm bar}/(1-\Gamma)$ in between (GRUT §22.9 [32]).

The regime scales of draft 1—numbers multiplying 10^{-10} m s^{-2} for spirals, dwarfs and clusters—depend on the choice of the metre and second and are archived (Appendix F).

18 Cascade Structure and the Closure Weights

DERIVED Two perpendicular arcs enclose one region, so a chain of N_a arcs encloses $N_s = N_a - 1$. The symmetry-preserving ladder adds two arcs per level, $N_a = 2j+1$; the full ladder is

$(N_a, N_s) = (k+2, k+1)$. On the symmetric ladder the exponent sequence

$$n_j = \frac{2j+1}{2j}\, 2^{-2(j-1)} = \frac{N_a}{N_s}\, 2^{-(N_a-3)} \tag{38}$$

takes the values 1.5, 0.3125, 0.0729, 0.0176, ... The closure weights $\kappa_k = 1/N_a N_s = 1/(k+1)(k+2)$ telescope:

$$\sum_{k=1}^{K} \kappa_k = \frac{1}{2} - \frac{1}{K+2} \to \frac{1}{2}, \qquad \sum_{k\geq 2} \kappa_k = \frac{1}{3}. \tag{39}$$

IDENTIFICATION Identifying the multipole with the sphere count, $\ell = N_s = k+1$, gives $\kappa_k = 1/\ell(\ell+1)$, the form of the Sachs–Wolfe plateau: a correspondence of form, not a derivation of amplitudes. Every node of a C_3 is itself a C_3, so depth d holds $(3^{d+1}-1)/2$ units. HYPOTHESIS The unit excess $N_a - N_s = 1$ at every level is read as the directional degree of freedom of space. DERIVED In the dynamic exponent of §17 the closure fraction N_a/N_s is the exposure order of a level and the factor $2^{-2(j-1)}$ its scale; a single system is exposed at a common scale, so the orders enter with the closure weights and the scale factor does not.

19 Scale Bridge and the Information–Energy Relation

DERIVED Equation (29) bridges the galactic and cosmic scales by one pure number, $a_0 = \alpha_{\mathrm{TRSD}}\, cH_0$. Every derived quantity scales with H_0, so the geometry fixes ratios, not H_0 itself (Theorem 12).

With $N = 2\pi R_H^2/\ell_{\mathrm{P}}^2$ cells on the effective horizon and the information of an angular content δ taken as $I = (\delta/2\pi)N$ IDENTIFICATION, the deficit map (34) gives

$$E = Mc^2 = \frac{\delta c^4 R_H}{8\pi G} = \frac{c^4 R_H}{8\pi G} \cdot \frac{2\pi I}{N} = \frac{\hbar c}{8\pi R_H}\, I. \tag{40}$$

I counts nats: for $\delta_{\mathrm{tot}} = \pi$, $I = N/2 = \pi R_H^2/\ell_{\mathrm{P}}^2$, the Bekenstein–Hawking entropy of the horizon sphere. The rate $\hbar c/8\pi R_H = \hbar H_0/8\pi$ per nat is one quarter of the Gibbons–Hawking energy $k_B T_{\mathrm{GH}} = \hbar H_0/2\pi$. The energy of the total defect over the cone volume $V_{\mathrm{cone}} = \pi R_H^3/3$ is

$$\frac{E(I = N/2)}{V_{\mathrm{cone}}} = \frac{c^4 R_H/8G}{\pi R_H^3/3} = \frac{3H_0^2 c^2}{8\pi G} = \rho_c c^2, \tag{41}$$

the full critical energy density (§24.1). (The draft-1 appendix divided by a hemisphere volume and obtained half of this; the cone volume is the one used throughout.)

20 Regeneration

HYPOTHESIS A vacuum fluctuation of the terminal state sources a potential whose gradient at the cycle scale is a_c, i.e. $x = 1$, where the active fraction is $f_1(1) = \frac{1}{2}$ (or $f_3(1) = 37/64$ for the three-arc law). The half-activated state seeds displacement from the vacuum, which the flow of §7 returns to the vacuum while structure forms. The cycle terminal state $\to$ fluctuation $\to$ $g \sim a_c \to$ activation $\to$ structure $\to$ terminal state is a model of self-closure, not yet a dynamical calculation.

21 Dark Sector

Reading note. This section and the next describe the radial excitation of the scalar field as a condensate of dark-matter particles, the condensate reading. In the rotation reading of §21.1 the cosmic cold share is the phase held at its rate, pressureless dust with no particle, and the particle mass scale, solitonic cores and particle bounds do not apply to it; the radial mode and its potential stand as derived. §21.7 belongs to the rotation reading.

21.1 The rotation of the universe and the local baryonic coupling

HYPOTHESIS The universe rotates as a whole, with a fixed frequency: slow, everywhere and coherent. The rotation lives in the phase θ of the conjugate pair, which winds around the C_3 defects; the winding n is an integer and the global frequency $\omega_0 = c\sqrt{\lambda_1} = 0.827 H_0$ is fixed by the C_3 eigenvalue. DERIVED The rotation is internal: the phase turns in time and space does not rotate, its flow $\partial_\mu\theta$ being a gradient with zero vorticity, so the CMB bound $(\sigma_V/H)_0 < 4.7\times10^{-11}$ [59] is met exactly. On the closed horizon the windings of a single-valued complex field sum to zero and the C_3 symmetry sets each to zero, so a global frame dragging is excluded; the extra gravity is carried by the rotation's medium bound around the baryons (GRUT §23). Alone it gives $v_0 = \omega_0 R_H \sin\chi$, not a flat curve. The baryons modulate the frequency, $\omega = \omega_0 + \delta\omega_b$, without changing n or ω_0: they change the local rate at which the phase is sampled. The modulated phase carries a momentum density; held at its rate by the constraint, the rotation's medium is pressureless and binds around the baryons, and with the coupling $\Gamma_b(g_b/a_c)$,

$$v(r) = [1-\Gamma_b]\,\omega_0 R_H \sin(r/R_H) + \Gamma_b\, v_{\rm flat}, \tag{42}$$

flat where the acceleration is weak and the tiny global mode, ordinary Newtonian behaviour, where it is strong. There is no dark particle and no new force: the observed extra gravity is the rotation's medium bound around the baryons, tuned, not generated, by them (GRUT §23 [32]). PREDICTION With the windings zero, the local inertial frame shows no rotation against the quasar frame [53]; a slowly rotating universe is independently viable [52, 51]. The coupling has two parts. DERIVED The metric part: on the curved background the phase equation forces a uniform coordinate frequency, and the baryons modulate the locally measured one, $\Gamma_b^{(g)} = -\Phi/c^2$, the redshift of the core. The geometric part, by matching to the field equation: $\Gamma_b \to (a_0/g_b)^{1/2}$ in the deep regime, so $v_{\rm flat}^4 = a_0 G M_b$. HYPOTHESIS Its source is the cone each baryonic system carries: deficit π at every scale, so an equilateral axial section, with slant $L_\pi = 8GM/c^2$ from the defect relation of Section 16 (it returns R_H for the horizon's cone); the coupling length is $\ell = \sqrt{GM_b/a_0} = \sqrt{L_\pi c^2/8a_0}$ (GRUT §49.5–49.7). DERIVED Beyond the crossover the field of an isolated mass is two-dimensional Poisson gravity, $g = 2GM/(L_{\rm red} r)$; with the crossover at a_0, $L_{\rm red} = 2\ell$ and $L_{\rm red}^2 = L_\pi R_{a_0}$, where $R_{a_0} = c^2/2a_0$ is the radius of the horizon whose surface gravity is a_0, so $v_{\rm flat}^4 = a_0 G M_b$ follows (GRUT §49.7). PREDICTION The zero point lies 0.7σ from gas-rich galaxies, $a_0 = 1.24 \pm 0.14$ [54]; the law acts where the rotation's modulation is bound, so wide binaries, which bind none, are Newtonian [56]; applied to the total field it would give the Solar System a quadrupole 16σ above Cassini's. The anomaly current of the rotating phase grows only on scales of 2,181 Hubble radii, so magnetic fields are set by ordinary dynamos at equipartition, $\sqrt{\mu_0\rho}\, v_t \simeq 5\ \mu$G (GRUT §49.9). DERIVED The cosmic share is the rotation held at its rate by a constraint, $\int \sqrt{-g}\,\lambda(X - X_0)$: pressureless dust with zero sound speed, no particle and no vacuum energy [58, 57] (GRUT §23.1).

21.2 Vacuum

DERIVED The potential $V = \frac{1}{2}(\Xi - 2)^2 + V_0$ has its minimum on the ring $\Xi = \phi^2 = 2$, with $d^2V/d\Xi^2 = 1$ and, as a function of ϕ, $V''(\sqrt{2}) = 6\phi^2 - 4 = 8$. The vacuum stress is $-\varepsilon_V V_0\, g_{\mu\nu}$,

an energy density $\varepsilon_V V_0$ with $w = -1$.

DERIVED The vacuum ring has a geometric meaning. $|\Phi| = \sqrt{2}$ is the director circle of the unit circle that carries the three generators' phases $e^{2\pi ik/3}$: from every point of the vacuum the generators' circle subtends a right angle, the apex angle of the light cone $h = r$. Taking each generation's amplitude as the rise of its generator's invariant when the pair is added, $A_k = |e^{2\pi ik/3} + \Phi|^2 - 1 = \Xi + 2\sqrt{\Xi}\cos(\theta - 2\pi k/3)$, the triple has $\sum_k A_k^2/(\sum_k A_k)^2 = (\Xi+2)/3\Xi$, which is Koide's $\frac{2}{3}$ exactly at the vacuum, and the charged leptons read $\Xi = 2.000013 \pm 0.000031$: the lepton cone of the companion theory is this vacuum (GRUT §42.8 [32]).

21.3 The vacuum constant must vanish

Theorem 11. DERIVED *With the potential scale fixed by the dark-matter mass (9), a vacuum constant V_0 of order one overshoots the observed vacuum energy by more than 10^{21}; the dark energy cannot be carried by V_0.*

Proof. At $\delta_{\rm tot} = \pi$, $f_\delta^2 = 1.513 \times 10^{43}$ N. For the draft-1 calibration $L_* = 0.180$ pc, $\varepsilon_V = f_\delta^2/L_*^2 = 4.9 \times 10^{11}$ J m^{-3}, so $V_0 = 2$ gives 9.8×10^{11} J m^{-3} against $\rho_\Lambda c^2 = 5.2 \times 10^{-10}$ J m^{-3}, a factor 1.9×10^{21}. The Lyman-α bound (§21.6) shortens L_* and raises ε_V above 1.9×10^{16} J m^{-3}, so $V_0 < 3 \times 10^{-26}$. □

TRSD therefore takes $V_0 = 0$, the choice of §29, and the vacuum energy is carried by the horizon defect (§24.1).

21.4 Radial mode

DERIVED By (8) and Theorem 2 the radial fluctuation has

$$m_\delta = \frac{\sqrt{8}\,\hbar}{c\,L_*}, \tag{43}$$

and the phase is massless.

21.5 Self-interaction

DERIVED The nonrelativistic coupling of a real scalar with quartic λ is $g_\chi = \lambda\hbar^3/8m_\delta^2 c$ [17]. With the physical coupling (10),

$$g_\chi = \frac{3\pi}{2\delta}\,\frac{\hbar^3}{m_{\rm P}^2 c} \overset{\delta=\pi}{=} \frac{3\hbar^3}{2m_{\rm P}^2 c} = 1.24 \times 10^{-95}\ \text{J m}^3, \tag{44}$$

independent of m_δ. The draft-1 value $3\hbar^3/2m_\delta^2 c$ used the field-unit coefficient 12 in place of the physical $12\,m_\delta^2/M_{\rm P}^2$ and was larger by $(m_{\rm P}/m_\delta)^2 \sim 10^{100}$. In a halo of density 10^{-21} kg m^{-3} the self-interaction energy $g_\chi n$ is 2×10^{-9} of the gradient energy at 1 kpc: the condensate is effectively free, as in fuzzy dark matter.

21.6 Scale duality and the mass bound

IDENTIFICATION (I6) The dark-sector length is the arc removed by the deficit of a mass M, $L_* = \delta R_H = 8\pi GM/c^2$. With (43),

$$M\,m_\delta = \frac{\sqrt{8}}{8\pi}\,m_{\rm P}^2. \tag{45}$$

CALIBRATED $M = 1.5 \times 10^{11} M_\odot$ gives $m_\delta c^2 = 1.00 \times 10^{-22}$ eV and $L_* = 0.180$ pc (the draft-1 calibration). STANDARD The Lyman-α forest requires $m_\delta > 2 \times 10^{-20}$ eV [19], so $M < 7.5 \times 10^8 M_\odot$ and $L_* < 9.0 \times 10^{-4}$ pc: the 10^{-22} eV benchmark is excluded and the calibration moves to dwarf-galaxy masses.

DERIVED The bound applies to the radial mode. It assumes [19] a scalar that is all the dark matter and evolves linearly as a free field of mass m. Here (i) the condensate is the whole of Ω_c in the budget (§24.5); (ii) about the ring the potential is $4\eta^2 + 2\sqrt{2}\,\eta^3 + \frac{1}{2}\eta^4$ in field units (η the displacement over f_δ), whose anharmonic terms shift the oscillation frequency by $\Delta\omega/\omega = -\frac{3}{8}\eta^2$, below 4×10^{-5} for the initial displacements $\eta \sim 10^{-3}$–10^{-2} that give the relic density, and falling as a^{-3} thereafter—far from the large-misalignment regime that changes the transfer function [37]; (iii) the self-coupling is gravitational in strength (§21.5); and (iv) a radial oscillation at fixed phase does not populate the massless Goldstone mode. The linear transfer function is therefore that of the canonical free scalar. If the condensate were only a fraction of the dark matter the bound would weaken [35, 36]. The relic density follows from misalignment [17]; the required initial displacement is 10^{-3}–10^{-2} of f_δ across the allowed masses [32].

21.7 The dark sector as one geometry

This subsection formalizes a proposal made for this release. Dark energy is spacetime, the geometry itself; what is called dark matter is the same spacetime in its excited state, bound to baryonic structure; and what lensing infers as dark matter is the dilation—the excess of the inferred mass over the baryonic mass, smallest at the centre, growing toward the edge, and varying from system to system. There is spacetime and there are baryons: dark matter is not a separate entry. The vacuum is the capacity of the final horizon (§24.11), the cosmic share is the rotation held at its rate (§21.1), and the gravity law is the threshold response (§11, §17); each statement follows from them.

DERIVED *Spacetime, unexcited and excited.* The spin's quantum is $\hbar\omega$ with $\omega = H_\infty$, and the charge of the final horizon is $L_\infty = \hbar N_\infty/4\pi$ (eq. (79)): $N_\infty/4\pi$ quanta, whose energy is the Hubble mass of the final horizon, $c^5/2GH_\infty$. Spread over its volume they are the vacuum,

$$n_*\,\hbar H_\infty = \rho_\Lambda c^2, \qquad n_* = \frac{N_\infty/4\pi}{\frac{4\pi}{3}R_\infty^3} = \frac{3}{8\pi\ell_{\rm P}^2 R_\infty}, \tag{46}$$

the capacity of §24.11 counted in quanta, three per area step $8\pi\ell_{\rm P}^2$ per final radius. Dark energy is spacetime *unexcited*: the spin's quanta held by the horizon, $w = -1$, constant by the Bianchi identity. The rotation carries quanta of the same kind, each one quantum of the spin ($j = 1$, eq. (80)), and that is spacetime *excited*: dust, $n \propto a^{-3}$ by the rotation's Noether current, $w = 0$, cold on every scale, clustering and binding to baryonic structure. One stress tensor holds both,

$$\boxed{T_{\mu\nu}^{\rm dark} = \hbar H_\infty\big(n\,u_\mu u_\nu - n_*\,g_{\mu\nu}\big) = 2X_0\big(\lambda\,u_\mu u_\nu - \lambda_*\,g_{\mu\nu}\big), \qquad 2\lambda_* = \frac{f_\delta^2(\pi)}{\delta_{\rm gen}} = \frac{3c^4}{8\pi G},} \tag{47}$$

with $u_\mu = \partial_\mu\theta/\sqrt{X_0}$, $X_0 = \Lambda/3 = R_\infty^{-2}$, and $\lambda = n\hbar c^2/2H_\infty$ the multiplier of the constraint of §21.1. In the normalization of the defect map the vacuum is the generator-defect tension, $\delta_{\rm gen}^{-1} f_\delta^2(\pi) = 2\lambda_*$, over the square of the final radius: draft 1's $\delta_{\rm gen}^{-1} f_\delta^2(\pi)/R_H^2$, which §24.1 showed to be the critical total today, is the vacuum exactly at the final horizon, where the total and the vacuum coincide. The rotation's tension today is $\lambda_0 = 2.80 \times 10^{42}$ N against $\lambda_* = 7.22 \times 10^{42}$ N, in the ratio 1/2.580 of the resonance budget. Quanta, energy and information—2π nats per quantum (§24.15)—are proportional in both states, so the split is the same in the three currencies of §24.4. The inventory is spacetime, 0.684 of the critical density unexcited and 0.265 excited, 0.949 together, and the baryons, 0.050, with the neutrinos and the radiation.

DERIVED *The equation of state of the whole.* From (47),

$$\boxed{w_{\rm dark} = -1 + \Theta, \qquad \Theta = \frac{n}{n + n_*} = \frac{\rho_\chi}{\rho_\chi + \rho_\Lambda},} \tag{48}$$

the excited fraction. This is the proposal's $w = -1 + \Theta$, with Θ derived rather than posited. Over cosmic time $n/n_* = (\Omega_c/\Omega_\Lambda)\, a^{-3}$, so $\Theta(a) = 1/[1 + (a/a_{\rm eq})^3]$ with $a_{\rm eq} = (\Omega_c/\Omega_\Lambda)^{1/3}$: a step in $\ln a$ from 1 to 0, centred on dark-sector equality. At the resonance budget (§24.16) equality falls at $z = 0.372$, the dark sector alone stops decelerating ($w_{\rm dark} = -\frac{1}{3}$) at $z = 0.728$, and today $\Theta_0 = 0.279$, $w_{\rm dark,0} = -0.721$. Locally $\Theta = n(\mathbf{x})/[n(\mathbf{x}) + n_*]$: $\Theta \simeq 1$ in haloes, 0.07 in voids at $\delta = -0.8$, and $\frac{1}{2}$ at the vacuum's own density, $\rho_\chi = \rho_\Lambda$. At the cosmic ratio that is a baryon density $(\Omega_b/\Omega_c)\,\rho_\Lambda$, 0.65 protons per cubic metre, 2.58 times the mean: the proposal's threshold in ρ_b, located. A region is bound inside its zero-acceleration radius, where the mean density is $2\rho_\Lambda$, at matter overdensity $2\Omega_\Lambda/\Omega_m = 4.33$; there $\Theta \geq 0.63$.

DERIVED *Excited once, gathered since.* The excitation is not made where baryons are dense. The two states are separately conserved: n_* by the capacity and the Bianchi identity, n by the Noether current, and nothing in the action exchanges energy with the vacuum ($V_0 = 0$, §21.3; the constraint adds none, §24.15). Nor could the vacuum in a halo's own volume supply it: inside r_{200} the mean density is $292\,\rho_\Lambda$, so the vacuum there is 0.34% of the halo, and the dark density near the Sun, 0.4 GeV cm^{-3}, is $1.2 \times 10^5 \rho_\Lambda$. The excitation was set once, at the beginning, as the integration constant of §24.16, and gravity has gathered it onto baryonic structure since: the infall configuration of §15.2, Ω_c/Ω_b times the baryons. With each state conserved the expansion is the resonance budget's; no distance, growth rate or parameter moves. STANDARD Gravity measures only the total dark stress-energy [81], and one multiplier field for both components was proposed in [82]; here the multiplier carries no potential, and the vacuum is the horizon's, tied to the dust by the one constant X_0. MEASURED Fits that allow an exchange find at most a mild one: with DESI DR2 the simplest coupling is preferred at slightly more than 1σ [83], and a non-parametric reconstruction finds a coupling that reverses sign with time, at about 2σ at either end [84]. The prediction of none stands or falls with $w = -1$ (register row 9).

DERIVED *The dilation.* Lensing and dynamics infer mass with the flat relation between field and source, Poisson's equation: a field g at radius r is read as the mass gr^2/G. The dilation is the ratio of the inferred mass to the baryonic mass, $\mathcal{S} \equiv M_{\rm inf}(<r)/M_b(<r) = g/g_{\rm bar}$, and the excess $(\mathcal{S} - 1)M_b$ is what is called dark matter. The geometry does not respond flatly: in a relaxed system the response is the field equation (28), $\mu g = g_{\rm bar}$ exactly in spherical symmetry (§13), and in the Torridity form whose exponent runs with the acceleration (§17) the response is a cosine,

$$\boxed{\mathcal{S} = \frac{1}{\mu} = \frac{1}{\cos_n \theta}, \qquad \tan\theta = \frac{a_0}{g}, \qquad \cos_n \theta \equiv \left(1 + \tan^n \theta\right)^{-1/n},} \tag{49}$$

exact at every acceleration, with $n = n(z)$ the dynamic exponent of §17. Take the local field g and the horizon's acceleration a_0 as the two components of one field, tilted by θ from the local direction. Projecting the measured field through the tilt gives the baryons' field, $g_{\rm bar} = g \cos_n \theta$; the flat inference takes no tilt and reads the whole measured field as the baryons', so the mass it infers is dilated by the secant. Equivalently $g^2 = g_{\rm bar}\,(g^n + a_0^n)^{1/n}$: the measured field is the geometric mean of the baryons' field and the whole field—two projections through one angle, as the angular reduction of §11 projects the load and then its radial response. In the Euclidean plane, $n = 2$, the cosine is the ordinary one, $\mathcal{S} = \sqrt{1 + a_0^2/g^2} = 1/\cos\theta$ exactly: the proposal's form, whose values 2, 3.86 and 11.5 at $\theta = 60°$, 75° and 85° are the dilation where the field has fallen to 0.58, 0.27 and $0.087\,a_0$. (The proposal's first table attached these angles to the heights $R/2$, $0.87R$ and $0.98R$ of a sliced sphere, where $1/\cos$ is 1.15, 2.03 and 5.03.) The cascade sets the exponent between 1.01 and 1.10 across galaxies (§17), near the ℓ^1 plane, $\mathcal{S} \simeq 1 + a_0/g$; at

those tilts the dilation is 2.67, 4.67 and 12.4. A fixed exponent near the Euclidean one is too sharp at the transition: the rotation curves put $2\pi/3$ at $\Delta\chi^2/S^2 = 397$ (§17).

Smallest at the centre, growing toward the edge. Where $g \gg a_0$, $\theta \to 0$ and $\mathcal{S} \to 1$: Newton. Where $g \ll a_0$, around an isolated mass, $\mathcal{S} \simeq a_0/g = r/\ell$ with $\ell = \sqrt{GM_b/a_0}$ the coupling length of §21.1. Around $10^{11} M_\odot$ of baryons, $\ell = 11.0$ kpc and the dilation is 2 at 16 kpc, 4 at 39 kpc and 11 at 116 kpc—the range the proposal gave—and 91 at 1 Mpc; at $g_{\rm bar} = 0.1$, 1, 10 and $30\,a_0$ it is 3.67, 1.58, 1.067 and 1.019 (GRUT §22.9). *The tilt is the horizon's:* $a_0 = (\sqrt{3}/\pi^2)\,cH$ (29), and at the present $(2/3\pi)\,c^2\sqrt{\Lambda/3}$ (§24.15). Without a horizon $a_0 = 0$, $\theta = 0$ and $\mathcal{S} = 1$ at every radius: baryons alone, nothing to call dark matter. The dilation is the effect of dark energy, the geometry, on the lensing of the baryons—the proposal's statement, derived.

It varies: through each system's field, θ rising outward; through $n(z)$; through capture, $\mathcal{S} = 1/[1 - \Gamma(1-\mu)]$ (§17), which is 1 for the Solar System and wide binaries (§15.1); and through the relaxation state (§15.2), a relaxed galaxy carrying the dilation of the threshold law and an unrelaxed cluster its gathered share. MEASURED Weak lensing extends the radial acceleration relation by two decades of baryonic acceleration, in agreement with the MOND-type law, yet early- and late-type galaxies of the same stellar mass differ by at least 6σ [89], which no function of the acceleration alone can give; clusters depart from the galactic relation [85], one CLASH analysis finding the deep slope at a scale seventeen times higher [91]. PREDICTION The dilation differs between systems at the same acceleration as their capture and relaxation differ (register row 33).

What carries it. DERIVED In general relativity light bends only by energy and momentum, and spacetime's excitation carries them, as a gravitational wave does, without any matter. The dilation looks like mass and is no matter: in a relaxed system the excited geometry is distributed exactly as the phantom density of the field equation, $(4\pi G)^{-1}\nabla\cdot[(1-\mu)\nabla\Phi]$ (GRUT §46.6)—the mass the flat inference sees and no matter supplies—and since the completion lenses by the dynamical potential (§27), the dilation of lensing and of the rotation curves is one. *How far it grows.* OPEN If the dilation draws only on the excitation bound to a system, it stops where that is spent, at the r_t of §15.1: 59 kpc around $10^{11} M_\odot$ holding its cosmic share. MEASURED Weak lensing finds the circular velocities of isolated galaxies flat with no clear decline to 1 Mpc [90], a dilation near 90 there: either an isolated galaxy binds 30–100 times its baryons in excited spacetime, or the dilation is not drawn from the bound store; the data so far favour a dilation that keeps growing (register row 34).

DERIVED *The dilation on the sky.* Lensing sees the dilation through a second one of the same form, the slant of each shell to the line of sight:

$$\Sigma(b) = 2\int_b^\infty \rho(r)\,S_r(b)\,dr, \qquad S_r(b) = \frac{r}{\sqrt{r^2-b^2}} = \frac{1}{\cos\psi}, \qquad \int_0^\infty \Sigma(b)\,2\pi b\,db = M, \tag{50}$$

with ψ the angle between the line of sight and the shell's normal where it crosses: 1 at the centre of the shell's disc, rising to the limb. For a thin shell of surface density σ this is the proposal's $\kappa = (\Sigma_{\rm surf}/\Sigma_{\rm crit})\,S$ for each crossing, and $2\sigma S_R(b)/\Sigma_{\rm crit}$ for the two. Over the sky each crossing's mean is 2, so a closed shell's mean surface density is 4σ: Cauchy's surface-to-shadow ratio, which the proposal's first form took as the mean dilation, and with its base the deficit-π cone of every system (§21.1), with its equilateral axial section, gives 3. The four is TRSD's own: a ball is the union of cones of height R from its centre to its surface, $V = \frac{1}{3}RA$, and the cone of §24.3 stands on the shadow, $\frac{1}{3}R\,\pi R^2$, so the closure $M_{\rm Hubble}/M_{\rm total} = 4$ is Cauchy's ratio. The last equation of (50), since $\int_0^r 2\pi b\cdot 2S_r\,db = 4\pi r^2$, says the sky's slant redistributes toward the limbs and keeps the total, so the excess lensing measures is the field's dilation, carried onto the sky intact; in an aperture the cylinder takes in the outer shells, by $\pi/2$ on the isothermal plateau beyond r_* (§24.7). Nor is the excess one number: 6.35 for the universe as a whole, $1+\Omega_c/\Omega_b$, reached at R_{200} in clusters, and growing without a fixed mean in galaxies. At the splashback radius,

where the infalling streams of the excited geometry turn round, the density drops sharply [87], and the slant turns that edge into the steepening measured around clusters [88].

MEASURED *The Bullet Cluster.* The lensing peaks of 1E 0657−56 lie within 1σ of the galaxies' light and 8σ from the gas, the dominant baryonic component, 29–48 times the stars at the gas peaks; at the subcluster $\kappa = 0.20 \pm 0.05$ at the brightest galaxy and 0.02 ± 0.06 at the gas peak, where the visible mass is 1.8 times larger [28]. DERIVED A dilation computed from the baryons present would follow the gas; so would the phantom density of the field equation alone (§27). The dilation follows the excited geometry that carries it, and that is collisionless: with $X = X_0$ its flow is geodesic, $u^\nu \nabla_\nu u_\mu = \nabla_\mu(\partial\theta \cdot \partial\theta)/2X_0 = 0$; it is pressureless, and the constraint contains no interaction, so it scatters only gravitationally, $\sigma/m = 0$. A cluster, about four crossing times old, holds its infall configuration (§15.2), and a merger of two is the passage of two excited geometries through each other with their galaxies, while the gas is shocked and held back: the galaxies sit where the dilation is large and the gas where it is small, as the proposal had it. A system holding its cosmic share is dilated by $1 + \Omega_c/\Omega_b = 6.35$, $f_b = 0.158$, with the derived baryon ratio (§24.5). MEASURED The X-COP clusters reach the cosmic baryon fraction at R_{200} [85], and in the Bullet Cluster the dark-plus-galaxy convergence is about seven times the gas's [28]. PREDICTION $\sigma/m = 0$, against $\sigma/m < 0.63\ \mathrm{cm^2\,g^{-1}}$ (95%) from eleven double-radio-relic mergers [86]; no offset between the excess and the galaxies; and the splashback edge of cold matter around clusters. The merger is worked in §15.3. Register rows 30–32 and 41–42.

DERIVED *The dilation field.* The proposal's second form writes the dilation as a field, $S = e^{-\psi}$, with $S \simeq 1 - 2\Phi/c^2$ in the weak field: the refractive index of the geometry. With $\gamma = 1$ (§27) light obeys Fermat's principle in the optical metric $d\ell_{\rm opt} = S\, d\ell$, so S dilates the measured path, delays the light by $\int (S-1)\, d\ell/c$ and bends it by $\nabla_\perp \int \ln S\, d\ell$. It is the conformal factor of the optical metric, not of spacetime, as the proposal says; what it records is spacetime's dilation along the light path: in the weak field time is dilated by $1 - \Phi/c^2$ and space by $1 - \Phi/c^2$, and light, crossing both, is slowed by their product, $S \simeq 1 - 2\Phi/c^2$—half from time and half from space, the two halves that double the bending of light over Newton's. Hence the names: S is the dilation field and $\mathcal{S}$, its gradient against the baryons', the dilation. Its value differs from one by $2|\Phi|/c^2$, a few parts per million in galaxies and 10^{-5}–10^{-4} in clusters; its gradient carries the field, $\nabla \ln S = 2\mathbf{g}/c^2$. So the dilation above is a ratio of gradients, $\mathcal{S} = \partial_r \ln S/\partial_r \ln S_b$, with S_b the index the baryons alone would make. In these variables the field equation (28) reads

$$\boxed{\nabla \cdot \Big[\mu\Big(\frac{c^2|\nabla \ln S|}{2a_c}\Big)\, \nabla \ln S\Big] = -\frac{8\pi G}{c^2}\,\rho_b, \qquad R_{a_0}\,|\nabla \ln S| = \frac{g}{a_0},} \tag{51}$$

with $R_{a_0} = c^2/2a_0 = 3.91 \times 10^{26}$ m the radius of the horizon whose surface gravity is a_0 (§21.1), and with capture as in §17. Where the path's dilation changes by more than one e-fold per R_{a_0} the geometry is stiff, and the equation is the proposal's $(\nabla^2 - \mu_{\rm Y}^2) \ln S = -(8\pi G/c^2)\rho$ at $S \simeq 1$; where it changes more slowly, the geometry softens and dilates further. The proposal's self-coupling therefore acts through the gradient: the value of $\ln S$ is a few parts per million throughout a galaxy, while its gradient in units of $1/R_{a_0}$ runs over decades. Outside a source the proposal's solution $S = \exp[(2GM_S/c^2)\, e^{-\mu_{\rm Y} r}/r]$ is exact for a point source. Inside, Gauss fixes the gradient, $\partial_r \ln S = -2GM_S(<r)/c^2 r^2$, while the value adds the shells outside r. The screening scale has one value in the rotation reading, the horizon's, $\mu_{\rm Y} = 1/R_\infty$; it changes κ by less than 10^{-7} inside 1 Mpc. PREDICTION The lensing excess has no exponential cut inside the horizon (register row 35).

DERIVED *The switch.* The proposal's coupling $\alpha(\rho) = \alpha_0(\rho/\rho_c)^n/[1 + (\rho/\rho_c)^n]$ is the capture fraction of §15.1. A system of radius R in a field of dispersion σ binds $\Gamma = \mathrm{erf}\, X - (2X/\sqrt{\pi})\, e^{-X^2}$ of the excited geometry streaming past it, with $X^2 = \rho/\rho_c$ and $\rho_c = 3\sigma^2/4\pi G R^2$. So $\Gamma \simeq (4/3\sqrt{\pi})(\rho/\rho_c)^{3/2}$ below ρ_c and $\Gamma \to 1$ above it: a switch with $n = 3/2$ and $\alpha_0 = 1$, derived, whose ρ_c is set by the system's size and the field around it, not by nature. A galaxy forming

in the cold cosmic field captures fully; the Sun at its MOND radius has $\rho/\rho_c = 4.5 \times 10^{-6}$ and $\Gamma = 7 \times 10^{-9}$. The switch gates the excess, not the baryons: what has captured nothing lenses as its baryons, never as nothing. DERIVED In the second form's galaxy estimate the exponential is $1 + O(10^{-6})$, so the whole excess would be α_0. A constant α_0 at high density is shared by the laboratory, the Sun and the stars, and is absorbed into G. A coupling that rises with density puts the excess where it is absent, in galaxy centres and the Solar System, and removes it where it is largest, in the outskirts and in low-surface-brightness galaxies [10, 11]. $\alpha \to 0$ would switch off the gas's own gravity, and a shock compresses gas. What the second form asks of α in the Bullet Cluster—galaxies lensing, gas not—the collisionless excited geometry gives (above).

DERIVED *Scales and the background.* The proposal's last part asks that each scale be the coarse-grained output of those below and the fixed input of those above. An intermediate scale B is fixed by two outer scales $A \cong C$ jointly, as neither their product nor their union, and the background solves its own consistency condition. The background's condition is the closure law (73): the contents' defects summed are the horizon, $\sum r_s = R_H$, and the horizon in turn sets, through a_0, the dilation of everything inside it—above and below at once, with no external stage. The rule that does not change with scale is the spinning-arc rule, $\omega R = c$, obeyed by the final horizon, black holes and every particle's sphere (§24.14–24.15): the fixed point of the coarse-graining. And the middle scale is the geometric mean of the outer two, their midpoint on the logarithmic scale:

$$\ell_P^2 = \tfrac{1}{2}\, r_s \lambda_C, \qquad \ell^2 = \frac{GM}{a_0} = r_s\, R_{a_0}, \qquad g^2 = g_{\rm bar}\, a_0 \quad \text{(deep regime)}. \tag{52}$$

The Planck length lies between the gravitational and quantum lengths of every mass (§24.12). The coupling length lies between the system's own horizon and the horizon whose surface gravity is a_0: 7,200 AU for the Sun, between 3 km and 13 Gpc, and 8.5 kpc for the Milky Way's baryons. The measured field lies between the baryons' field and the horizon's, and at every acceleration $g^2 = g_{\rm bar}(g^n + a_0^n)^{1/n}$. OPEN A sheaf or renormalization-group formulation of the coarse-graining is not supplied.

DERIVED *The dilation on the unbound web.* A homogeneous universe has no gradient of the dilation field, so $\mathcal{S} = 1$ and the background—the Friedmann equation, the sound horizon, the CMB's $H_0 = 67.35$—cannot move. The distance ladder measures the expansion inside our own neighbourhood, $0.023 < z < 0.15$, and continuity gives the rate inside a region of mean contrast δ growing at $f = d\ln\delta/d\ln a$:

$$\frac{H_{\rm loc}}{H} = 1 - \frac{f\,\delta}{3}. \tag{53}$$

The local distance network's 73.50 ± 0.81 [92] needs $f|\delta| = 0.275 \pm 0.036$; undilated growth, $f = 0.528$, needs a region 52% underdense. The web is deep: its rms peculiar field, $\frac{1}{2}\Omega_m H_0^2 \sigma(R) R$, is 0.0010–0.0024 a_0 from 12 to 500 Mpc and 0.0079 a_0 at a point, whose baryons' share the law dilates 28.9-fold, at most 4.5 times the cosmic dilation $1 + \Omega_c/\Omega_b$ that the share already supplies. The field cannot tell $8\,h^{-1}$ Mpc (0.0019 a_0) from 300 Mpc (0.0014 a_0), so a dilation set by it alone raises S_8 to 0.89 for the smallest boost, 6.4σ above DES Y6's 0.789 ± 0.012 [97]: excluded. DERIVED The dilation keeps a clock's causes. A clock runs at $\Theta = N\sqrt{1 - v^2/c^2}$, $N^2 = 1 - r_s/R - R^2/r_\Lambda^2$ (GRUT §43B [32]), slowed by its well, its motion and the horizon; in a bound system the dilation is the horizon's tilt of the light path, on the unbound web its setting of the clocks. *When:* the horizon sets a region's clocks only beyond its balance surface, where its mean density is below $2\rho_\Lambda$: the universe as a whole from $q = 0$, $z = 0.630$, the onset of acceleration; a region 20% underdense today from $z = 0.713$, one 80% overdense from $z = 0.378$. At recombination, with matter at 3×10^8 times $2\rho_\Lambda$ and the baryons moving with the photon fluid at $0.45c$, it is off. *Where:* a region holds the medium where its well slows time more than the medium's peculiar motion does, $v^2 < 2|\Phi|$, the capture fraction, with the potential carrying the scale the field lacks: rms $|\Phi|/c^2 = 0.9 \times 10^{-6}$ at 12 Mpc and 1.7×10^{-5} at 300 Mpc, flat

beyond the equality horizon $1/k_{\rm eq} = 96$ Mpc. All of the motion counts—the flows the dilation drives and the haloes' virial motion, 264 km s^{-1} over 77% of the mass—and the law acts on the field the matter moves in, so the dilation exceeds the share where $\Gamma > 0.873$. Dispersion and cut close on themselves at $\sigma_{1D} = 494$ km s^{-1} and 68 Mpc, with boosts 1.6–4.2 above it; no input remains. *Light:* de Sitter is conformally flat, so the horizon dilates clocks and rulers by one factor, $\tilde{g}_{\mu\nu} = \Omega^2 g_{\mu\nu}$. The optical index, their ratio, is unchanged, a photon picks up Ω only at its two ends, and only massive bodies feel the pull:

$$\frac{\tilde{g}_{ij}}{|\tilde{g}_{00}|} = \frac{g_{ij}}{|g_{00}|}, \qquad \ddot{\mathbf{x}} = -\nabla\Phi - c^2\nabla \ln \Omega. \tag{54}$$

Had light crossed the web dilated, the late Sachs–Wolfe signal would rise 4,200-fold at $\ell = 10$: excluded. As derived it falls, to 0.28, 0.54 and 0.71 of the background's at $\ell = 2$, 10 and 20. MEASURED With the cut, σ_8 rises by 0.16% today and less at the lensing epochs, cosmic shear is unchanged from $\ell = 100$ and CMB lensing from $L = 40$: DES Y6 stays 2.8σ below, as for the background, and the lensing amplitude at the framework's own neutrino mass, $S_8^{\rm CMBL} = 0.822$, is the $0.825^{+0.015}_{-0.013}$ measured jointly by ACT, SPT and Planck [100], leaving no excess for a negative neutrino mass [96]; the floor's pull rests on DESI's BAO matter density [25]. The Cosmicflows-4 bulk flows, 387 ± 28 and 419 ± 36 km s^{-1} within 150 and 200 h^{-1} Mpc, below 0.03% and 0.003% in ΛCDM [98], fall at the 78th and 90th percentiles of the dilated web's distribution, and the 628 ± 82 km s^{-1} dipole of the 120–160 h^{-1} Mpc shell [99] at its 98th. PREDICTION The ladder's H_0 is the background's. A clock or ruler is shifted by $|\Phi|/c^2 \lesssim 10^{-5}$, so the dilation cannot make a ladder read 9% more, and a local void cannot carry it either: read as a matter deficit, the K-band luminosity density, 3.09 ± 0.19 below $z = 0.07$ against 5.40 ± 0.27 beyond $z = 0.09$ in $10^8\, h_{70} L_\odot\,{\rm Mpc}^{-3}$ [94], is a void 37% deep whose interior would expand at 74.2 ± 0.6, but the deficit ends by $z \approx 0.09$, beyond it the rate would fall back by 8.3%, and the supernovae show no change of H_0 across $0.023 < z < 0.15$ to 0.60% [95]. The dilation widens the local rate's scatter to 0.88% on 300 Mpc (0.41% in ΛCDM). The framework therefore has one expansion rate, 67.35, and predicts that distance ladders calibrated alike converge on 67.3 within that scatter: the network lies 6.1σ away, SH0ES 4.8σ, the red-giant and carbon-star ladder 1.5σ; JWST distances to the supernova hosts settled at the network's value at 5σ would falsify it. HYPOTHESIS GRUT as built keeps the web undilated; the dilation as geometry rather than store, the branch isolated galaxies' lensing favours (register row 34), lets the web take it from the horizon on a clock's terms. Unaffected either way: TM1 and $a_0(z)$ (GRUT §53.9 [32]; register rows 36–37).

DERIVED *What this changes.* Derived: dark energy as spacetime unexcited and the cosmic share as spacetime excited (47), with no separate dark matter, the vacuum the generator-defect tension over the square of the final radius; $w = -1 + \Theta$ with the vacuum's density as the threshold (48); the dilation $\mathcal{S} = 1/\cos_n \theta$, $\tan\theta = a_0/g$ (49), 1 at the centre and r/ℓ toward the edge, set by the horizon and varying with capture and relaxation, its excess carried by the excited geometry as the phantom density; the dilation field as the index of the optical metric, its gradient carrying the law (51); the switch as the capture fraction; the sky's slant, which keeps the total (50), and the closure's four as Cauchy's ratio; the Bullet Cluster's offset from the collisionless excitation; and the background as the closure law, with the middle scales as geometric means (52). OPEN How far the dilation grows is open (register row 34). Unchanged: the background expansion, which has no gradient and so no local dilation, and with it the CMB's H_0, TM1 and the acceleration scale at $z \simeq 1$. Changed in the proposal's reading: the growth of the web above 68 Mpc after $z = 0.63$, on clocks and not on light (54), which carries the measured bulk flows and lowers the late Sachs–Wolfe signal, with S_8 and CMB lensing at the background's values. Not changed: the distance-ladder H_0. Corrected: the first form of this paragraph, which held the test run's tensions unchanged—only the background is; its second form, which read the ladder's H_0 as the rate inside a local void, against the supernova Hubble diagram, and cut the web's

dilation at 29–80 Mpc by the motions it counted; the heights of the proposal's first table, and its single mean dilation $\langle \mathcal{S} \rangle = 4$, which is Cauchy's ratio, not the measured excess; in the second form, the excess assigned to the value of the dilation field rather than its gradient, a coupling that rises with density rather than with capture and multiplies the baryons rather than the excess, and the enclosed-mass form of $\ln S$ inside a source.

22 Quantum Condensate Limit

STANDARD For occupation $N \gg 1$ the field is replaced by its coherent expectation value, and the nonrelativistic envelope $\chi_{\rm cl} = (\hbar/\sqrt{2m_\delta})[\psi e^{-im_\delta c^2 t/\hbar} + \text{c.c.}]$ obeys the Gross–Pitaevskii equation

$$i\hbar\,\partial_t\psi = -\frac{\hbar^2}{2m_\delta}\nabla^2\psi + m_\delta\Phi\,\psi + g_\chi|\psi|^2\psi, \qquad \rho_\chi = m_\delta|\psi|^2, \tag{55}$$

with g_χ from (44). Quantum pressure $Q = -(\hbar^2/2m_\delta^2)\nabla^2\sqrt{\rho_\chi}/\sqrt{\rho_\chi}$ supports cores of size the de Broglie length. For 10^{-22} eV the reduced Compton length is 0.064 pc and $\lambda_{\rm dB} = 2\pi\hbar/m_\delta v = 0.60$ kpc at 200 km s^{-1}; at the Lyman-α bound, 2×10^{-20} eV, $\lambda_{\rm dB} = 3.0$ pc, so the predicted solitonic cores are parsec-scale.

23 The Coupled System

DERIVED The condensate (55), the field equation with the three-arc law,

$$\nabla\cdot\big[\mu_n(|\nabla\Phi|/a_c)\,\nabla\Phi\big] = 4\pi G\big(\rho_b + m_\delta|\psi|^2\big), \tag{56}$$

and the phantom density $\rho_{\rm ph} = (4\pi G)^{-1}\nabla\cdot[(1-\mu_n)\nabla\Phi]$ form the coupled system; the effective dark matter is $m_\delta|\psi|^2 + \rho_{\rm ph}$. DERIVED The covariant completion is general relativity with the conjugate-pair scalar held at its rotation rate by $\lambda(X - X_0)$ (GRUT §23.1 [32]); for bound systems its weak-field limit is the field equation above, through the polarization law of the bound medium (IDENTIFICATION).

Component	Expression	Status
Vacuum energy	horizon defect, $V_0 = 0$	DERIVED (Theorem 11)
Condensate density	$m_\delta\|\psi\|^2$, $w = 0$	DERIVED
Self-interaction	$g_\chi = 3\hbar^3/2m_{\rm P}^2 c$	DERIVED
Gravity law	$\mu_n(\|\nabla\Phi\|/a_c)$, $1.62 \le n \le 1.91$	DERIVED given I3; n DATA-CONSTRAINED
Phantom density	$(4\pi G)^{-1}\nabla\cdot[(1-\mu_n)\nabla\Phi]$	DERIVED
Dark-matter mass	$\sqrt{8}\,\hbar/cL_*$, $m_\delta > 2\times10^{-20}$ eV	CALIBRATED
Spacetime, two states	$\hbar H_\infty(n\,u_\mu u_\nu - n_* g_{\mu\nu})$, $w = -1 + \Theta$	DERIVED (§21.7)
Dilation	$\mathcal{S} = 1/\mu = 1/\cos_n\theta$, $\tan\theta = a_0/g$	DERIVED (§21.7)
Dilation on the web	on clocks, not light, above 68 Mpc	HYPOTHESIS (§21.7)

24 Cosmological Implications

24.1 Flatness

DERIVED The mass of a ball of radius R_H at the critical density is $M_{\rm Hubble} = \frac{4}{3}\pi R_H^3\rho_c = c^3/2GH_0$, and its Schwarzschild radius is exactly R_H. Conversely, $2GM/c^2 = R_H$ forces $\rho = \rho_c$, which for

a Friedmann universe is $\Omega_{\rm tot}=1$. IDENTIFICATION draft 1 wrote the same density as the generator-defect tension, $\delta_{\rm gen}^{-1}f_\delta^2(\pi)/R_H^2=3c^4/8\pi GR_H^2=\rho_c c^2$, and called it $\rho_{\rm DE}$; it is the total, of which the vacuum is the fraction Ω_Λ. Eliminating H_0 with $a_0=\alpha_{\rm TRSD}cH_0$,

$$\rho_c c^2=\frac{\pi^3}{8}\,\frac{a_0^2}{G},\qquad \rho_\Lambda c^2=\Omega_\Lambda\,\frac{\pi^3}{8}\,\frac{a_0^2}{G}. \tag{57}$$

Planck and BAO give $\Omega_K=0.0007\pm0.0019$ [21].

24.2 Horizon totals

DERIVED Over the cone volume $V_{\rm cone}=\pi R_H^3/3$:

$$E_{\rm total}=\frac{c^5}{8GH_0},\qquad M_{\rm total}=\frac{c^3}{8GH_0},\qquad I_{\rm total}=\frac{N}{2}=\frac{\pi c^5}{\hbar GH_0^2}\ \text{nats}, \tag{58}$$

that is 1.916×10^{69} J, 2.132×10^{52} kg and 1.929×10^{122} nats (2.783×10^{122} bits) for $H_0=73.04$, with $N=3.858\times10^{122}$. The information of a direction is taken as the full azimuth, $I_F=N$, twice the scalar content $I_E=N/2$ IDENTIFICATION, and the corresponding force per unit length of horizon is

$$F=\frac{\hbar c}{8\pi R_H^2}\,I_F=\frac{\hbar c\,N}{8\pi R_H^2}=\frac{c^4}{4G}=3.026\times10^{43}\ \text{N}, \tag{59}$$

the maximum force of one horizon [15]. Solving $2\pi R_H^2=N\ell_{\rm P}^2$ for G gives $G=2\pi c^3R_H^2/\hbar N$: this is the definition of N read backwards, not an independent determination of G (the draft-1 "agreement with CODATA to 0.001%" is withdrawn as circular).

24.3 The Hubble-sphere mass

DERIVED The critical energy density over the full ball gives

$$E_{\rm Hubble}=\frac{c^5}{2GH_0}=7.664\times10^{69}\ \text{J},\qquad M_{\rm Hubble}=\frac{c^3}{2GH_0}=8.527\times10^{52}\ \text{kg}\quad(H_0=73.04), \tag{60}$$

with $r_s=R_H$. The ratio $M_{\rm Hubble}/M_{\rm total}=4$ is the ratio of the sphere to the cone of height R_H, and it is the closure of Theorem 9. Multiplying the fractions of §24.5 by $M_{\rm Hubble}$ gives the component masses in the astronomical convention: baryons 4.254×10^{51}, condensate 2.315×10^{52}, all matter 2.740×10^{52} and vacuum 5.787×10^{52} kg.

24.4 Three currencies

DERIVED Mass, energy and information are one amount at fixed exchange rates:

$$\frac{E_{\rm total}}{M_{\rm total}}=c^2,\qquad \frac{E_{\rm total}}{I_{\rm total}}=\frac{\hbar H_0}{8\pi}=9.932\times10^{-54}\ \text{J nat}^{-1}, \tag{61}$$

$$\frac{M_{\rm total}}{I_{\rm total}}=\frac{\hbar H_0}{8\pi c^2}=\frac{M_\star}{8\pi}=1.105\times10^{-70}\ \text{kg nat}^{-1}, \tag{62}$$

with the natural mass unit $M_\star=\hbar H_0/c^2=2.777\times10^{-69}$ kg ($H_0=73.04$).

24.5 The cosmic budget

Reading note I5 is withdrawn: with the present at the resonance the dark ratio is 2.580 (§24.16). The 5/2 budget is kept here in full, and the numbers of this version are quoted at it so that each can be traced.

IDENTIFICATION (I5) The dark-energy-to-condensate ratio is identified with the defect power ratio of §25, $\Omega_\Lambda/\Omega_c = C_3/C_2 = 5/2$. DERIVED The baryon ratio is draft 1's product of the scale coefficient and the Torridity exponent, $\Omega_b/\Omega_c = \alpha_{\mathrm{TRSD}}\, n$. The exponent is dynamic (§17), and the universe reads it at the load of its own horizon: the surface gravity $g_H = c^2/2R_H = R_H/2t_H^2$, the size over twice the square of the age, which in units of the scale is the pure number $z_H = \pi^2/2\sqrt{3} = 2.8491$ IDENTIFICATION. There $n = 1.06540$, and

$$\frac{\Omega_b}{\Omega_c} = \frac{\sqrt{3}}{\pi^2}\, n\Big(\frac{\pi^2}{2\sqrt{3}}\Big) = 0.18697, \tag{63}$$

$+0.3\sigma$ from Planck; draft 1's $\sqrt{3}/3\pi$, with $n = \pi/3$, lay at -1.2σ. With $\Omega_\Lambda/\Omega_c = 5/2$ (the identification I5, withdrawn in §24.16) the budget was $\Omega_c = 0.27123$, $\Omega_b = 0.05071$, $\Omega_\Lambda = 0.67806$, $\Omega_m = 0.32194$, and matching the acoustic angle gave $H_0 = 67.12$ (GRUT §25.7 [32]); the framework's budget is the resonance budget of §24.16, and every current number is quoted there. The calibrated budget of draft 1, with $\mathcal{W} = 1 + \sqrt{3}/3\pi + 5/2 = (21\pi + 2\sqrt{3})/6\pi = 3.683776$,

$$\Omega_c = \frac{6\pi}{21\pi + 2\sqrt{3}} = 0.27146, \qquad \Omega_b = \frac{2\sqrt{3}}{21\pi + 2\sqrt{3}} = 0.04989, \tag{64}$$

$$\Omega_\Lambda = \frac{15\pi}{21\pi + 2\sqrt{3}} = 0.67865, \qquad \Omega_m = 0.32135. \tag{65}$$

PREDICTION Against Planck 2018 ($\Omega_m = 0.3153 \pm 0.0073$, $\Omega_\Lambda/\Omega_c = 2.589 \pm 0.088$, $\omega_b/\omega_c = 0.1864 \pm 0.0022$) [21] that budget's pulls were $+0.9\sigma$, -1.0σ and $+0.3\sigma$. Fitting the CMB acoustic angle at it with CAMB [26] gave

$$H_0 = 67.12 \pm 0.06\,(\theta_*) \pm 0.04\,(\nu)\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{66}$$

0.4σ below Planck's ΛCDM value and 5.7σ below SH0ES [22], with $\omega_m = 0.14504$ ($+1.9\sigma$) and $\omega_b = 0.02275$ ($+2.5\sigma$). The baryon ratio agrees with Planck's; the absolute baryon density does not, because $\Omega_\Lambda/\Omega_c = 5/2$ raises Ω_b and Ω_c together, by 2.4% at fixed ratio. PREDICTION The CMB measures ω_b from the acoustic peak heights: a full-spectrum fit of the budget that excludes $\omega_b = 0.02275$ at 5σ removes the identification I5. A calibrated Planck-like fit of the full spectra (GRUT §25.9 [32]) disfavours 5/2 at 4.6σ with the baryon ratio kept, preferring $\Omega_\Lambda/\Omega_c = 2.61 \pm 0.02$: close to, but short of, the 5σ threshold; the full Planck likelihood decides, and the baryon ratio passes. With the measured ratio in place of 5/2 the budget reproduces Planck's densities, $\omega_b = 0.02239$ and $\omega_c = 0.1198$, and lies 1.9σ from DESI's matter density, as ΛCDM does (GRUT §25.10). The 2026 DESI Lyman-α full-shape analysis, independent of the CMB, measures $\Omega_m = 0.325 \pm 0.018$ and, with a nucleosynthesis prior, $H_0 = 66.5 \pm 1.3$ [38]: -0.2σ and $+0.5\sigma$ from the budget. The budget also fixes $q_0 = -0.517$, the onset of acceleration at $z = 0.615$, and the epoch at which the horizon energy grows at exactly the maximum power $c^5/4G$, $z = 0.017$ [32].

24.6 The dual of the Hubble mass

DERIVED Evaluating the duality (45) at M_{Hubble},

$$m_U c^2 = \frac{\sqrt{8}}{8\pi}\,\frac{m_{\mathrm{P}}^2 c^2}{M_{\mathrm{Hubble}}} = \frac{\sqrt{2}}{2\pi}\,\hbar H_0 = 0.2251\,\hbar H_0, \tag{67}$$

3.5×10^{-34} eV for $H_0 = 73.04$: the dual of the whole horizon has the Hubble mass scale. (draft 1 quoted $\sim 10^{-33}$ eV.)

24.7 The transition radius

PREDICTION Around a mass M the acceleration equals a_c at

$$r_* = \sqrt{\frac{GM}{\mu(1)\, a_c}}, \qquad \mu_1(1) = 0.4206, \quad \mu_2(1) = 0.4623, \quad \mu_3(1) = 0.4790. \tag{68}$$

For $H_0 = 67.35$ and $n = 1.75$ ($\mu(1) = 0.4557$): 0.060 pc for the Sun, 1.9 kpc for $10^9 M_\odot$, 19 kpc for $10^{11} M_\odot$ and 0.60 Mpc for $10^{14} M_\odot$. DERIVED The phantom mass per unit radius, $dM_{\rm ph}/dr$, rises monotonically across r_* to its deep-regime plateau; computed for μ_1 and μ_3, it has no maximum, so there is no ring-shaped excess. The lensing signature is the transition itself: stacked excess surface density turns from Newtonian to isothermal ($\rho_{\rm ph} \propto r^{-2}$) at r_*. This replaces the draft-1 "horizon resonance ring".

24.8 Limits and the freeze

DERIVED Every dimensionful quantity is a pure number times a monomial in c, $\hbar$ and R_H and a power of N (§26). Three kinds of bound follow: constant limits, with no horizon in them (c, F, the force $c^4/4G$ between two bodies, the power $c^5/4G$); ceilings and floors, which contain the horizon (the lengths R_H and $\ell_{\rm P}$, the masses $c^2 R_H/2G$ and $\hbar/cR_H$, the forces F and $\hbar c/R_H^2$), whose ratios are $\sqrt{N/2\pi}$, $N/4\pi$ or $N/2\pi$; and clock readings, built from a particular event, which fall as $1/t$. Since $N = 2\pi R_H^2/\ell_{\rm P}^2 \propto H^{-2}$ and $H \to H_\infty = H_0\sqrt{\Omega_\Lambda}$ for $w = -1$,

$$\frac{N(t)}{N_\infty} = \Omega_\Lambda(t) \tag{69}$$

exactly: the universe holds 68.4% of its final cells, $N_\infty = 6.63 \times 10^{122}$, and the horizon stops at 17.56 Gly. The horizon energy grows at the maximum power $c^5/4G$ when $\Omega_\Lambda = \frac{2}{3}$, 0.38 Gyr ago, and the aeon's conformal time is finite, $4.327\, c/H_0 = 62.8$ Gly, of which 73.4% has elapsed—0.7310 ± 0.0062 from the DESI Lyman-α Ω_m [38], $+0.5\sigma$ (GRUT §48.1–48.5 [32]). PREDICTION All three require $w = -1$ and fail with evolving dark energy at 5σ [25].

24.9 Constants and clock readings

DERIVED The cells do not stretch: with a fixed cell count $G \propto R_H^2$ would drift at $2(1+q)H = 6.5 \times 10^{-11}$ yr^{-1}, against the $(7.1 \pm 7.6) \times 10^{-14}$ yr^{-1} of lunar laser ranging [46], so the Planck length is fixed and $N \propto R_H^2$. The fine-structure constant is not a clock reading, $\dot\alpha/\alpha = (1.0 \pm 1.1) \times 10^{-18}$ yr^{-1} [47], but it sets when light escapes: in the Saha balance for hydrogen, whose binding energy is $\frac{1}{2}\alpha^2 m_e c^2$, $d\ln(1+z_{\rm rec})/d\ln\alpha = 2.07$ and $d\ln(t_0/t_{\rm rec})/d\ln\alpha = 3.54$.

24.10 First starlight

HYPOTHESIS If the first starlight marks the present, $t_\star = \alpha t_0$: cosmic dawn, the onset of the global 21-cm coupling, at 100.6 Myr, $z = 29.6$, 46.4 MHz; stated between physical epochs, $t_\star = \alpha t_{\rm mp} =$ 97.9 Myr, $z = 30.2$, 45.6 MHz. PREDICTION Coupling detected below 45.5 MHz excludes it. A 3% timing gives the age, $t_0 = 137.036\, t_\star$, and H_0 to ± 2.0 km s^{-1} Mpc^{-1} independently of the distance ladder; with the condensate it requires $m_\delta \gtrsim 4 \times 10^{-19}$ eV. A chance match within

today's estimated range has probability 0.53; within 3%, stated in advance, 0.04 (GRUT §48.8–48.10). HYPOTHESIS With the present selected by the resonance (§24.13), the relation no longer selects it; it relates two physical epochs, cosmic dawn and the resonance, $t_\star = \alpha\, t_{\rm res}$, and needs no observer: at the resonance budget (§24.16) 100.6 Myr, $z = 29.6$, 46.4 MHz, the ±3% window 45.5–47.3 MHz. It is not derived, since the onset depends on the primordial amplitude and the first stars. No starlight before it is known: the earliest confirmed galaxy is at $z = 14.44$ [79], and the claimed $z \simeq 32$ candidate is a Y dwarf in the Galaxy [80]. DERIVED A frequency is a number of cycles per second, so the onset is read as a ratio of counts, the cycles received per cycle emitted, $\nu_\star/\nu_{21} = 1/(1+z_\star)$ with $\nu_{21} = 1420.405751768$ MHz, and the second cancels. With the present at the resonance,

$$\frac{\nu_\star}{\nu_{21}} = \Big[\frac{\sqrt{4\pi^2-27}}{3\sqrt{3}}\,\sinh\Big(\alpha\,\operatorname{artanh}\frac{3\sqrt{3}}{2\pi}\Big)\Big]^{2/3} = \frac{1}{30.81} \tag{70}$$

(46.10 MHz; 1/30.61, 46.41 MHz, with radiation). The birefringence counts turns of the same phase weighted by α, $\beta = (\kappa\alpha/2\pi)\,H_\infty(t_0 - t_{\rm rec})$ (§24.14, §25.3), so α cancels between the two, and with it the present's phase:

$$\boxed{H_\infty t_\star = \frac{2\pi}{\kappa}\,\beta = \frac{\pi}{3}\,\beta, \qquad \frac{\nu_\star}{\nu_{21}} = \Big[\sqrt{\frac{\Omega_m}{\Omega_\Lambda}}\,\sinh\frac{\pi\beta}{2}\Big]^{2/3}.} \tag{71}$$

PREDICTION The measured $0.277° \pm 0.057°$ [50] places the onset at $42.7^{+5.6}_{-6.0}$ MHz ($z = 32.2$), 0.6σ from the closed form; it moves as $\beta^{0.66}$, so a twist known to ±0.01° places it to ±1.0 MHz. An onset and a twist off (71) by 5σ together exclude the hypothesis with the rotation's birefringence. No clock, no α and no Hubble constant enter; the relation tests the hypothesis and does not derive it (GRUT §48.8 [32]).

24.11 The capacity of the horizon

DERIVED With $V_0 = 0$ (§21.3) and cells that do not stretch, the dark energy is the energy of the horizon's finite capacity,

$$\rho_\Lambda = \frac{3}{4}\,\frac{\rho_P}{N_\infty} = 1.130\times10^{-123}\rho_P, \qquad \rho_P = \frac{c^7}{\hbar G^2}, \tag{72}$$

exactly: the number of the cosmological constant problem [43] is the inverse of the final cell count. A vacuum of Planck density would exceed the horizon's content by exactly $4N/3$ and put it inside its own Schwarzschild radius; gravity couples to every field alike, so the horizon condition caps the vacuum energy of all fields together at the critical density [44, 45]. In the information–energy relation of Section 19, each bit of the final horizon carries $\hbar c/8\pi R_\infty = \frac{1}{4}k_B T_\infty$, with $T_\infty = \hbar H_\infty/2\pi k_B = 2.18\times10^{-30}$ K. DERIVED The value of N_∞ is the boundary datum (§26: no internal condition fixes it). The suppression of high-frequency vacuum modes follows mode by mode: a field theory in the horizon is consistent only up to the wavenumber at which its zero-point energy reaches the critical density, $k_{\rm max} = (6\pi)^{1/4}/\sqrt{\ell_{\rm P} R_H}$, a wavelength of 0.14 mm [44] (GRUT §48.11).

24.12 The closure law

DERIVED By the defect map, $\delta = 8\pi GM/(c^2 R_H) = 4\pi r_s/R_H$, every energy inside the horizon is a defect on it, and the horizon is a closed surface, $\chi = 2$. Gauss–Bonnet fixes the total, as in 2+1 gravity, where a closed universe has total deficit exactly 4π [14, 62]:

$$\sum_i r_{s,i} = \frac{\chi}{2}\,R_H, \qquad \Omega_{\rm total} = \frac{\chi}{2} = 1, \tag{73}$$

a topological form of the flatness derived above (Planck: 1.0007 ± 0.0019). The $\mathbb{Z}_2$ cut of the deficit π halves energy, defect and information together; iterated, it gives the scale ladder $E_n = E_H/2^n$ from the Hubble mass, $E_H = c^4 R_H/2G$, down to one Planck cell at $n = \log_2(4\pi R_H^2/\ell_P^2) = 408.5$, whose energy $\hbar H_0/8\pi$ is the energy per bit of the information–energy relation. Its fixed point is the Planck energy, where the gravitational and quantum lengths cross, $r_s\lambda_C = 2\ell_P^2$; the dark-energy scale sits three quarters of the way down; and independently formed masses show no preference for rungs (GRUT §50 [32]).

24.13 Now: where the horizon meets the final rate

HYPOTHESIS The rotation of the horizon's phase (§21.1) is held at a constant rate, and the one constant rate of the late universe is the final expansion rate H_∞; the rotation turns at H_∞. DERIVED The lowest mode of the C_3 cycle on the horizon (Section 9) has frequency $c\sqrt{\lambda_1} = (3\sqrt{3}/2\pi)H(t)$, which falls as the universe expands, and it meets the rotation's rate once:

$$\frac{3\sqrt{3}}{2\pi}\,H_{\text{now}} = H_\infty, \qquad \Omega_\Lambda(\text{now}) = \left(\frac{H_\infty}{H_{\text{now}}}\right)^2 = \frac{27}{4\pi^2} = 0.6839. \tag{74}$$

That moment is the present: the future's rate meeting the present horizon. It fixes when now is, $t_{\text{now}} = 0.786/H_\infty$, and with the baryon ratio of the budget it gives $\Omega_\Lambda/\Omega_c = 2.580$ in place of $5/2$, with the neutrinos counted inside Ω_m as everywhere in the budget (release 1.6.0 left them out and quoted 2.568); the full budget is §24.16. Passing through resonance turns the relative phase by π, so the present is a sign flip of that phase, as the exchange of time and radius is at a black hole's horizon and the gluing of one aeon's future to the next one's past is at the reset. Planck gives $\Omega_\Lambda = 0.6847 \pm 0.0073$, 0.1σ from (74); the full-spectrum fit with the derived baryon ratio prefers 2.611 ± 0.024, 1.3σ (GRUT §36.7, §25.11 [32]). The identification of the rotation's rate with H_∞ was made with the measured value in view and is recorded as a hypothesis, tested as the CMB and BAO sharpen. DERIVED The constraint makes the rate a constant of the solution, so it can be read in the final state, where the only scale is H_∞. IDENTIFICATION The rate is the final horizon's own frequency, its surface gravity over c, $\kappa_\infty/c = H_\infty$: the period of its thermal circle, with $T_\infty = \hbar H_\infty/2\pi k_B$, and the lowest conformal mode on its spatial sphere of radius c/H_∞ both give it (GRUT §36.8 [32]). The equilibrium of the phase with the final horizon is the premise not yet derived from the action. DERIVED It is derived up to an integer by the quantum of the spin, and the integer is measured to be one (§24.15).

24.14 The spinning-arc principle

DERIVED (A hypothesis in release 1.4.0; derived for the final horizon in §24.15.) The horizon is not a static sphere: it is swept out by C-arcs spinning about perpendicular axes, vertical and horizontal, crossing one another (the geometric origin of Section 3). The spin is physical. Nothing acts on the arcs, so their rate is fixed, and on a horizon, a surface receding at the speed of light, the arcs' rims move at c:

$$\boxed{\omega R = c.} \tag{75}$$

DERIVED *The cosmic sphere.* The universe's horizon grows toward the radius at which the fixed spin's rims reach light speed, and no further: $R_\infty = c/\omega$, so $H_\infty = \omega$ and

$$\Lambda = \frac{3\omega^2}{c^2}. \tag{76}$$

The cosmological constant is the signature of the spin. The internal rotation is the spin itself (I7, derived in §24.15), so it turns at H_∞, with no equilibrium premise, and the present is the moment the current horizon's C_3 cycle has slowed to the spin, $\Omega_\Lambda = 27/4\pi^2$ (§24.13).

DERIVED *The wobble meets the spin.* The threefold wobble of the arcs, which creates the three generator defects (Section 3; its count is the beat of §2.4), is the horizon's C_3 mode: its frequency $(3\sqrt{3}/2\pi)\,c/R$ falls as the horizon grows, while the spin is fixed. The present is the moment the wobble falls into step with the spin.

The spin is measured. The birefringence of the CMB is the spin's phase accumulated since last scattering, $\beta = \kappa\alpha\,\omega(t_0 - t_{\rm rec})/2\pi$ (§25.3), so the measured angle measures the spin: $0.277^\circ \pm 0.057^\circ$ gives $\omega = (0.73 \pm 0.15)H_0$, against $H_\infty = 0.827H_0$ from the expansion, 0.6σ. Equivalently the twist of the oldest light measures the cosmological constant, $\beta = (\kappa\alpha/2\pi)\sqrt{\Lambda c^2/3}\,(t_0 - t_{\rm rec})$.

DERIVED *The quantum sphere.* The same rule holds for a particle. A sphere swept at frequency ω has radius c/ω; with energy $\hbar\omega = mc^2$ that radius is $\hbar/mc$, the reduced Compton wavelength, and with the zero-point energy $\hbar\omega/2$ it is half that. The internal clock de Broglie gave every particle, at $mc^2/\hbar$ [63], and the light-speed trembling Schrödinger found in the electron's motion, at $2mc^2/\hbar$ and radius $\hbar/2mc$ [64, 65], are this rule at the smallest scale. From the electron to the edge of the universe each closed sphere is swept by arcs whose rims move at light speed, and the Hubble quantum at the foot of the ladder, $\hbar H_\infty$, is the cosmic sphere's quantum of spin (§24.12).

24.15 The spin derived

STANDARD In the final state the vacuum alone remains ($V_0 = 0$, §21.3; $w = -1$), and the horizon is the de Sitter horizon of radius $R_\infty = c/H_\infty$. With ϱ the distance from a static observer, its thermal form, static time $T \to -i\tau$, is the four-sphere of radius R_∞ [66],

$$X_0 + iX_4 = \sqrt{R_\infty^2 - \varrho^2}\; e^{iH_\infty \tau}, \qquad \vec{X} = \varrho\,\hat{n}, \qquad X_0^2 + X_4^2 + |\vec{X}|^2 = R_\infty^2. \tag{77}$$

DERIVED Static time is the angle of a turn, and the turn is the spinning-arc picture exactly. Each line of sight, from a horizon point through the observer to the opposite horizon point, is a C-arc of radius R_∞ spinning about the axis through its ends; lines of sight at right angles spin about perpendicular axes, and all of them cross at the observer. A point at distance ϱ moves at $c\sqrt{1 - \varrho^2/R_\infty^2}$, the rate of a static clock there: light speed on the rim, which is the observer, and rest at the horizon. At proper distance σ from the horizon the plane of the turn is $\sigma^2(H_\infty d\tau)^2 + d\sigma^2$, which closes without a conical defect only for the period $2\pi/H_\infty$: smoothness where the arcs meet the horizon fixes the rate. Hence

$$\boxed{\omega = H_\infty, \qquad \omega R_\infty = c, \qquad \Lambda = \frac{3\omega^2}{c^2}, \qquad k_B T_\infty = \frac{\hbar\omega}{2\pi}.} \tag{78}$$

STANDARD The zeroth law makes the rate the same over the whole horizon [67], and the Bianchi identity, with matter separately conserved, forces $\partial_\nu \Lambda = 0$; in unimodular gravity Λ is the constant conjugate to the four-volume [68]. DERIVED Nothing in the action exchanges energy with the vacuum ($V_0 = 0$; the rotation's constraint adds none, §21.1), so the spin does not change. It is the boundary datum read as a rate, $\omega\ell_{\rm P}/c = \ell_{\rm P}/R_\infty = \sqrt{2\pi/N_\infty} = 9.66 \times 10^{-62}$ (§26).

The charge. STANDARD The entropy of a Killing horizon is 2π times the Noether charge of its Killing field at unit surface gravity [69]. DERIVED For a horizon of radius R_H turning at its own rate c/R_H that charge is the Hubble mass carried around the horizon at light speed, $L = M_{\rm Hubble}\,cR_H = \hbar N/4\pi$, and

$$I_{\rm total} = \frac{2\pi L}{\hbar}, \qquad E_{\rm total} = \frac{\omega L}{4} = \frac{\hbar\omega}{8\pi}\,I_{\rm total}, \qquad \omega = \frac{c}{R_H}: \tag{79}$$

the information–energy relation (40) is the energy of the turn, and its 8π is one turn, 2π, shared over the four cones of the closure, $M_{\rm Hubble} = 4M_{\rm total}$ (§24.3). STANDARD A body with mechanical

spin s has the lower entropy bound $2\pi\sqrt{(ER/c)^2 - s^2}/\hbar$ [70]. DERIVED The horizon meets the Bekenstein bound with equality, $2\pi E_{\rm Hubble} R_H/\hbar c = \pi c^5/\hbar G H_0^2 = I_{\rm total}$ (§24.2), so it cannot spin mechanically: its spin is the turn of its time, and the universe's rotation is internal (§21.1). Read as a spin, the Bekenstein bound is $S \le 2\pi L_{\rm max}/\hbar$ with $L_{\rm max} = ER/c$, reached only on a rim moving at light speed; with $I = 2\pi L/\hbar$ and $E = \omega L$ the horizon saturates it exactly when $\omega R = c$.

The quantum. DERIVED The turn is an angle of period 2π, so its charge is $L = n\hbar$: $I = 2\pi n$ nats, $N = 4\pi n$, an area step $8\pi\ell_{\rm P}^2$ ($4\pi\ell_{\rm P}^2$ of effective area, the surface of one Planck sphere, which carries $\hbar$), and $\hbar\omega$ per quantum. STANDARD This is the spacing Bekenstein proposed [71] and Maggiore found in the quasinormal modes [72]; here it is the quantum of the spin. The same holds for every horizon: for a black hole the turn's rate is $c^3/4GM$, 846 rad s^{-1} at 60 $M_\odot$, and the LIGO–Virgo ringdowns give the area quantum $15.6^{+20.5}_{-13.3}\,\ell_{\rm P}^2$ [73], $+0.5\sigma$ from 8π. For a particle the rule $L = E/\omega$ gives $\hbar$ at the Compton radius and, at the zero-point energy $mc^2 = \frac{1}{2}\hbar\omega$, $\hbar/2$ at the trembling radius $\hbar/2mc$ [65].

What it predicts. IDENTIFICATION (I7) The phase of §21.1 turns with the spin, $X_0 = \Lambda/3$: one radian per e-fold of the final expansion. The birefringence gives $\omega/H_\infty = 0.88 \pm 0.18$.

The quantum of the rotation DERIVED The quantum of the spin derives I7 up to an integer. On the rotation the constraint vanishes, so the dust's energy is its Hamiltonian, $E = p_\theta\dot\theta = \Omega_0 Q$, with Q the momentum conjugate to the phase; the phase is an angle, so $Q = n\hbar$ and the dust comes in quanta of rest energy $\hbar\Omega_0$, the Josephson–Anderson relation [75]. STANDARD A comoving quantum carries a Killing energy equal to its rest energy, and every one but the observer's own leaves the static patch; energy carried across the cosmological horizon raises its entropy by $\delta E/k_B T_\infty$ [66]. DERIVED That entropy steps by 2π, so the horizon takes up energy only in multiples of $\hbar\omega$, and

$$\boxed{\Omega_0 = j\,H_\infty, \qquad X_0 = j^2\,\frac{\Lambda}{3}, \qquad j = 1, 2, 3, \dots} \tag{80}$$

MEASURED The birefringence, with $\kappa = 6$, puts $j = 1$ at 0.7σ and excludes $j = 2$ at 6.2σ and $j = 0$ at 4.9σ; with the present at the resonance, Ω_Λ gives 0.9994 ± 0.0053. So $j = 1$: each quantum of the rotation is one quantum of the spin, at the floor mass $\hbar/cR_\infty$, and I7 is derived, with the integer measured. The dust's density is the constraint's multiplier, which carries no gradient energy, so the quantum sets no Jeans length and the dust stays cold on every scale. A scalar condensate in equilibrium with a rotating black hole is locked the same way, $\omega = m\Omega_{\rm H}$ with m an integer [76] (GRUT §36.14 [32]). PREDICTION With the present at the resonance of §24.13 the phase turned since the beginning is $\theta_0 = \frac{2}{3}\,\mathrm{artanh}(3\sqrt{3}/2\pi) = 0.7857$ rad (0.78545 with radiation; it lies within 5×10^{-5} of $\pi/4$, a coincidence noticed after the computation, with a chance of 0.15%), and the birefringence angle is

$$\boxed{\beta = \frac{2\alpha}{\pi}\,\mathrm{artanh}\,\frac{3\sqrt{3}}{2\pi} = 0.3137^\circ,} \tag{81}$$

$+0.6\sigma$ from $0.277^\circ \pm 0.057^\circ$ [50], with no Hubble constant in it. The acceleration scale is set by the cosmological constant alone, $a_0 = (\sqrt{3}/\pi^2)\,cH_{\rm now} = (2/3\pi)\,c^2\sqrt{\Lambda/3}$, 1.149×10^{-10} m s^{-2} with Planck's Λ: the spin fixes the factor of order one in Milgrom's $a_0 \approx c^2(\Lambda/3)^{1/2}$ [74] (GRUT §36.10–36.13 [32]).

24.16 The budget at the resonance

DERIVED The amount of the cosmic share is an integration constant of the rotation's constraint (§21.1), and in a flat universe with a constant vacuum $\Omega_\Lambda(t) = H_\infty^2/H(t)^2$ exactly: the constant

decides when the resonance falls, not the budget there. With the rotation at the spin (§24.15) and the present at the resonance (§24.13, the opening principle), $\Omega_\Lambda = 27/4\pi^2$ whatever that constant is, and flatness, the derived baryon ratio and the neutrino masses fix the rest. Computed with CAMB with the acoustic angle matched, on the scale of the budgets of GRUT §25.10:

	5/2 (I5)	resonance	data
Ω_Λ/Ω_c	2.500	2.580	2.611 ± 0.024 (CMB spectra)
H_0 ($\mathrm{km\,s^{-1}\,Mpc^{-1}}$)	67.04	67.35	67.36 ± 0.54 [21]
ω_b	0.02279	0.02248	0.02237 ± 0.00015
ω_c	0.12188	0.12022	0.1200 ± 0.0012
Ω_m	0.3234	0.3160	0.2975 ± 0.0086 (DESI) [25]
S_8	0.846	0.833	0.832 ± 0.013
a_0 (10^{-10} $\mathrm{m\,s^{-2}}$)	1.143	1.148	$1.04 \pm 0.04 \pm 0.10$ (§14)

PREDICTION The resonance budget lies 1.3σ below the CMB's preferred ratio, $+0.7\sigma$ in ω_b, $+0.2\sigma$ in ω_c, and $+2.2\sigma$ and -2.1σ from the DESI DR2 and DES-Y5 matter densities, where ΛCDM's own CMB fit lies; the material variances of the 5/2 budget, the baryon density at $+2.5\sigma$ and DESI at $+3.0\sigma$, were the identification's. It gives $\eta = 6.16 \times 10^{-10}$, $+0.5\sigma$ from the baryon density of deuterium burning [78], $N_\infty = 6.63 \times 10^{122}$, $\rho_\Lambda = \frac{3}{4}\rho_P/N_\infty = 1.130 \times 10^{-123}\rho_P$, and 73.4% of the conformal time elapsed. DERIVED Hence I5 is withdrawn: the dark ratio is not the defect power ratio, whose value 5/2 stays the single-triple limit of the sky pattern (§25). The value 2.568 of release 1.6.0 left the neutrinos out of Ω_m. The earlier numbers of this version are quoted at the 5/2 budget, so that each can be traced (GRUT §25.11 [32]).

24.17 The acceleration scale tracks the horizon

DERIVED The critical acceleration is one activation per period of the current horizon's lowest C_3 mode (I2, Section 9), so $a_c \propto H(t)$ and $a_0(z) = a_0 E(z)$ with $E = H(z)/H_0$; the spin is fixed, and the two meet only at the resonance. The relations that equate them, $a_0 = (2/3\pi)\,c^2\sqrt{\Lambda/3}$ and $\beta = \frac{9}{2}\alpha\, a_0 t_0/c$, hold at the present, where they are tested, and the load $z_H = \pi^2/2\sqrt{3}$ of the baryon ratio is a pure number only because a_0 tracks H. MEASURED The radial acceleration relation of 79 galaxies at $0.33 < z < 1.44$ [77] has a scale rising from 1.99 to 2.71×10^{-10} $\mathrm{m\,s^{-2}}$ across four bins near $z = 0.47$–1.27, so a constant scale is excluded; $a_0 E(z)$ gives 1.49–2.40. The data lie 13–33% above and rise by 1.36 where $H(z)$ rises by 1.61, with the survey's own fit extrapolating to 1.00 ± 0.04 at $z = 0$. PREDICTION A constant a_0, or one off $a_0 E(z)$ at 5σ in a single method spanning $z = 0$ to 1.5, falsifies the reading (GRUT §36.15 [32]).

24.18 The present in Planck cells

HYPOTHESIS The resonance fixes the present in the expansion history (§24.13); the horizon's size in Planck lengths is the framework's one boundary datum (§26). The hypothesis H4 of GRUT §48.12 [32] fixes it by the electric coupling:

$$\ln \frac{R_H}{\ell_\mathrm{P}} = \frac{1}{\alpha} + \ln 8\pi + \frac{3\pi}{2}\,\alpha, \tag{82}$$

the horizon's solid angle over the double cover of the fold, and the three arcs times the pair's quarter turn at first order in α: the form of dimensional transmutation, $R_H = 8\pi\ell_\mathrm{P} e^{1/\alpha}$, with the coupling run through $1/\alpha$ e-folds from one Planck cell. It selects the epoch and is not a law for α: tied to the growing horizon, α would drift by 2.4×10^{-13} $\mathrm{yr^{-1}}$, against $(1.0 \pm 1.1) \times 10^{-18}$ $\mathrm{yr^{-1}}$ from atomic clocks [47]. PREDICTION With CODATA's α it gives $H_0 = 67.3765 \pm 0.0007$ (the

error is G's), 0.4σ from the resonance budget's CMB fit, 67.35 ± 0.06; with the resonance, $\Lambda \ell_{\mathrm{P}}^2 = (81/256\pi^4)\, e^{-2/\alpha - 3\pi\alpha} = 2.843 \times 10^{-122}$ and $\rho_\Lambda/\rho_P = 1.131 \times 10^{-123}$. The constants were read after the value was known; H4 fails if H_0 settles away from 67.38 at 5σ or if α varies (register row 38).

25 The Conical Horizon on the CMB Sky

25.1 Harmonic expansion

DERIVED Model the three generator defects as point sources of potential on the equator, $\hat{n}_j = (\pi/2,\, 2\pi(j-1)/3)$, each weighted by $\delta_{\mathrm{gen}} = \pi/3$: $\Phi(\hat{n}) = A \sum_j \delta_{\mathrm{gen}}\, \delta^{(2)}(\hat{n} - \hat{n}_j)$, with Sachs–Wolfe temperature $\Delta T/T = -\Phi/3$. Then $b_{\ell m} = A \delta_{\mathrm{gen}} \sum_j Y^*_{\ell m}(\hat{n}_j)$, and $\sum_j e^{-im\phi_j} = 3$ for $3 \mid m$ and 0 otherwise. The plane of the defects is reflection-symmetric, and $Y_{\ell m}(\pi - \theta, \phi) = (-1)^{\ell+m} Y_{\ell m}(\theta, \phi)$, so

$$b_{\ell m} = 0 \quad \text{unless} \quad m \equiv 0 \pmod 3 \ \text{ and } \ \ell + m \text{ even}. \tag{83}$$

Summing the addition theorem over pairs of defects gives the closed form

$$C_\ell = \frac{1}{2\ell+1} \sum_m |b_{\ell m}|^2 = \frac{3A^2\delta_{\mathrm{gen}}^2}{4\pi} \Big[1 + 2P_\ell\big(-\tfrac{1}{2}\big) \Big], \tag{84}$$

so $C_1 = 0$, and with $P_2(-\frac{1}{2}) = -\frac{1}{8}$ and $P_3(-\frac{1}{2}) = \frac{7}{16}$,

$$\frac{C_3}{C_2} = \frac{1 + \frac{7}{8}}{1 - \frac{1}{4}} = \frac{5}{2}. \tag{85}$$

The quadrupole is pure $(2, 0)$ and the octupole pure $(3, \pm 3)$ in the defect frame. The normalization $1/(2\ell+1)$ is essential: the bare amplitude ratio is $|b_{33}|/|b_{20}| = \sqrt{7}/2$. Further ratios are $C_4/C_2 = 0.5625$, $C_5/C_2 = 1.0938$, $C_6/C_2 = 2.1953$.

25.2 What the sky can test

STANDARD For a Gaussian full sky the cosmic variance of C_ℓ is $\sqrt{2/(2\ell+1)}$, so the ratio C_3/C_2 carries an irreducible 83% (1σ) uncertainty. The observed ratio also depends on the component-separation map and mask. The draft-1 comparison of the number $2\pi^{\pi/3}/3 + \frac{1}{12} = 2.2940$ HYPOTHESIS with an observed 2.2887 is therefore not a test at the stated 0.23%: the sky constrains C_3/C_2 only to within a factor of about two. The value 5/2 is a property of the assumed three-point source configuration; it becomes an observational prediction only after projection through the radiation transfer function, which is not yet done.

The selection rule (83) is testable. About an axis $\hat{n}$, the share of power in the admissible modes has, under isotropy, mean $f_\ell = |\mathcal{M}_\ell|/(2\ell+1) = 0.111, 0.182, 0.231, 0.133, 0.176, 0.211, 0.143$ for $\ell = 4, \ldots, 10$; a defect component of fraction ε raises it to $\varepsilon + (1-\varepsilon) f_\ell$. The parity-odd admissible set is an internal null channel. MEASURED The search at $\ell = 4$–10 about the octopole axis on the Planck Commander map returned no excess over f_ℓ: ε is consistent with zero, so by criterion F7 the point-source imprint is not detected. The pattern after projection through the radiation transfer function—smoothed, not point-like—is the next target.

25.3 Handedness

For a real sky $a_{\ell,-m} = (-1)^m a^*_{\ell m}$, so $|a_{3,3}| = |a_{3,-3}|$ in every frame. The draft 1 handedness prediction $|a_{3,+3}| \neq |a_{3,-3}|$ is impossible and is withdrawn. A handed structure appears instead

in the phase of a_{33} and in the parity-odd EB correlation, where the Planck data show a cosmic birefringence angle $\beta = 0.34° \pm 0.09°$ [27]; DERIVED in the rotation reading its size is $\beta = \kappa\alpha\,\Omega_0(t_0 - t_{\rm rec})/2\pi = 0.052°\kappa$, with $\kappa = \sum N_c Q^2 = 3+3 = 6$ the anomaly coefficient of the two cone triples, the charged leptons and (c, b, t): $\beta = 0.312°$, 0.6σ above the joint value $0.277° \pm 0.057°$ [50], and its sign is the sense of the rotation (GRUT §49.12). DERIVED The rotation's rate is also the acceleration scale's: $a_c = c\,\Omega_0/2\pi$ and $a_0 = \frac{4}{3}a_c$, so $\Omega_0 = (3\pi/2)\,a_0/c$, and eliminating it,

$$\beta = \frac{3\kappa\alpha}{4}\,\frac{a_0 t_0}{c} = \frac{9\alpha}{2}\,\frac{a_0 t_0}{c} \qquad (\kappa = 6). \tag{86}$$

Galaxy rotation curves predict the polarization rotation of the CMB: the radial acceleration relation's $a_0 = (1.20 \pm 0.24) \times 10^{-10}\ \mathrm{m\,s^{-2}}$ [10] with $t_0 = 13.80$ Gyr gives $\beta = 0.33° \pm 0.07°$, 0.6σ from the measured angle. Milgrom's coincidence $2\pi a_0 \approx cH_0$ [61] ties a_0 to the expansion; this ties it, through α, to the polarization of the oldest light (GRUT §49.13). With the rate set by the spin and the present at the resonance, the angle takes the closed form $\beta = (2\alpha/\pi)\,\mathrm{artanh}(3\sqrt{3}/2\pi) = 0.3137°$ (§24.15).

26 Parameter Structure

Theorem 12 (Input counting). DERIVED *A theory whose primitives are dimensionless—angles, incidence counts, ratios—determines no dimensionful constant. The unit-rescaling group of mechanics is three-dimensional, so three dimensionful anchors are conventions and carry no predictive content.*

Proof. The rescalings of mass, length and time form $(\mathbb{R}_{>0})^3$; the primitives are invariant, so every function of them is, and no dimensionful constant is. Fixing three independent dimensionful quantities fixes the units; by Buckingham's theorem every other quantity is then a pure number times a monomial in them. □

The triple $(c, \hbar, R_H)$ spans the units: length R_H, time R_H/c, mass $\hbar/cR_H$. The remaining input is the pure number $N = 2\pi R_H^2/\ell_{\rm P}^2 = 2\pi c^5/\hbar G H_0^2$, which enters only through $\ell_{\rm P}$: G, $\ell_{\rm P}$, $m_{\rm P}$ and F carry powers of N, and no prediction of the gravity law, the budget or the selection rule does. N is a boundary datum, measured, not fitted. The closure $F/M_{\rm Hubble}a_0 = \pi^2/2\sqrt{3} = 2.8491$ holds for every N, so the horizon quantities cannot determine N; a condition from outside—plausibly at the reset of §29—would be needed.

Quantity	MOND + ΛCDM	TRSD draft 2
a_0	fitted	derived, $\alpha_{\rm TRSD} cH_0$
interpolation function	chosen	derived family; tail order constrained by data
Ω_Λ/Ω_c	fitted	identified (I5)
Ω_b/Ω_c	fitted	derived (§24.5)
H_0	fitted	from the CMB angle given the budget
dark-matter scale	–	calibrated (L_*)
boundary datum	–	N

26.1 Classification of every number

Class	Members
unit conventions	c, $\hbar$, and a length (Theorem 12)
boundary datum	N (equivalently H_0)
derived	a_c/cH_0, a_0/cH_0, $\frac{3}{4}$, the family μ_n, f_δ^2, $\varepsilon_V = f_\delta^2/L_*^2$, $V_0 = 0$, g_χ, $M_{\text{Hubble}} = 4M_{\text{total}}$, m_U, the source-model C_3/C_2
identifications	I1–I6 (§0.4)
data-constrained	the tail order n, $1.62 \le n \le 1.91$ (§14–15)
calibrated	L_*, ϕ_i (Ω_b/Ω_c, formerly calibrated, is derived in §24.5)
observational inputs	Planck θ_* (for H_0), the SPARC data (for n and the tests)

26.2 Dependency of the predictions

Prediction	Depends on	Data already used	Independent test
$a_0 = 0.1755\,cH_0$	C_3 spectrum, $\frac{3}{4}$, I2, H_0	Planck θ_* via the budget	SPARC likelihood (§14.3)
gravity-law shape	I3, n	SPARC and ephemerides constrain n	new rotation curves, lensing, the external-field quadrupole
$\Omega_m = 0.3160$	H3 (the resonance), Ω_b/Ω_c from the dynamic exponent, $\sum m_\nu$	–	supernovae, DESI BAO and Lyα
$H_0 = 67.35$	Ω_m, θ_*, $\sum m_\nu$	Planck θ_*, ω_b/ω_c	distance ladders, DESI Lyα + BBN
Ω_b/Ω_c	dynamic exponent at the horizon load	–	Planck ω_b/ω_c
H_0 against Planck ΛCDM	shares θ_*	Planck	none (consistency)
$w = -1$, $V_0 = 0$	$\varepsilon_V = f_\delta^2/L_*^2$, Ly$\alpha$ bound	Lyα, ρ_Λ	DESI $w(z)$
$m_\delta > 2 \times 10^{-20}$ eV	I6, applicability (§21.6)	Lyα	cores, other small-scale probes
selection rule	three-point source model	Commander $\ell = 4$–10 (null)	smoothed, projected pattern

The gravity law, its SPARC confrontation and the scalar-sector closure are shared with GRUT [32]; the two papers are one theory in two notations, and their agreement on these is consistency, not independent confirmation.

27 Discussion

MOND and AQUAL. MOND [1] and AQUAL [2] take μ and a_0 as inputs. TRSD derives both, and with H_0 from the CMB the prediction meets the SPARC relation with no parameter (§14).

Covariant completion. TeVeS [3] and bimetric MOND [4] are covariant and add fields. TRSD's gravity law is the nonrelativistic AQUAL form and TRSD does not yet supply covariant field equations; lensing, the post-Newtonian parameters, cosmological perturbations, gravitational waves and energy–momentum conservation therefore remain open. DERIVED A completion must (1) reduce to (28) with μ_n in the window in the quasistatic weak field; (2) lens by the same potential that governs dynamics; (3) propagate tensor modes at c, as GW170817 requires [40]; (4) give $\gamma = \beta = 1$ in the solar system at the Cassini level; (5) conserve energy–momentum through the Bianchi identity; (6) reproduce the CMB and matter spectra at linear

order; and (7) be free of ghost and gradient instabilities. Such theories exist for other interpolation functions: AeST meets (1)–(3), (6) and (7) [39]. GRUT's completion, general relativity with the conjugate-pair scalar held at its rotation rate by $\lambda(X - X_0)$, meets (2)–(6): minimally coupled, it propagates tensor modes at c, conserves energy–momentum through the Bianchi identity, lenses by the dynamical potential for pressureless sources, reduces to general relativity where no medium is bound, as in the Solar System, and evolves as cold dust in cosmology; (7) holds at linear order; and (1) follows for bound systems from the polarization law of the bound medium (IDENTIFICATION).

Emergent gravity. Verlinde's construction gives $a_0 = cH_0/6 = 0.1667\,cH_0$ [8]; TRSD gives $0.1755\,cH_0$. The two differ by 5%, inside the present ±20% systematic of galaxies; a measurement of a_0 to 2% with H_0 fixed would separate them.

Clusters and the CMB. Phantom mass alone does not account for the mass of clusters [28] or the third acoustic peak; the condensate is the clustering component, and its share is fixed by the budget. In the rotation reading the clustering component is spacetime excited, geodesic and collisionless, and what lensing infers as dark matter is its dilation, $\mathcal{S} = 1/\cos_n\theta$; in the Bullet Cluster the dilation passed through with the galaxies (§21.7).

Open problems. The absolute value of N; the covariant completion; the CMB spectrum from the defect sector with transfer functions; the dark-sector length L_* from the theory; and the external-field test of μ_2.

28 Conclusions

1. DERIVED The C_3 eigenvalue and the isotropic slope $\frac{3}{4}$ give $a_0 = (\sqrt{3}/\pi^2)cH_0$, independent of the exposure law.
2. DERIVED A Poisson threshold response gives the family μ_n in closed form; the deep slope $\frac{3}{4}$ holds for every exposure law with finite mean, and the linear hazard is required by the Tully–Fisher slope.
3. MEASURED The full likelihood of 153 rotation curves with 490 nuisance parameters gives $a_0 = 1.04 \pm 0.04 \pm 0.10$ across the allowed shapes; the prediction 1.148 is 1.0σ high. On the RAR points, with nothing fitted, every member meets the relation to 0.025 dex in the mean.
4. DATA-CONSTRAINED Galaxies fix the transition and planets the tail. The one-exponent Torridity form cannot meet both for any exponent, and draft 1's disk-galaxy exponent $2\pi/3$ is excluded; the threshold family meets both for $1.62 \le n \le 1.91$, and the draft-1 screening is retired.
5. DERIVED Canonical normalization gives $\varepsilon_V = f_\delta^2/L_*^2$, $m_\delta = \sqrt{8}\hbar/cL_*$, a vanishing vacuum constant and a gravitational-strength self-coupling $3\hbar^3/2m_{\rm P}^2c$.
6. DERIVED The horizon mass is the closure mass of four tetrahedral deficits of π, and its dual is a field of mass $(\sqrt{2}/2\pi)\hbar H_0/c^2$.
7. PREDICTION The resonance budget gives $\Omega_m = 0.316$ and $H_0 = 67.35 \pm 0.06$, within 1σ of Planck's densities and of Pantheon+, 2.2σ from DESI DR2 BAO and 2.1σ from DES-Y5 (where ΛCDM's own CMB fit lies), and predicts that distance ladders calibrated alike converge on 67.3.
8. MEASURED The CMB point-source imprint is not detected at $\ell = 4$–10; the smoothed pattern is next.
9. DERIVED The spinning arcs are the thermal geometry of the final horizon: the rate H_∞ is fixed by smoothness at the horizon, the rim moves at c, the information is 2π times the

spin's charge over $\hbar$, and the quantum of the spin is the area step $8\pi\ell_{\mathrm{P}}^2$; PREDICTION with the present at the resonance, $\beta = (2\alpha/\pi)\,\mathrm{artanh}(3\sqrt{3}/2\pi) = 0.3137°$, $+0.6\sigma$.

10. DERIVED The rotation's quanta are the spin's: $\Omega_0 = jH_\infty$ with $j = 1$ measured. PREDICTION With the present at the resonance the dark ratio is 2.580 and the budget lies within 1σ of Planck's densities; the identification 5/2 is withdrawn. The acceleration scale tracks the horizon, $a_0(z) = a_0 E(z)$.

11. PREDICTION Counted in cycles, cosmic dawn is fixed by the CMB's twist, $H_\infty t_\star = (\pi/3)\beta$, with no clock and no α: the measured angle places the 21-cm onset at $42.7^{+5.6}_{-6.0}$ MHz.

12. DERIVED The dark sector is one geometry: spacetime unexcited, the vacuum, which is the generator-defect tension over the square of the final radius, and spacetime excited, the cosmic share. Its equation of state is $w = -1 + \Theta$ with Θ the excited fraction, equality at $z = 0.372$. What lensing infers as dark matter is the dilation, $\mathcal{S} = 1/\cos_n\theta$ with $\tan\theta = a_0/g$: 1 at the centre, r/ℓ toward the edge, varying with capture and relaxation; in the Bullet Cluster it passed through with the galaxies. OPEN How far it grows is open. The background expansion is unchanged. The dilation keeps a clock's causes: it reaches the unbound web once the horizon sets its clocks, at the onset of acceleration, holds above 68 Mpc, and acts on clocks, not light; the web then carries the measured bulk flows with S_8, cosmic shear and CMB lensing unchanged, while the Hubble tension stays the background's.

29 Cosmic Reset and the Complete Ending

This section is HYPOTHESIS throughout: a proposed end state and transition, not yet derived from the field equations.

Terminal state. With $V_0 = 0$ (Theorem 11) the potential is $\frac{1}{2}(\Xi - 2)^2$ with zero vacuum energy; the vacuum energy is carried by the horizon defect. If the dark-sector length tracks the horizon in the terminal phase, $L_*(t) = R_H(t) = c/H(t)$, then $m_\delta = \sqrt{8}\,\hbar/cR_H \to 0$ as $H \to 0$, the condensate sits at $\Xi = 2$, and every remaining degree of freedom is massless.

Conformal boundary. STANDARD A massless scalar with the coupling $-\frac{1}{12}R\chi^2$ and a quartic term is classically conformally invariant in four dimensions. Near a boundary where $g_{\mu\nu} = \Omega^2\eta_{\mu\nu}$ with $\Omega \to 0$, the Einstein–Hilbert term scales as Ω^2. The trace anomaly, $\langle T^\mu{}_\mu\rangle = (16\pi^2)^{-1}(c_W C^2 - a_E E_4)$ with $c_W = 1/120$ and $a_E = 1/360$ for one conformal scalar, breaks the symmetry at the quantum level; through $R = -8\pi G\langle T^\mu{}_\mu\rangle/c^4$ it sources curvature, the scalar acquires $m_{\mathrm{eff}}^2 = R/6$, and time dependence produces particles.

Seed of matter. Photon collisions produce pairs: nonlinear Breit–Wheeler production was observed at SLAC E-144 [29], and the linear process in ultraperipheral heavy-ion collisions by STAR at RHIC [30].

Defects and closure. In 3+1 dimensions a point mass does not produce a conical deficit; strings and the reduced 2+1 geometry do (§16). A closed spatial section requires the deficits to sum to 4π (Theorem 9), realized minimally by four C_3 apexes of π. The condensate of the defect network is described by the action (4).

Gluing. Following Penrose's conformal cyclic cosmology [31], the future boundary of one aeon is identified with the past boundary of the next, $\tilde{g}_{\mu\nu}^{(\mathrm{next})} = \Omega^2 g_{\mu\nu}^{(\mathrm{old})}$.

Finiteness. The horizon entropy is $S = N/2$ nats, so the number of distinguishable boundary configurations is $e^{N/2} = 2^{N/2\ln 2}$ (draft 1 wrote 2^N), finite and astronomically large.

30 Completed Thermodynamic and Geometric Reset Closure

The reset is a closed thermodynamic–geometric transition when the following distinction is kept exact: the old aeon does not undergo a global contraction of its physical scale factor. Before the junction, $a = b^{-1}$ and $b \to 0^+$, so $a \to \infty$; after the junction the new physical scale factor is $\tilde{a} = |b| \to 0^+$. Thus the reset is not a global compressive shock of the old aeon. It is a conformal identification at which the old physical scale loses its operational meaning and the new aeon begins with a vanishing physical scale. This is the geometry already fixed by (159) and $\tilde{H} = 1/(2\tilde{\tau})$.

R1. Matter-sector conservation and thermodynamic closure

On every regular matter branch let N^μ denote a conserved matter-number current and $T^{\mu\nu}_{\rm m}$ the total matter stress tensor. In the absence of a localized junction source, the finite-flux conditions are

$$\nabla_\mu N^\mu = 0, \qquad \nabla_\mu T^{\mu\nu}_{\rm m} = 0, \qquad \boxed{[N^\mu n_\mu]^+_- = 0}, \qquad \boxed{[T^{\mu\nu}_{\rm m} n_\nu]^+_- = 0}. \tag{87}$$

These equations express material and energy–momentum continuity without requiring that every old composite object survive as the same bound state. Stars, galaxies and black holes are not conserved objects; their conserved charges and stress–energy are redistributed among the degrees of freedom of the thermal state.

Thermalization is represented by the entropy current s^μ. The second law gives

$$\nabla_\mu s^\mu = \sigma_s \geq 0. \tag{88}$$

Consequently the reset may destroy macroscopic structure while preserving conserved currents and increasing coarse-grained entropy. The radiation-dominated opening state is therefore a high-entropy thermal state, but it is not identified with exact global maximum entropy or exact equilibrium: GRUT's reset is explicitly out of equilibrium and its subsequent cooling and baryon asymmetry calculation rely on that fact.

For the relativistic plasma stage,

$$p = \frac{1}{3}\rho c^2, \qquad \rho = a_R T^4, \qquad s = \frac{4\rho}{3T}, \qquad \tilde{H} = \frac{1}{2\tilde{\tau}}, \tag{89}$$

so the new aeon has the required radiation-like equation of state and cools continuously as it expands. The thermodynamic closure is therefore thermalization plus conservation plus the entropy inequality, not a postulated instantaneous equilibrium.

R2. The collisionless phase survives the reset

The phase sector is governed by the already closed junction theorem,

$$\boxed{\mathcal{T}_\chi = 1}, \qquad \boxed{\mathcal{M}_{\mathcal{R}}(k) = 1 \text{ for the adiabatic superhorizon mode}}, \qquad \boxed{\mathcal{M}_{\mathcal{I}}(k) = 1}. \tag{90}$$

Thus the phase current crosses the conformal junction without a junction-localized source, and the specified superhorizon curvature correlations and conformal information modes are transmitted unchanged. The Bullet Cluster provides the intended local dynamical analogy: collisional plasma is shock-heated while the gravitating component is much less affected by drag. It is an analogy for the separation of collisional and collisionless sectors, not a separate derivation of the cosmological junction.

R3. No additional Stückelberg transition is required

The reset does not invoke a Stückelberg mass cancellation or restoration of a gauge $U(1)$. In the existing framework, the dark-sector phase θ is a global phase/clock variable and carries no gauge field. The transition is instead the conformal continuation

$$g_{\mu\nu} = b^{-2}\hat{g}_{\mu\nu} \quad \longrightarrow \quad \tilde{g}_{\mu\nu} = b^2\hat{g}_{\mu\nu}, \qquad b(\eta_c) = 0, \tag{91}$$

with the same regular conformal geometry and $b'(\eta_c) = -\omega$. No new vector mass or gauge-symmetry-breaking sector is introduced by the reset.

R4. Recombination and the next structure hierarchy

After the reset, $\tilde{a}$ grows, the radiation density falls as $\tilde{a}^{-4}$, and the baryonic component cools. The transferred adiabatic phase correlations provide the initial gravitational seed through $\mathcal{M}_{\mathcal{R}} = 1$. Baryons subsequently respond to the resulting gravitational potentials and, after neutral recombination, decouple from the radiation and participate in structure formation.

The statement that the *same phase correlations* seed the next aeon is therefore a closed consequence of the junction transport equations. The stronger statement that every individual baryon or bound object in the old aeon is literally the same particle in the new aeon is not part of the conserved-field description and is not required by the framework. In particular, GRUT's baryogenesis sector permits baryon number violation through electroweak anomaly processes while conserving the appropriate anomaly-free combination; material continuity must therefore be understood as continuity of the conserved field degrees of freedom and fluxes, not identity of every composite particle.

The CMB then has its standard role within the framework's sequence: the radiation field evolves through the hot plasma era and the observed microwave background is the relic radiation released as the baryonic plasma recombines. Structure formation occurs later from the inherited primordial perturbations and the subsequent matter dynamics. No independent creation of an abstract information carrier is required: the conformal information channel and phase correlations are transported by the already derived junction map, while the horizon-cell count itself resets according to $N_\infty \to 0 \to N_\infty$.

R5. Completed reset statement

The physical reset is therefore a single closed sequence:

$$\boxed{\text{terminal dilution} \;\to\; \text{conformal junction} \;\to\; \text{radiation-dominated new aeon} \;\to\; \text{cooling and recombination} \;-} \tag{92}$$

with phase-current transmission, adiabatic curvature transmission and conformal information transmission fixed by their stated junction assumptions. Separately, the finite conformal-volume remainder of Auxoviscausal strain matches by (138) when the common conformal spatial metric and identified flow tube are regular; the raw logarithmic strain does not. The reset is thermodynamically irreversible at the coarse-grained level through $\nabla_\mu s^\mu \geq 0$, while the specified conformal transport channels are lossless. This is the scope of the thermodynamic and geometric reset closure used by TRSD/GRUT.

31 The Four Interactions from the Coil

DERIVED The coil of §2 has four features, and each carries one interaction (GRUT §54 [32]): the cone of the tangent bicone, with its deficit (§3.3, §16), carries gravity, of strength $G = c^4/F$; the

phase of the three crossings, $\mathbb{Z}_3 \subset U(1)$ made continuous and acting on the fermions, carries electromagnetism, of strength α; the two conjugate lobes, glued into $S^3 \cong SU(2)$ and of opposite handedness, carry the weak interaction and its chirality; the three crossings of the beat (§2.4), the triangle C_3, carry colour, $SU(3)$ with centre $\mathbb{Z}_3$. The phase θ of $\phi_\pm = \phi e^{\pm i\theta}$ is a different, global phase: held at its rate it is the rotation that carries the dark sector (§21.1), and it carries no gauge field (GRUT §31.1, §41.7). The lobes' $\mathbb{Z}_2$ and the beat's $\mathbb{Z}_3$ generate the quotient

$$G_{\rm SM} = \frac{SU(3) \times SU(2) \times U(1)}{\mathbb{Z}_6}, \qquad \frac{t}{3} + T + Y \in \mathbb{Z}, \tag{93}$$

for every field of triality t, isospin T and hypercharge Y [48]: every colour singlet has integer charge, and quarks carry $\frac{2}{3}$ and $-\frac{1}{3}$.

DERIVED The features write one action: the axiom (4) with the rotation held at its rate, the Yang–Mills terms of the group (93), three generations of fermions on the cone's double cover, and the pair as one Higgs doublet (GRUT §54.5). Its content fixes the running, $b_3 = -7$, $b_2 = -\frac{19}{6}$ and $b_Y = \frac{41}{6}$ at one loop, from three colours, three generations and one doublet (GRUT §54.6): the triangle's coupling grows toward low energy and closes in strong coupling at the nucleon, and the phase and the pair are one electroweak coupling, $\alpha_w = \alpha/\sin^2\theta_W$.

DERIVED The strengths lie on one chain of scales. With H4 (§24.18) the horizon is $8\pi e^{1/\alpha+3\pi\alpha/2}$ Planck lengths, the balance mass of GRUT §50.8 has $(m_*/m_{\rm P})^3 = \ell_{\rm P}/2R_H$, and the nucleon is $m_p = Km_*$, $K = 19.7566 \pm 0.0002$, the strong datum. Then

$$\frac{G\hbar H_0^2}{c^5} = \frac{e^{-2/\alpha-3\pi\alpha}}{64\pi^2}, \qquad \frac{Gm_p^2}{\hbar c} = \frac{K^2}{(16\pi)^{2/3}}\, e^{-2/3\alpha-\pi\alpha} = 5.906 \times 10^{-39}\,: \tag{94}$$

gravity's weakness is exponential in $1/\alpha$, because the present is $e^{1/\alpha}$ Planck lengths across. Both rungs are transmutations from one Planck cell, the horizon of the electric coupling (H4) and the nucleon of the strong coupling with $b_0 = 7$, $m_p \propto m_{\rm P} e^{-2\pi/7\alpha_s(m_{\rm P})}$. Two strengths are fixed by the geometry, α and G at the present; the strong and electroweak strengths are each one number at the closure of their feature, the nucleon's K and the pair's $v_{\rm EW}$ and θ_W, measured today and the next closures to compute (GRUT §54.7).

PREDICTION The group (93) is a product, not a subgroup of a simple group: there is no scale at which the couplings must meet and no gauge boson that turns quarks into leptons, so the proton is stable against gauge-mediated decay, which Super-Kamiokande bounds at $\tau/B(p \to e^+\pi^0) > 2.4 \times 10^{34}$ yr [49]. Gauge-mediated proton decay, or a free fractionally charged colour singlet, falsifies the section (register rows 39–40).

32 Collision Lag, Geometric Recoil, and CHIRP

This section derives the collision–relaxation chain from the existing TRSD action, merger map, metric equations, and horizon definitions. The key distinction is between (i) the displacement of the excited component relative to collisional baryons and (ii) the expansion of a proper spatial volume. They are connected by the phase current and the metric, but they are not the same velocity. No external streaming-velocity amplitude, new force, or additional matter component is introduced.

1. Collision separates baryonic motion from phase motion

The existing constrained phase sector is

$$S_{\rm rot} = \int d^4x\sqrt{-g}\,\lambda(X - X_0), \qquad X = -g^{\mu\nu}\nabla_\mu\theta\nabla_\nu\theta = X_0. \tag{95}$$

Variation with respect to θ gives $\nabla_\mu(\lambda\nabla^\mu\theta) = 0$; variation with respect to the metric gives

$$T^{(\chi)}_{\mu\nu} = 2\lambda\nabla_\mu\theta\nabla_\nu\theta + g_{\mu\nu}\lambda(X - X_0) \xrightarrow{X=X_0} 2\lambda\nabla_\mu\theta\nabla_\nu\theta. \tag{96}$$

With $u_\mu = \nabla_\mu\theta/\sqrt{X_0}$, the fixed-norm condition gives $u^\mu u_\mu = -1$ and $u^\nu\nabla_\nu u_\mu = 0$. The phase component therefore follows geodesics and has no gas-shock drag term. In the non-relativistic limit its mass current obeys

$$\partial_t\rho_\chi + \nabla\cdot(\rho_\chi\mathbf{v}_\chi) = 0, \qquad (\partial_t + \mathbf{v}_\chi\cdot\nabla)\mathbf{v}_\chi = -\nabla\Phi. \tag{97}$$

The collisional baryonic gas instead obeys

$$(\partial_t + \mathbf{v}_b\cdot\nabla)\mathbf{v}_b = -\nabla\Phi - \frac{\nabla p_b}{\rho_b} + \mathbf{a}_{\rm coll}. \tag{98}$$

Writing $\mathbf{w} = \mathbf{v}_\chi - \mathbf{v}_b$ and subtracting while retaining the distinct convective derivatives gives

$$\boxed{(\partial_t + \mathbf{v}_\chi\cdot\nabla)\mathbf{w} = \frac{\nabla p_b}{\rho_b} - \mathbf{a}_{\rm coll} - (\mathbf{w}\cdot\nabla)\mathbf{v}_b.} \tag{99}$$

The common gravitational acceleration cancels at the same event. Thus the gas can lose bulk momentum in a shock while the phase component continues on its own geodesic; the previous phase density and its stress-energy are not algebraically erased by a change in the baryonic trajectory. The gas terms are those already present in its fluid solution, not fitted velocity inputs.

2. The existing relaxation law derives the component drift

At core passage, the framework resets the relaxation clock. With $t = \Delta t \geq 0$,

$$f(t) = 1 - e^{-t/t_{\rm rel}}, \qquad q(t) = 1 - f = e^{-t/t_{\rm rel}}, \qquad \dot f = \frac{q}{t_{\rm rel}}, \quad \dot q = -\frac{q}{t_{\rm rel}}. \tag{100}$$

The existing merger map is

$$\rho_\chi(\mathbf{x}, t) = q\rho_{\rm inf}(\mathbf{x}, t) + f\rho_{\rm pol}(\mathbf{x}, t). \tag{101}$$

For equal conserved masses in the normalized profiles, let their centroids be $\mathbf{x}_{\rm inf}$ and $\mathbf{x}_{\rm pol}$, and define $\mathbf{D} = \mathbf{x}_{\rm pol} - \mathbf{x}_{\rm inf}$. The mixture centroid is $\mathbf{x}_\chi = q\mathbf{x}_{\rm inf} + f\mathbf{x}_{\rm pol}$, hence $\Delta\mathbf{x}_\chi \equiv \mathbf{x}_\chi - \mathbf{x}_{\rm inf} = f\mathbf{D}$. Differentiating gives

$$\boxed{\mathbf{v}_{\rm rel} = \frac{d\Delta\mathbf{x}_\chi}{dt} = \frac{q}{t_{\rm rel}}\mathbf{D} + f\dot{\mathbf{D}}.} \tag{102}$$

For approximately fixed $\mathbf{D}$,

$$\boxed{\mathbf{v}_{\rm rel} = \frac{e^{-t/t_{\rm rel}}}{t_{\rm rel}}\mathbf{D}, \qquad \Delta\mathbf{x}_\chi = (1 - e^{-t/t_{\rm rel}})\mathbf{D}.} \tag{103}$$

This velocity is derived from the existing displacement and relaxation time; no streaming-speed amplitude is inserted. The framework already fixes $t_{\rm rel} = 4\pi^2/[9\sqrt{3}H(z)]$. For the Bullet Cluster, $t_{\rm rel} \simeq 31.4$ Gyr and $f(0.15\,{\rm Gyr}) \simeq 0.0048$, so the relaxation displacement is 0.48% of D at that epoch. This is a centroid drift of the excess distribution, not yet the velocity of space itself. Substitution of the framework's own horizon-set clock gives the closed form, for fixed $\mathbf{D}$,

$$\boxed{\mathbf{v}_{\rm rel}(t) = \frac{9\sqrt{3}}{4\pi^2}H(z)\exp\left[-\frac{9\sqrt{3}}{4\pi^2}H(z)t\right]\mathbf{D}.} \tag{104}$$

The velocity scale is therefore derived from the framework's $H(z)$, its fixed relaxation law, and the existing separation $\mathbf{D}$; it is not an externally supplied streaming amplitude.

3. Mass conservation fixes the local relaxation flux

Differentiate the density mixture and use the continuity equations for each profile, $\partial_t \rho_a + \nabla \cdot (\rho_a \mathbf{v}_a) = 0$ for $a \in \{\mathrm{inf}, \mathrm{pol}\}$. Directly,

$$\partial_t \rho_\chi + \nabla \cdot (q\rho_{\mathrm{inf}} \mathbf{v}_{\mathrm{inf}} + f\rho_{\mathrm{pol}} \mathbf{v}_{\mathrm{pol}}) = \dot{q}\rho_{\mathrm{inf}} + \dot{f}\rho_{\mathrm{pol}} \tag{105}$$

$$= \dot{f}(\rho_{\mathrm{pol}} - \rho_{\mathrm{inf}}). \tag{106}$$

Therefore the transfer flux required by mass conservation obeys

$$\boxed{\partial_t \rho_\chi + \nabla \cdot \mathbf{J}_\chi = 0, \quad \mathbf{J}_\chi = q\rho_{\mathrm{inf}} \mathbf{v}_{\mathrm{inf}} + f\rho_{\mathrm{pol}} \mathbf{v}_{\mathrm{pol}} + \mathbf{J}_{\mathrm{rel}}, \quad \nabla \cdot \mathbf{J}_{\mathrm{rel}} = -\dot{f}(\rho_{\mathrm{pol}} - \rho_{\mathrm{inf}}).} \tag{107}$$

Under spherical symmetry, regularity at $r = 0$ fixes the radial component:

$$\boxed{J_{\mathrm{rel},r}(r,t) = -\frac{\dot{f}}{r^2} \int_0^r [\rho_{\mathrm{pol}}(s,t) - \rho_{\mathrm{inf}}(s,t)] s^2 \, ds.} \tag{108}$$

Define the relaxation part of the coarse-grained transport velocity by $v_{\mathrm{rel},r} = J_{\mathrm{rel},r}/\rho_\chi$. Its local divergence is then not guessed but follows from the product rule:

$$\boxed{\vartheta_{\mathrm{rel}} \equiv \nabla \cdot \mathbf{v}_{\mathrm{rel}} = -\frac{\dot{f}(\rho_{\mathrm{pol}} - \rho_{\mathrm{inf}})}{\rho_\chi} - \mathbf{v}_{\mathrm{rel}} \cdot \nabla \ln \rho_\chi.} \tag{109}$$

The sign is determined by the existing profiles and their enclosed-mass difference. Mass conservation allows outward transport in some regions and inward transport in others; it does not imply a positive uniform expansion by itself. Compatibility with the phase action requires the total current $\mathbf{J}_\chi = \rho_\chi \mathbf{v}_\chi$ to satisfy the geodesic Euler equation (97). That equation is the consistency check, not an extra adjustable velocity law.

4. Geometric recoil is the trace of spatial deformation

Let h_{ij} be the proper spatial metric on slices orthogonal to the phase flow. The expansion tensor is $B_{ij} = \frac{1}{2}\mathcal{L}_u h_{ij}$; its symmetric decomposition is $B_{ij} = \sigma_{ij} + \frac{1}{3}\vartheta h_{ij}$, where $\vartheta = \nabla_\mu u^\mu$. A connecting separation ξ^i between neighboring flow lines obeys

$$\frac{D\xi^i}{D\tau} = B^i{}_j \xi^j, \qquad \frac{d \ln V_{\mathrm{proper}}}{d\tau} = \mathrm{tr}\, B = \vartheta. \tag{110}$$

The volume identity follows from differentiating the determinant of three independent connecting vectors: $d \ln \det(\xi^i{}_a)/d\tau = \mathrm{tr}\, B$. Therefore

$$\boxed{V_{\mathrm{proper}}(\tau) = V_{\mathrm{proper}}(\tau_0) \exp\left[\int_{\tau_0}^{\tau} \vartheta(\tau')\, d\tau'\right].} \tag{111}$$

This is the exact geometrical definition of stretching or recoil of a proper spatial volume. It is not the centroid drift in (102). In a locally isotropic patch, $h_{ij} = a^2 \delta_{ij}$, $\vartheta = 3H$ and the corresponding recession/recoil speed at proper radius R is

$$\boxed{v_{\mathrm{space}}(R) = HR = \frac{R}{3}\vartheta.} \tag{112}$$

For the relaxation contribution, replace ϑ by the component ϑ_{rel} only when the transfer current is the corresponding part of the phase congruence. The derived v_{space} is therefore a metric-deformation velocity, not the incorrect force impulse $-t_{\mathrm{rel}}\mathbf{g}$.

5. The existing action supplies the metric response

Varying the Einstein–Hilbert plus existing field action gives

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T^{\text{eff}}_{\mu\nu}, \tag{113}$$

where $T^{\text{eff}}_{\mu\nu}$ is the stress-energy of the baryons and the already-present scalar/phase sector. Thus the post-collision metric is sourced by the surviving excited geometry as well as the baryonic gas. In the weak-field, linear regime, harmonic gauge gives

$$\Box \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4}\,\delta T_{\mu\nu}, \qquad \bar{h}_{\mu\nu}(t,\mathbf{x}) = \frac{4G}{c^4}\int \frac{\delta T_{\mu\nu}(t-|\mathbf{x}-\mathbf{x}'|/c,\mathbf{x}')}{|\mathbf{x}-\mathbf{x}'|}\,d^3x'. \tag{114}$$

The retarded solution depends on the source's past light cone, so the metric response is not an algebraic instruction to erase the pre-collision geometry when the gas shocks. The continuing phase stress-energy and the retarded metric response are the two existing carriers of the lag. The proper-volume recoil is computed from that metric through (110); it is not set equal to a guessed multiple of g.

The same weak-field metric also gives the recoil velocity directly from the changing spatial dilation. With $S \simeq 1 - 2\Phi/c^2$ and $\Phi = \Psi$, the spatial metric coefficient is $h_{ij}/a^2 \simeq S\delta_{ij}$ to first order, while the physical length factor is $a_s = \sqrt{S} = e^{-\psi/2}$. This uses S as the weak-field spatial metric coefficient, not as a conformal rescaling of all spacetime. At a proper radius R corresponding to a fixed coordinate separation, to the weak-field order retained here,

$$\frac{\dot{a}_s}{a_s} = \frac{1}{2}\partial_t \ln S = -\frac{1}{2}\dot{\psi} \simeq -\frac{\dot{\Phi}}{c^2}, \qquad \boxed{v_{\text{rec}}(R,t) = R\frac{\dot{a}_s}{a_s} = \frac{R}{2}\partial_t \ln S = -\frac{R}{2}\dot{\psi} \simeq -\frac{R}{c^2}\dot{\Phi}.} \tag{115}$$

This is the spatial-recoil velocity: the time derivative of proper spatial length as the sourced metric changes. It is different from the centroid drift v_{rel} and does not assume the incorrect impulse $-t_{\text{rel}}\mathbf{g}$.

For the leading nonrelativistic phase-density contribution to the retarded potential, write $r = |\mathbf{x}-\mathbf{x}'|$ and

$$\Phi_\chi(t,\mathbf{x}) = -G\int \frac{\rho_\chi(t-r/c,\mathbf{x}')}{r}\,d^3x'. \tag{116}$$

Differentiating at fixed field point, then applying the already-derived continuity equation $\partial_t\rho_\chi + \nabla'\cdot\mathbf{J}_\chi = 0$ and integrating by parts (with the existing localized-source boundary condition), gives

$$v_{\text{rec},\chi}(R,t) = -\frac{R}{c^2}\partial_t\Phi_\chi = \frac{GR}{c^2}\int \frac{\partial_t\rho_\chi(t-r/c,\mathbf{x}')}{r}\,d^3x' \tag{117}$$

$$= \boxed{\frac{GR}{c^2}\int \mathbf{J}_\chi(t-r/c,\mathbf{x}')\cdot\frac{\mathbf{x}-\mathbf{x}'}{|\mathbf{x}-\mathbf{x}'|^3}\,d^3x'.} \tag{118}$$

This expression derives the leading weak-field recoil from the actual phase current generated by the framework's geodesic motion and relaxation map. The complete relativistic answer retains the full retarded $\delta T_{\mu\nu}$ in equation (114); the displayed scalar-source integral is its nonrelativistic phase-density contribution. No streaming-speed amplitude is added. The sign and magnitude follow from the source current and the point at which the metric is evaluated.

For geodesic flow, the Raychaudhuri equation gives the evolution of the same expansion scalar:

$$\boxed{\frac{d\vartheta}{d\tau} = -\frac{\vartheta^2}{3} - \sigma_{\mu\nu}\sigma^{\mu\nu} + \omega_{\mu\nu}\omega^{\mu\nu} - R_{\mu\nu}u^\mu u^\nu.} \tag{119}$$

Tracing (113) in the stated stress-energy units gives $R_{\mu\nu}u^\mu u^\nu = \frac{8\pi G}{c^4}(T_{\mu\nu}u^\mu u^\nu + T/2) - \Lambda$. For the existing pressureless phase sector, this is fixed by its density; it is not an independent recoil force. The sign of the volume response is determined by the metric solution, its initial congruence, and the stress-energy. A positive integrated ϑ leaves a larger proper volume; a negative one contracts it. This is the mathematical distinction between a residual spatial stretch and a label for a moving mass peak.

6. Dilation recoil as a domain expansion rate

In the same weak-field slicing, the isotropic spatial metric including the background scale factor has $dV_{\rm proper} = a^3 S^{3/2} d^3x$. Differentiating its determinant gives the expansion rate of a domain D:

$$\boxed{H_D \equiv \frac{1}{3}\frac{d\ln V_{\rm proper}}{dt} = H + \frac{1}{2}\langle \partial_t \ln S\rangle_D = H - \frac{1}{2}\langle \dot{\psi}\rangle_D \simeq H - \frac{\langle \dot{\Phi}\rangle_D}{c^2}.} \tag{120}$$

The brackets denote the average with the domain proper-volume measure $S^{3/2}d^3x$ in this weak-field slicing (equivalent to coordinate-volume averaging at first order in the weak-field perturbation). This is a kinematic identity linking the existing dilation field to proper-volume growth; the field equation and the retarded source-current expression above determine its value. The relaxation centroid alone is not inserted into this equation as a space velocity.

Using the two expansion rates already tabulated in GRUT, $H_{\rm local} = 73.04$ and $H_{\rm bg} = 67.35$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, their difference is $\Delta H = 5.69$ $\mathrm{km\,s^{-1}\,Mpc^{-1}} = 1.84 \times 10^{-19}$ $\mathrm{s^{-1}}$. A homogeneous account of this difference would require

$$\boxed{\delta\langle\vartheta\rangle_D = 3\Delta H = 17.07\ \mathrm{km\,s^{-1}\,Mpc^{-1}} \simeq 5.53\times10^{-19}\ \mathrm{s^{-1}}, \qquad \langle\partial_t \ln S\rangle_D = 2\Delta H \simeq 3.69\times10^{-19}\ \mathrm{s^{-1}}.} \tag{121}$$

These are derived target values from the framework's stated rates, not fitted recoil inputs. The theory's prediction is the source-current integral's value; it explains the difference only if that calculation returns the required domain average.

7. Horizon information density and the expansion law

The existing Hubble-horizon definitions give

$$I_H = \frac{\pi c^5}{\hbar G H^2}, \qquad V_H = \frac{4\pi}{3}\left(\frac{c}{H}\right)^3, \qquad k_B T_H = \frac{\hbar H}{2\pi}. \tag{122}$$

Dividing information by volume and eliminating H yields

$$\boxed{\iota_H \equiv \frac{I_H}{V_H} = \frac{3c^2 H}{4\hbar G}, \qquad H = \frac{4\hbar G}{3c^2}\iota_H, \qquad V_H = \frac{9\pi c^9}{16\hbar^3 G^3}\,\iota_H^{-3}.} \tag{123}$$

Consequently, the horizon's logarithmic rates obey

$$\boxed{\frac{\dot{\iota}_H}{\iota_H} = \frac{\dot{H}}{H}, \qquad \frac{\dot{V}_H}{V_H} = -3\frac{\dot{\iota}_H}{\iota_H}, \qquad \frac{\dot{T}_H}{T_H} = \frac{\dot{\iota}_H}{\iota_H}.} \tag{124}$$

This is the exact horizon form of the proposed cooling/recoil relation: as H and T_H fall, the horizon information density falls and its Hubble volume grows. It is not an independently fitted streaming amplitude. For the flat-FLRW background already used in TRSD,

$$\dot{H} = -4\pi G\left(\rho + \frac{p}{c^2}\right), \tag{125}$$

so

$$\boxed{\frac{\dot{\iota}_H}{\iota_H} = -\frac{4\pi G}{H}\left(\rho + \frac{p}{c^2}\right), \qquad \frac{\dot{V}_H}{V_H} = \frac{12\pi G}{H}\left(\rho + \frac{p}{c^2}\right).} \tag{126}$$

For $H > 0$ and $\rho + p/c^2 \geq 0$, the Hubble-horizon volume grows while its information density and Gibbons–Hawking temperature fall. Separately, the proper volume of a comoving FLRW region obeys $\dot{V}_{\rm com}/V_{\rm com} = 3H$, and the same horizon identity gives

$$\boxed{\frac{\dot{V}_{\rm com}}{V_{\rm com}} = 3H = \frac{4\hbar G}{c^2}\,\iota_H.} \tag{127}$$

Thus the framework derives the exact horizon identities: H is proportional to the Hubble-horizon information density and the Hubble-horizon volume is proportional to its inverse cube. When the solved background has decreasing H, the horizon volume grows while its information density and temperature fall. These identities are not a standalone field equation for H. Auxoviscausality, derived below, identifies the phase-flow integral of the proper-volume expansion as retained logarithmic strain; Raychaudhuri and Einstein determine its local load-dependent rate. The collision/relaxation sector supplies the local metric response through (113)–(119); its contribution to volume is the volume average of the resulting ϑ, not the centroid speed substituted for it.

8. The derived recoil chain

The complete equation chain is

$$\begin{gathered}
\text{gas collision} \;\Rightarrow\; \text{baryonic momentum loss but geodesic phase flow,} \\
f = 1 - e^{-t/t_{\rm rel}} \;\Rightarrow\; \mathbf{v}_{\rm rel} = \dot{f}\mathbf{D} + f\dot{\mathbf{D}}, \\
\rho_\chi = q\rho_{\rm inf} + f\rho_{\rm pol} \;\Rightarrow\; \nabla\cdot\mathbf{J}_{\rm rel} = -\dot{f}(\rho_{\rm pol} - \rho_{\rm inf}), \\
T^{\rm eff}_{\mu\nu} \;\Rightarrow\; g_{\mu\nu} \;\Rightarrow\; \vartheta = \frac{d\ln V_{\rm proper}}{d\tau}, \\
I_H, V_H, T_H \;\Rightarrow\; \iota_H \propto H, \quad V_H \propto \iota_H^{-3}.
\end{gathered}$$

No step adds a recoil force or external velocity. The sign and magnitude of a particular system's net proper-volume change are obtained by solving the already-defined profile/current and metric equations with that system's initial and boundary data; the framework does not permit assigning a universal positive sign to every local transfer region independently of those equations.

9. Auxoviscausality: retained geometric memory

Definition and scope. Auxoviscausality is the phase-flow property by which a local gravitationally sourced change in proper spatial volume is recorded as accumulated logarithmic deformation rather than being returned to a reference value by a restoring term. The operational variable is the logarithmic volume strain of a small comoving flow tube, not a new matter density and not a count of newly created particles or topological cells. This definition uses the existing fixed-norm phase field and adds no field, coupling, force, or independent velocity.

On a regular branch, let u^μ be the unit, geodesic, hypersurface-orthogonal phase-flow congruence already defined in equation (95), and write $D \equiv u^\mu\nabla_\mu$ for differentiation along one of its integral curves. Let $V_{\rm prop}$ be the infinitesimal proper three-volume spanned by three neighboring flow lines, with reference value V_*. The volume-determinant identity already derived in equation (110) is

$$D\ln V_{\rm prop} = \vartheta, \qquad \vartheta \equiv \nabla_\mu u^\mu. \tag{128}$$

We therefore define the dimensionless auxoviscausal strain by

$$\boxed{\mathcal{A}_{\rm aux} \equiv \frac{1}{3}\ln\!\left(\frac{V_{\rm prop}}{V_*}\right), \qquad D\mathcal{A}_{\rm aux} = \frac{1}{3}\vartheta.} \tag{129}$$

The factor $1/3$ makes $\mathcal{A}_{\rm aux}$ the logarithmic linear scale associated with a three-volume. Integrating along the same phase-flow worldline gives the causal-memory form

$$\boxed{\mathcal{A}_{\rm aux}(\tau) = \mathcal{A}_{\rm aux}(\tau_0) + \frac{1}{3}\int_{\tau_0}^{\tau} \vartheta(\tau')\, d\tau', \qquad V_{\rm prop}(\tau) = V_* e^{3\mathcal{A}_{\rm aux}(\tau)}.} \tag{130}$$

Only the past segment $\tau' \le \tau$ on that phase-flow curve contributes: causal order is carried by the existing phase flow, not by an added nonlocal force or an acausal boundary condition.

Derivation of the load equation. The phase constraint already makes the congruence geodesic and hypersurface-orthogonal, so its acceleration and vorticity vanish. The existing Raychaudhuri equation, (119), then becomes

$$D\vartheta = -\frac{1}{3}\vartheta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} - R_{\mu\nu}u^\mu u^\nu, \tag{131}$$

where $\sigma_{\mu\nu}$ is the shear of the phase congruence. Define the signed Ricci-focusing load by $\mathcal{L}_g \equiv R_{\mu\nu}u^\mu u^\nu$, so positive $\mathcal{L}_g$ is a focusing contribution in Raychaudhuri. Substituting $\vartheta = 3D\mathcal{A}_{\rm aux}$ into Raychaudhuri yields the covariant second-order memory equation

$$\boxed{D^2\mathcal{A}_{\rm aux} + (D\mathcal{A}_{\rm aux})^2 + \frac{1}{3}\sigma_{\mu\nu}\sigma^{\mu\nu} = -\frac{1}{3}\mathcal{L}_g.} \tag{132}$$

This is the rigorous local link to gravitational load supplied by the existing geometry: the Ricci projection enters through Einstein's equation and anisotropic deformation enters through the already-defined shear. The Ricci projection alone does not capture arbitrary anisotropic or tidal deformation; the phase congruence records that part through its shear. Positive Ricci load focuses the congruence; the cosmological constant contributes with the opposite sign. The covariant equations use the framework's normalized phase-flow convention (natural units in Raychaudhuri); explicit factors of c are restored in the FLRW source relation. No independent proportionality coefficient between load and strain is introduced.

For a homogeneous, isotropic FLRW branch, $\sigma_{\mu\nu} = 0$, $V_{\rm prop} \propto a^3$, and $\vartheta = 3H$. Hence

$$\boxed{\mathcal{A}_{\rm aux} = \ln\frac{a}{a_0}, \qquad \dot{\mathcal{A}}_{\rm aux} = H, \qquad \mathcal{L}_g = -3(\dot{H} + H^2) = -3\frac{\ddot{a}}{a}.} \tag{133}$$

With mass density ρ_m, pressure p, and cosmological constant Λ (inverse-length-squared units), the standard Einstein–Friedmann acceleration equation gives

$$\boxed{\mathcal{L}_g = 4\pi G\left(\rho_m + \frac{3p}{c^2}\right) - \Lambda c^2.} \tag{134}$$

Thus the gravitational load drives the expansion history through the existing Einstein/Raychaudhuri system, and Auxoviscausality records that solved history as $\mathcal{A}_{\rm aux}$. It does not replace the field equations with an algebraic proportionality between strain and the instantaneous Ricci projection.

The horizon identities already derived in equation (123) further imply, on the FLRW branch,

$$\boxed{\dot{\mathcal{A}}_{\rm aux} = H = \frac{4\hbar G}{3c^2}\,\iota_H, \qquad \mathcal{A}_{\rm aux}(t) - \mathcal{A}_{\rm aux}(t_0) = \frac{4\hbar G}{3c^2}\int_{t_0}^{t} \iota_H(t')\, dt'.} \tag{135}$$

This is the downstream link between the framework's horizon-information density and retained expansion strain; $\dot{V}_{\rm prop}/V_{\rm prop} = 3H$ remains exactly the previously derived volume law.

Retention condition and limits. Equation (129) contains no restoring term of the form $-\mathcal{A}_{\rm aux}/\tau_r$. If the expansion rate settles to zero after a period of positive expansion, then $D\mathcal{A}_{\rm aux} = 0$ and the previously accumulated value remains; it is not driven back to zero. On an expanding FLRW branch with $H \geq 0$, $\mathcal{A}_{\rm aux}$ is nondecreasing. In a locally contracting region with $\vartheta < 0$, however, the signed geometric strain decreases with the actual proper volume; the present framework does not artificially forbid real contraction. A strictly one-way microscopic damage/plasticity variable, a yield threshold, and positive material entropy production do not follow from this geometric identity and would require an additional irreversible constitutive sector. They are not needed for the Auxoviscausal geometric-memory closure.

9.1. Auxoviscausal memory at the conformal junction

The bulk law (129) is defined using physical proper volume on a regular aeon branch. It must not be matched by setting the two limiting values of $\mathcal{A}_{\rm aux}$ equal: those values diverge with opposite signs at the stated conformal junction. The existing metric identification (159) instead supplies a finite, geometric matching prescription for the conformal-volume remainder, provided the regular conformal spatial metric is nondegenerate on the junction and the same comoving flow tube is identifiable on its two sides.

Let $\hat{h}_{ij}$ be the induced spatial metric of the common conformal representative, and let $\hat{V}$ be the volume of that same coordinate-labelled tube computed with $\hat{h}_{ij}$. The physical induced metrics and volumes are

$$h^{(-)}_{ij} = b^{-2}\hat{h}_{ij}, \qquad h^{(+)}_{ij} = b^{2}\hat{h}_{ij}, \qquad V^{(-)}_{\rm prop} = b^{-3}\hat{V}, \qquad V^{(+)}_{\rm prop} = |b|^{3}\hat{V}. \tag{136}$$

Using the same reference volume V_* in the branchwise definition therefore gives

$$\mathcal{A}^{(-)}_{\rm aux} = -\ln b + \widehat{\mathcal{A}}_{\rm aux}, \qquad \mathcal{A}^{(+)}_{\rm aux} = \ln|b| + \widehat{\mathcal{A}}_{\rm aux}, \qquad \widehat{\mathcal{A}}_{\rm aux} \equiv \frac{1}{3}\ln\left(\frac{\hat{V}}{V_*}\right). \tag{137}$$

The finite matching variable is thus fixed by the existing conformal-volume determinant; it is not an additional field or a new transmission coefficient. Explicitly, under the regularity and tube-identification conditions stated above,

$$\boxed{\lim_{\eta\to\eta_c^-}\left(\mathcal{A}^{(-)}_{\rm aux} + \ln b\right) = \widehat{\mathcal{A}}_{{\rm aux},c} = \lim_{\eta\to\eta_c^+}\left(\mathcal{A}^{(+)}_{\rm aux} - \ln|b|\right)}, \qquad \boxed{[\widehat{\mathcal{A}}_{\rm aux}]^{+}_{-} = 0}. \tag{138}$$

This is the Auxoviscausal junction rule: the divergent isotropic scale contribution is reinitialized by the reciprocal scale-factor continuation ($-\ln b$ on the old branch and $+\ln|b|$ on the new), while the finite conformal-volume remainder is matched. It does *not* make the unregularized $\mathcal{A}_{\rm aux}$ continuous, nor does it imply that every matter degree of freedom or every microscopic record crosses the boundary.

The scope of the result matters. In exactly homogeneous FLRW for a fixed comoving tube, $\hat{V}$ is constant, so $\widehat{\mathcal{A}}_{\rm aux}$ is only the reference-normalized conformal-volume datum; the accumulated $\ln a$ of the old aeon is not preserved as a separate finite scalar. In an inhomogeneous branch, the finite remainder can encode the volume determinant of the regular conformal geometry, but it still is not an independent ledger of the entire past expansion integral. Retaining that entire integral as an extra cross-aeon state would require a new history variable and matching dynamics absent from the present action. The derived result here is deliberately narrower: the geometric volume remainder matches when the existing conformal geometry and the identified flow tube match.

Accordingly, the closed result is the phase-flow law (129), its causal history integral (130), and its load-source form (132). The word *viscoelastic* is an analogy for retained geometric response here; it is not a claim that the existing conservative phase action has derived microscopic viscosity or a material hysteresis law.

9.2. Local Auxoviscausal strain field and the Bullet Cluster

Purpose. The scalar $\mathcal{A}_{\rm aux}$ is local in space-time: it is evaluated on each worldline of the existing phase-flow congruence. It is not a spatially uniform cosmological number, and it is not an independent field with a freely chosen potential. This subsection specifies the local calculation required for a merging cluster without adding a new force or coupling.

Let $h_{ij}(x,\tau)$ be the induced proper spatial metric on hypersurfaces orthogonal to the phase flow, and let $dV_{\rm prop} = \sqrt{h}\, d^3q$ for a fixed infinitesimal coordinate cell d^3q. Relative to the same cell and a reference metric determinant h_*, the local strain is

$$\boxed{\mathcal{A}_{\rm aux}(x,\tau) = \frac{1}{6}\ln\!\left(\frac{h(x,\tau)}{h_*(x)}\right), \qquad D\mathcal{A}_{\rm aux} = \frac{1}{3}\vartheta.} \tag{139}$$

The determinant formula is the local form of the proper-volume definition, not an extra field equation. Along each phase-flow worldline the evolution is fixed by

$$\boxed{D^2\mathcal{A}_{\rm aux} = -\left(D\mathcal{A}_{\rm aux}\right)^2 - \frac{1}{3}\sigma_{\mu\nu}\sigma^{\mu\nu} - \frac{1}{3}\mathcal{L}_g, \qquad \mathcal{L}_g \equiv R_{\mu\nu}u^\mu u^\nu.} \tag{140}$$

This equation fixes the sign convention. $D\mathcal{A}_{\rm aux} > 0$ means the flow-tube volume is currently expanding and the accumulated strain is increasing; $D\mathcal{A}_{\rm aux} < 0$ means it is contracting and the signed strain is decreasing. Positive $\mathcal{L}_g$ contributes negatively to $D\vartheta$ and focuses the congruence, but it does *not* by itself imply that $\mathcal{A}_{\rm aux}$ is already negative: the sign of the strain depends on its reference value and prior expansion history. Shear contributes a non-positive term to $D\vartheta$. A cosmological-constant contribution enters through the signed Ricci projection and has the opposite sign to positive matter focusing under the framework's stated Einstein convention.

In the weak-field, flow-aligned, approximately isotropic slicing already used for the dilation, $h_{ij} \simeq a^2 S\delta_{ij}$, so the same determinant gives

$$\boxed{\mathcal{A}_{\rm aux}(x,t) - \mathcal{A}_{\rm aux,*} = \ln\!\left(\frac{a}{a_*}\right) + \frac{1}{2}\ln\!\left(\frac{S(x,t)}{S_*}\right), \qquad S \simeq 1 - \frac{2\Phi}{c^2}.} \tag{141}$$

To first order, the spatially varying part is $\delta\mathcal{A}_{\rm aux} \simeq -\delta\Phi/c^2$, and its local rate is $D\mathcal{A}_{\rm aux} \simeq H - \dot{\Phi}/c^2$ only when the chosen coordinates follow the same congruence to the retained weak-field order. This reproduces the sign in the existing recoil relation $v_{\rm rec}/R \simeq -\dot{\Phi}/c^2$; it does not justify treating a static potential offset as a continuing expansion rate. For a general anisotropic metric, equation (139), not the isotropic S approximation, is the definition.

Bullet Cluster prescription. The local strain contrast between a phase-flow tube through a lensing/galaxy peak and a comparison tube through the X-ray gas is

$$\boxed{\Delta\mathcal{A}_{\rm aux}(\tau) = \Delta\mathcal{A}_{\rm aux}(\tau_0) + \frac{1}{3}\int_{\tau_0}^{\tau}\left[\vartheta_{\rm peak}(\tau') - \vartheta_{\rm gas}(\tau')\right] d\tau'.} \tag{142}$$

The two expansion scalars must be computed from the same solved metric and the corresponding phase-flow trajectories. The metric source must include the observed baryonic gas, galaxies, and the framework's existing phase/scalar stress-energy without double-counting any component.

The observed lensing convergence and galaxy distribution constrain the gravitating map; X-ray surface brightness and temperature constrain the shocked gas; merger geometry and line-of-sight projection constrain initial and boundary data. The gas is collisional and shock-heated, while the phase congruence is treated according to its existing geodesic equation. The strain is therefore a derived local diagnostic of the solution, not an additional mass component and not a substitute for lensing.

This closes the *definition and sign-consistent evolution equation* for the local Auxoviscausal strain field. A numerical Bullet Cluster strain map is not claimed here: it requires digitized observational maps, a specified merger initial-value reconstruction, and a solved metric/phase-flow solution. In particular, the centroid relaxation fraction $f(\Delta t)$ cannot be inserted as ϑ; centroid drift and proper-volume expansion are different observables. The present equations state exactly what must be computed and what data determine it.

9.3. Cosmological matter, radiation, and thermodynamic closure

The cosmological model must evolve physical matter and radiation through changing thermodynamic regimes, not only a background expansion and a pressureless phase sector. This closure uses the standard baryonic, gauge, fermion, Higgs, electromagnetic, and radiation content already admitted by the framework's total action; it introduces no new fundamental field. Its purpose is to make the background and perturbation equations explicit and to prevent the radiation-era relation $p = \epsilon/3$ or the pressureless phase equation of state from being incorrectly applied to every component and epoch.

Write the total source schematically as $T^{\mu\nu}_{\rm tot} = T^{\mu\nu}_{b} + T^{\mu\nu}_{r} + T^{\mu\nu}_{\chi} + T^{\mu\nu}_{\rm other}$, with each existing scalar/defect contribution included exactly once in $T^{\mu\nu}_{\rm other}$. The metric field equation and the Bianchi identity require

$$\boxed{\nabla_\mu T^{\mu\nu}_{\rm tot} = 0, \qquad \nabla_\mu T^{\mu\nu}_{i} = Q^{\nu}_{i}, \qquad \sum_i Q^{\nu}_{i} = 0.} \tag{143}$$

The Q^{ν}_{i} terms describe transfer among already-present sectors and must be obtained from their interaction terms; they are not adjustable exchange functions. Where the action contains no interaction, the corresponding transfer is zero. The existing separately conserved phase current remains separately conserved unless the action is explicitly amended and re-derived.

For a homogeneous FLRW background, using energy density ϵ_i and pressure p_i, the continuity equations are

$$\boxed{\dot{\epsilon}_i + 3H(\epsilon_i + p_i) = Q_i, \qquad \sum_i Q_i = 0,} \tag{144}$$

where Q_i is the energy-transfer rate per proper volume in the comoving frame. The total equation of state is $p_i = p_i(\epsilon_i, T, n_j, \{x_j\})$ as appropriate to the regime, rather than a universal constant w_i. Radiation in the relativistic limit obeys $p_r = \epsilon_r/3$; a nonrelativistic collisionless phase component obeys its already-derived dust limit; baryons, photons, neutrinos, ionization states, and later neutral gas must use their applicable thermodynamic and kinetic closures.

Composition is tracked by number currents and reaction terms,

$$\boxed{\nabla_\mu (n_j u^\mu_j + J^\mu_j) = R_j,} \tag{145}$$

where J^μ_j is a diffusion/transport current and R_j is the net production rate from the specified reactions. The reaction network must conserve every exactly conserved charge; any allowed violation (for example, an anomaly-mediated process) must follow the existing particle-physics

sector rather than an arbitrary source. The equation of state, heat capacities, reaction rates, and transport coefficients are microphysical inputs to be sourced from established physics or derived from the framework's particle sector, not new free cosmological forces.

Radiation and relativistic species are evolved with a covariant kinetic equation,

$$p^\mu \frac{\partial f}{\partial x^\mu} - \Gamma^\alpha_{\mu\nu} p^\mu p^\nu \frac{\partial f}{\partial p^\alpha} = C[f], \tag{146}$$

where $f(x,p)$ is the distribution function and $C[f]$ is the collision operator for the relevant scattering, emission, absorption, and decay processes. In a fluid/radiation moment treatment, the same physics appears as emissivity, opacity, heat flux, and anisotropic stress; moment truncations must be stated and checked in their regime of validity. For thermalized components the entropy current obeys

$$\boxed{\nabla_\mu s^\mu = \sigma_s \geq 0,} \tag{147}$$

with entropy production attributed to actual collisions, diffusion, shocks, chemical reactions, or other specified irreversible processes—not inferred merely from the geometric strain $\mathcal{A}_{\rm aux}$.

The perturbation problem must use the same closure. In particular, pressure perturbations must separate adiabatic and non-adiabatic parts, $\delta p = c_s^2 \delta\epsilon + \delta p_{\rm nad}$, and the calculation must retain anisotropic stress and the relevant radiation/particle hierarchy. The framework's scalar/defect perturbations and the metric response must then be evolved with these matter perturbations to generate its own transfer functions. Using a precomputed ΛCDM transfer function can remain a comparison baseline, but it is not a derivation of the framework's perturbation predictions.

This is a closed statement of the conservation and kinetic structure, not a claim that every Q_i, reaction rate, opacity, or perturbation kernel has already been derived. The model becomes numerically closed only after each component's equation of state, collision/reaction data, and initial conditions are specified and the coupled background and perturbation system is integrated. Low-temperature molecular opacity and cloud-condensate physics are specialized applications of this same matter/radiation machinery, not ingredients to insert unchanged into the early Universe. No numerical cosmological parameter is changed by this patch.

Update supplement: derivations and the status of the open equations

This supplement integrates the twenty-principle update ledger while keeping a strict distinction between an algebraic consequence, a constitutive closure, an interpretation and a genuinely open matching problem. No field, coupling or boundary-transfer coefficient is added. The merger timeline below is an astrophysical analogy for local collision and relaxation; it is not a derivation of a cosmological junction.

A. The C_3 relaxation scale and the exponential response

From the framework's horizon mode and its one-third-cycle identification,

$$\omega_{C_3} = \frac{3\sqrt{3}}{2\pi} H, \qquad \omega_{C_3} t_{\rm rel} = \frac{2\pi}{3} \quad \Longrightarrow \quad t_{\rm rel} = \frac{4\pi^2}{9\sqrt{3}\, H}. \tag{148}$$

The numerical coefficient is therefore derived once the stated C_3 identification is accepted. The exponential response additionally uses a first-order relaxation closure, $\dot f = (1-f)/t_{\rm rel}$, with $f(0)=0$. Separation of variables gives

$$\int_0^{f(t)} \frac{df'}{1-f'} = \frac{t}{t_{\rm rel}} \quad \Longrightarrow \quad f(t) = 1 - e^{-t/t_{\rm rel}}, \qquad q(t) = 1 - f = e^{-t/t_{\rm rel}}. \tag{149}$$

Thus the exponential is the exact solution of the declared relaxation law, but it is not implied by geodesic flow or stress-energy conservation alone. For a bound system formed at t_f in matter domination, $t_f = 2/(3H_f)$ and the framework's formation-epoch value of $t_{\rm rel}$ give

$$\frac{t_{\rm rel}}{t_f} = \frac{2\pi^2}{3\sqrt{3}} \equiv x_H \simeq 3.80, \qquad f = 1 - \exp\left[-\frac{D_{\rm age} - 1}{x_H}\right], \quad D_{\rm age} \equiv \frac{t}{t_f} \geq 1. \tag{150}$$

For a merger, the clock is reset by definition at core passage and t in (149) means elapsed proper time after that event. The cross-scale use of the same $t_{\rm rel}H$ is a framework identification, not an independently established cluster microphysics result.

B. Early-aeon horizon equations and horizon power

For a spatially flat constant-w epoch, $a \propto \tau^{2/[3(1+w)]}$, hence $H = 2/[3(1+w)\tau]$. With $\mathsf{J} = 1/(2H^2 t_P^2)$ and $N = 4\pi\mathsf{J}$,

$$\mathsf{J} = \frac{9}{8}(1+w)^2\left(\frac{\tau}{t_P}\right)^2, \qquad N = \frac{9\pi}{2}(1+w)^2\left(\frac{\tau}{t_P}\right)^2. \tag{151}$$

Matter ($w = 0$) gives $\mathsf{J} = 9(\tau/t_P)^2/8$; radiation ($w = 1/3$) gives $\mathsf{J} = 2(\tau/t_P)^2$ and $N = 8\pi(\tau/t_P)^2$. Since $N_\infty = 2\pi/(\omega^2 t_P^2)$,

$$\frac{\widetilde{N}}{N_\infty} = (2\omega\widetilde{\tau})^2 \qquad \text{(radiation-dominated start)}. \tag{152}$$

The Hubble-horizon energy is $E_H = c^5/(2GH)$. Using $\dot{H} = -\frac{3}{2}(1+w_{\rm eff})H^2$ gives

$$\dot{E}_H = -\frac{c^5}{2G}\frac{\dot{H}}{H^2} = \frac{3}{4}(1+w_{\rm eff})\frac{c^5}{G} = 3(1+w_{\rm eff})P_{\rm max}, \qquad P_{\rm max} = \frac{c^5}{4G}. \tag{153}$$

Consequently the radiation-era value is $4P_{\rm max}$ and the matter-era value is $3P_{\rm max}$. These are consequences of the Friedmann power-law epoch and the horizon-energy definition; the interpretation as a literal cosmic collision is separate.

C. Local condensation: weak-field scope

For $\Phi = -GM/r$ and $r_s = 2GM/c^2$, the stated weak-field metric identification gives

$$S = e^{-\psi} \simeq 1 - \frac{2\Phi}{c^2} = 1 + \frac{r_s}{r} = 1 + \Omega\frac{R_H}{r}, \qquad \Omega \equiv \frac{r_s}{R_H}. \tag{154}$$

If $a_s = \sqrt{S}$, then to first order in $|\Phi|/c^2$, $\dot{a}_s/a_s \simeq -\dot{\Phi}/c^2$. For inward motion with speed $|\dot{r}|$, $\dot{\Phi} = -g|\dot{r}|$ and therefore

$$v_{\rm rec}(R) \equiv R\frac{\dot{a}_s}{a_s} \simeq \frac{g|\dot{r}|}{c^2}R. \tag{155}$$

Equations (154)–(155) are weak-field, quasi-static relations. They are not exact strong-field formulae, and $v_{\rm rec}$ is a metric-length rate in this approximation, not a new force or a signal velocity.

D. Hawking temperature and the conditional lifetime coefficient

With $R_s = 2GM/c^2$, $\mathsf{J} = R_s^2/(2\ell_P^2) = 2GM^2/(\hbar c)$ and $S = 2\pi\mathsf{J}$,

$$\frac{d(Mc^2)}{dS} = \frac{\hbar c^3}{8\pi GM} \quad \Longrightarrow \quad k_B T_H = \frac{\hbar c^3}{8\pi GM}. \tag{156}$$

For the ideal-blackbody approximation with two photon polarizations, $A = 4\pi R_s^2$ and $\sigma = \pi^2 k_B^4/(60\hbar^3 c^2)$ give

$$P_{\gamma,\,\mathrm{bb}} = A\sigma T_H^4 = \frac{\hbar c^6}{15360\pi G^2 M^2}, \qquad \dot{M} = -\frac{P}{c^2}. \tag{157}$$

Integrating from M_0 to zero yields

$$t_{\mathrm{ev}}^{\mathrm{bb}} = \frac{5120\pi G^2 M_0^3}{\hbar c^4}, \qquad \mathsf{J}(t) = \mathsf{J}_0\left(1 - \frac{t}{t_{\mathrm{ev}}^{\mathrm{bb}}}\right)^{2/3}, \qquad \dot{\mathsf{J}} = -\frac{1}{1920\pi}\frac{c}{R_s}. \tag{158}$$

The coefficient $1/(1920\pi)$ is therefore the perfect-blackbody, photon-only coefficient. It is not the exact universal Hawking coefficient: the physical flux depends on greybody transmission factors and on all particle species accessible at the black-hole temperature. The semi-classical derivation also fails when $\mathsf{J} = O(1)$, so the endpoint remains open.

E. Merger phases as a local analogue, not a cosmic time-lapse

The observations are interpreted with three physically distinct diagnostics: (i) before first core passage, subclusters approach and their gas is still associated with their own potentials; (ii) during/after core passage, the intracluster plasma shocks and is decelerated while galaxies and the lensing-inferred mass peaks can remain spatially offset; and (iii) later, the disturbed system evolves through orbital motion, shocks, and relaxation. The Bullet Cluster and MACS J0025.4−1222 are examples of lensing–gas separation; El Gordo is a high-redshift ongoing merger usually reconstructed as shortly after first core passage, with outgoing/returning solutions; Pandora's Cluster (Abell 2744) is a multi-subcluster collision, not a clean relaxed endpoint. Sources: NASA on the Bullet Cluster, NASA on MACS J0025, El Gordo dynamical reconstruction, and NASA on Abell 2744.

These objects are different systems observed at different cosmic times and viewing geometries; they are not successive frames of one merger. Projection and time-since-pericentre degeneracies complicate phase assignments (see the 2026 review, Merging Galaxy Clusters and the Search for New Physics of Dark Matter). The data support a collisionless gravitating component relative to the shocked gas under standard lensing/GR interpretations; they do not establish a conformal boundary, a sign flip of the scale factor, or information transfer between aeons.

F. The update-file open items: what is and is not derivable

Open item from the update ledger	Status after derivation	Reason / missing calculation
Whether ℓ_P is the smallest spatial unit	OPEN	The current action uses the Planck cell as a boundary datum; it contains no ultraviolet/quantum-gravity argument proving a minimum length.

Open item from the update ledger	Status after derivation	Reason / missing calculation
Independent temperature of the excited share	OPEN	The pressureless conserved phase sector has no supplied local thermal distribution or equilibrium condition from which a separate temperature can be defined.
Emergence of $[\hat{g}]$ from a more primitive relation	OPEN	The generation relation determines the conformal scale once $[\hat{g}]$ and θ are supplied; it does not derive the conformal class itself.
Auxoviscausal bulk memory	DERIVED (bulk)	The proper-volume determinant and phase-flow expansion give $D\mathcal{A}_{\mathrm{aux}} = \vartheta/3$; Raychaudhuri fixes its Ricci-load/shear evolution on each regular branch.
Cross-aeon conformal-volume matching	DERIVED (conditional)	If the common conformal spatial metric is nondegenerate and the same comoving flow tube is identified across the junction, the finite remainder $\widehat{\mathcal{A}}_{\mathrm{aux}} = \frac{1}{3}\ln(\hat{V}/V_*)$ matches. The raw strain and full FLRW expansion integral do not.
Microscopic yield-threshold plasticity	OPEN (material)	The conservative action has no internal damage variable or entropy-producing constitutive law. A one-way material yield mechanism is not implied by the geometric memory equation.
Field that causes and glues the conformal flip	PARTIAL	If $b(\eta_c) = 0$ and $H \to \omega$, then $b' = -H \to -\omega$ and the local linear slope follows. The existence of the zero, signed continuation and junction matching are not fixed by the action.
Transfer of S_∞ across the junction	OK (conformal channel)	The conformal carrier has continuous field and canonical momentum, giving $\alpha_k = 1$, $\beta_k = 0$ and $\mathcal{M}_{\mathcal{I}} = 1$; the cell count itself still resets $N_\infty \to 0 \to N_\infty$.
Adiabatic superhorizon perturbation transport	OK	Smooth conformal spatial geometry gives $[\mathcal{R}]^+_- = 0$; with $\delta p_{\mathrm{nad}} = 0$ on each regular branch, $\mathcal{M}_{\mathcal{R}} = 1$ for the specified mode.
Hawking coefficient and $\mathsf{J} = O(1)$ endpoint	PARTIAL	The photon-only blackbody limit yields (158); exact greybody/species factors and the quantum endpoint need additional QFT/quantum-gravity input.
Black holes changing ε_Λ	NOT GENERATED BY CURRENT ACTION	For fixed Λ, $\varepsilon_\Lambda = \Lambda c^4/(8\pi G)$ and $\dot{\varepsilon}_\Lambda = 0$. A dependence on black-hole history needs an extra coupling or field, which is absent by construction.
Whether na^3V_c crosses the flip	OK	The global phase action has no junction source; continuity of the canonical phase flux gives $[\sqrt{-g}J^\mu n_\mu]^+_- = 0$, hence $n_-a_-^3V_c = n_+a_+^3V_c$ and $\mathcal{T}_\chi = 1$.

The phase-current, curvature and conformal-information rows above are closed under their stated assumptions. The Auxoviscausal bulk law is separately closed in Section 32; its finite conformal-volume remainder has the conditional matching rule (138). The microscopic origin of the zero of b, non-adiabatic/subhorizon transfer, the independent thermal state of the excited share, and a genuinely dissipative plasticity sector remain separate questions.

Junction closure: phase current, curvature perturbation, and conformal information

The conformal junction is now closed at the level of the three transport channels used by the framework. The closure does not add a new field, force, or phenomenological transmission coefficient. It follows from the existing constrained phase action, the smooth conformal metric, and the absence of a junction-localized source or surface action. The physical metrics remain distinct on the two sides:

$$g_{\mu\nu} = b^{-2}\hat{g}_{\mu\nu}, \qquad \widetilde{g}_{\mu\nu} = b^{2}\hat{g}_{\mu\nu}, \qquad b(\eta_c) = 0, \qquad b'(\eta_c) = -\omega \neq 0. \tag{159}$$

All matching statements below are made in the regular conformal description and then translated back to the physical branches. This avoids incorrectly identifying the divergent pre-flip volume factor with the vanishing post-flip volume factor.

J1. Phase norm and geodesic flow

The phase constraint is

$$X \equiv -g^{\mu\nu}\partial_\mu\theta\partial_\nu\theta = X_0 = \frac{\omega^2}{c^2}, \qquad u_\mu = \frac{\partial_\mu\theta}{\sqrt{X_0}}. \tag{160}$$

Since the Hessian of a scalar is symmetric,

$$u^\nu\nabla_\nu u_\mu = \frac{1}{X_0}\partial^\nu\theta\,\nabla_\nu\partial_\mu\theta = \frac{1}{2X_0}\nabla_\mu(\partial^\nu\theta\partial_\nu\theta) \tag{161}$$

$$= -\frac{1}{2X_0}\nabla_\mu X_0 = 0. \tag{162}$$

Thus the phase congruence is exactly geodesic and hypersurface-orthogonal on every regular branch.

J2. Exact phase-current matching and $\mathcal{T}_\chi = 1$

The phase action already present in the framework is

$$S_\theta = \int d^4x\sqrt{-g}\,\lambda(X - X_0). \tag{163}$$

Variation with respect to θ gives

$$\nabla_\mu J^\mu = 0, \qquad J^\mu \equiv \lambda g^{\mu\nu}\partial_\nu\theta. \tag{164}$$

In a conformal-time foliation, the conserved canonical flux density is

$$\mathcal{F}_\theta(\eta,\mathbf{x}) \equiv \sqrt{-g}\,J^\eta, \qquad \partial_\eta\mathcal{F}_\theta + \partial_i(\sqrt{-g}\,J^i) = 0. \tag{165}$$

Integrate the field equation over an infinitesimal Gaussian pillbox around $\Sigma_c : \eta = \eta_c$. Because the action contains no junction-localized source or surface term, variation of the single global action requires continuity of the canonical momentum conjugate to θ:

$$\boxed{\left[\sqrt{-g}\,J^\mu n_\mu\right]_-^+ = 0} \qquad \Longleftrightarrow \qquad \boxed{\mathcal{F}_{\theta,+} = \mathcal{F}_{\theta,-}}. \tag{166}$$

This is the required junction condition; it is not an additional phenomenological transmission postulate.

For a homogeneous FLRW branch, $J^\mu = nu^\mu$ and $u^\eta \propto a^{-1}$. Hence

$$\mathcal{F}_\theta \propto na^3 V_c, \tag{167}$$

where V_c is a comoving volume. Therefore

$$n_- a_-^3 V_c = n_+ a_+^3 V_c \equiv N_\chi. \tag{168}$$

On the pre-flip branch $a_- = b^{-1}$ and $n_- \propto b^3$; on the post-flip branch $a_+ = b$ and $n_+ \propto b^{-3}$. The same conserved canonical flux is obtained from either side. Consequently the phase-quanta transmission coefficient is

$$\boxed{\mathcal{T}_\chi \equiv \frac{\mathcal{F}_{\theta,+}}{\mathcal{F}_{\theta,-}} = 1.} \tag{169}$$

Thus the former partial result $na^3V_c = \text{const}$ on each branch is promoted to an exact cross-junction result, provided the stated action is the complete junction action and no localized source is introduced.

J3. Curvature perturbation transport and $\mathcal{M}_\mathcal{R} = 1$

The previous Bardeen-potential argument is not used. The appropriate matching variable is the comoving curvature perturbation because it is directly the scalar perturbation of the spatial conformal geometry. Write the scalar spatial metric as

$$\gamma_{ij} = a^2 e^{2\mathcal{R}} \delta_{ij} + \cdots . \tag{170}$$

A homogeneous conformal rescaling changes a but not $\mathcal{R}$. Therefore the same regular conformal metric on the two sides has

$$\widehat{\gamma}_{ij} = \widehat{a}^{\,2} e^{2\mathcal{R}} \delta_{ij} + \cdots , \qquad [\widehat{\gamma}_{ij}]_-^+ = 0 \quad \Longrightarrow \quad \boxed{[\mathcal{R}]_-^+ = 0}. \tag{171}$$

There is no junction-localized stress tensor in the stated action, so the conformal spatial geometry is the quantity whose finite limit is matched. On each regular branch the pressureless adiabatic phase has

$$\dot{\mathcal{R}} = -\frac{H}{\rho + p}\, \delta p_{\rm nad} + O\!\left(\frac{k^2}{a^2 H^2}\right), \qquad \delta p_{\rm nad} = 0, \tag{172}$$

so the superhorizon adiabatic mode is constant away from the junction. Combining branch conservation with (171) gives the exact junction transfer law for that mode:

$$\boxed{\mathcal{R}_+(k) = \mathcal{R}_-(k)}, \qquad \boxed{\mathcal{M}_\mathcal{R}(k) = 1} \quad \text{(adiabatic, superhorizon mode).} \tag{173}$$

This does not assert that every subhorizon, entropy, anisotropic-stress, or additional-field mode has unit transfer. Those modes are outside the closed single-component matching sector.

J4. Conformal information transport and $\mathcal{M}_\mathcal{I} = 1$

The information channel is the conformal, massless sector already identified by the framework. For a conformally invariant scalar, the rescaled field $\widehat{\chi}$ has a regular action on the conformal geometry. In the locally flat conformal patch used for the junction calculation,

$$S_\chi = -\frac{1}{2} \int d^4x \sqrt{-\widehat{g}}\, \widehat{g}^{\mu\nu} \partial_\mu \widehat{\chi} \partial_\nu \widehat{\chi}, \qquad \widehat{\Box}\widehat{\chi} = 0. \tag{174}$$

For each Fourier mode, the junction has no delta-function potential because $\widehat{g}_{\mu\nu}$ is smooth and the action has no surface term. The variational junction conditions are therefore

$$[\widehat{\chi}_k]^+_- = 0, \qquad [\partial_\eta \widehat{\chi}_k]^+_- = 0. \tag{175}$$

Using the same global conformal mode basis on both sides,

$$\widehat{\chi}_k = A_k e^{-ik\eta} + B_k e^{ik\eta}, \tag{176}$$

these two equations form an invertible two-by-two system whose unique solution is

$$A_{k,+} = A_{k,-}, \qquad B_{k,+} = B_{k,-}. \tag{177}$$

Hence the local junction Bogoliubov map is

$$\alpha_k = 1, \qquad \beta_k = 0, \qquad \boxed{\mathcal{M}_{\mathcal{I}}(k) = 1}. \tag{178}$$

The map is unitary, so the von Neumann entropy of the complete conformal quantum state is unchanged by the junction. This statement concerns the conformal information channel; it does not identify the reset cell count itself with a conserved entropy, since the framework explicitly has $N_\infty \to 0 \to N_\infty$.

J5. Closed junction theorem

Under the framework's existing assumptions—(i) $b(\eta_c) = 0$ with a smooth conformal metric, (ii) the fixed-norm phase action, (iii) no junction-localized source or surface action, and (iv) a conformally invariant massless information carrier—the junction transport laws are therefore

$$\boxed{\mathcal{T}_\chi = 1, \qquad \mathcal{M}_{\mathcal{R}}(k) = 1 \text{ for the adiabatic superhorizon mode}, \qquad \mathcal{M}_{\mathcal{I}}(k) = 1.} \tag{179}$$

The conformal junction is consequently transparent to the conserved phase current, to the specified adiabatic curvature mode, and to the conformal information channel. The former OPEN/partial junction entries are closed at this level. The separate microscopic origin of the zero of b, any material dissipative plasticity law, and the transfer of non-adiabatic or sub-horizon perturbations remain outside this theorem. Auxoviscausal memory is governed by its bulk volume-flow law; its finite conformal-volume remainder has the conditional determinant-matching rule (138), not an independently postulated transmission coefficient.

J6. Gauge quotient and hierarchy: corrected scope

The global group used by GRUT/TRSD remains

$$G_{\rm SM} = \frac{SU(3)_C \times SU(2)_L \times U(1)_Y}{\mathbb{Z}_6}, \qquad \frac{t}{3} + T + Y \in \mathbb{Z}, \tag{180}$$

where T is the $SU(2)$ representation isospin, not T_3. The quotient is a global-form statement. Within the framework's geometric construction, the $\mathbb{Z}_2$ and $\mathbb{Z}_3$ features are identified with the quotient, but this identification is a model derivation rather than a general theorem of the Standard Model. The absence of a simple unified gauge group means there are no SU(5)-type X/Y gauge bosons and hence no SU(5)-type *gauge-mediated* proton-decay channel. The quotient does not, however, forbid the gauge-invariant dimension-six operator $QQQL/\Lambda^2$; such higher-dimensional baryon violation must be separately suppressed.

For the hierarchy, H4 and the footprint relation give

$$m_*^3 = \frac{\hbar^2 H_0}{2Gc}, \qquad R_H = 8\pi\ell_P \exp\left(\frac{1}{\alpha} + \frac{3\pi\alpha}{2}\right), \tag{181}$$

so

$$\boxed{\frac{Gm_p^2}{\hbar c} = \frac{K^2}{(16\pi)^{2/3}} \exp\left(-\frac{2}{3\alpha} - \pi\alpha\right)}. \tag{182}$$

Using the framework value $K = 19.7566 \pm 0.0002$ and $\alpha^{-1} = 137.036$ gives 5.9061×10^{-39}, consistent with the framework's quoted 5.906×10^{-39}. The earlier illustrative value $K = 19.759$ would instead give approximately 5.9076×10^{-39}; it is not the value used in the released framework. The hierarchy relation is therefore closed as a conditional consequence of H4 and the footprint balance, not as a consequence of cluster-merger phenomenology.

A Prediction register

#	Prediction	TRSD value	Current data	Falsified if
1	Galactic scale (§13)	$a_0 = 0.175493\, cH_0 = 1.148 \times 10^{-10}$ m s^{-2}	likelihood (window): $1.04 \pm 0.04 \pm 0.10$; RAR points 1.249 ± 0.075	a_0/cH_0 differs from 0.1755 at 3σ once the nuisance systematic is below 3%
2	Deep slope (§11)	$\frac{3}{4}$ for every exposure law	–	a deep-regime normalization inconsistent with $a_0 = \frac{4}{3}a_c$
3	Gravity-law shape (§14–15, §17)	threshold family, tail order $1.62 \le n \le 1.91$	galaxies $n = 1.38 \pm 0.20$; planets $n \ge 1.62$; Torridity: no exponent passes both	galaxies and planets disjoint in the threshold family as well
4	Planetary tail (§15)	$< 10^{-18}$ m s^{-2}	ephemeris residuals consistent with zero	a MOND-type radial anomaly above 10^{-15} m s^{-2}
5	Torridity exponent (§17)	$\pi/3$ = running exponent at galactic accelerations	running exponent 1.04–1.06 at 10^{-11} m s^{-2}; fixed-exponent fit 0.96 ± 0.04	the running exponent at galactic accelerations differs from $\pi/3$ at 3σ
6	Matter fraction (§24.16)	$\Omega_m = 0.3160$	Planck $+0.1\sigma$; DESI DR2 BAO $+2.2\sigma$; DESI Lyα -0.5σ	combined Ω_m 3σ from 0.3160
7	Hubble constant (§24.16, §21.7)	67.35 ± 0.06, one rate for every ladder calibrated alike	Planck 0.0σ; H0DN 2026 -7.6σ (-6.1σ with the local scatter); SH0ES -5.5σ; CCHP -1.6σ; DESI Lyα + BBN $+0.7\sigma$	the converged ladder H_0 more than 3σ from 67.35 beyond its 0.88% local scatter
8	Curvature (§24.1)	$\Omega_K = 0$	0.0007 ± 0.0019	$\lvert\Omega_K\rvert > 0.003$ at 3σ
9	Vacuum (§21.3)	$w = -1$ from the horizon defect, $V_0 = 0$	DESI 2026: 2.7σ (with CMB), 3.2σ (with SN) for evolution	$w \ne -1$ at 5σ

#	Prediction	TRSD value	Current data	Falsified if
10	Radial-mode mass (§21.6)	m_δ with $Mm_\delta = (\sqrt{8}/8\pi)m_{\rm P}^2$: the field's radial excitation, not the dark component (GRUT §23.4)	–	–
11	Radial-mode self-coupling (§21.5)	$g_\chi = 3\hbar^3/2m_{\rm P}^2 c$ (negligible)	–	a self-interaction above gravitational strength
12	Dwarf profiles (GRUT §23.1)	inner slope $-\frac{1}{2}$ inside the baryonic core, isothermal beyond	dwarf rotation curves	a cored or steep inner slope at 5σ
13	Transition radius (§24.7)	$r_* = \sqrt{GM/\mu(1)a_c}$; isothermal phantom beyond	–	stacked lensing without the transition at r_*
14	Selection rule (§25.2)	power in $3 \mid m$, $\ell + m$ even, about one axis	Commander $\ell = 4$–10: no excess	the smoothed pattern also absent at projected sensitivity
15	Handedness (§25.3)	in the phase of a_{33} and in EB	$\beta = 0.277^\circ \pm 0.057^\circ$ (ACT + Planck [50]), systematics open	a derived sign opposite to the measured β
16	Freeze (§24.8)	$N/N_\infty = \Omega_\Lambda(t)$; $N_\infty = 6.63 \times 10^{122}$	DESI w	evolving dark energy at 5σ
17	Finite aeon (§24.8)	73.4% of the conformal time elapsed	Ω_m rows	future deceleration
18	Cells do not stretch (§24.9)	$\dot{G}/G = 0$	LLR	$\dot{G}/G = 2(1+q)H$
19	First starlight (§24.10)	$t_\star = \alpha t_0$, $z = 29.6$, 46.4 MHz	–	21-cm coupling below 45.5 MHz
20	Capacity (§24.11)	$\rho_\Lambda = \frac{3}{4}\rho_P/N_\infty$	ρ_Λ	a vacuum above the critical density, or $w \neq -1$
21	Physical baryon density (§24.16)	$\omega_b = 0.02248$	Planck $+0.7\sigma$	a full-spectrum fit of the budget excluding 0.02248 at 5σ
22	Birefringence in closed form (§24.15)	$\beta = (2\alpha/\pi) \times \operatorname{artanh}(3\sqrt{3}/2\pi) = 0.3137^\circ$	$0.277^\circ \pm 0.057^\circ$ ($+0.6\sigma$), systematics open	β settled more than 5σ from 0.314°
23	Area quantum (§24.15)	$8\pi\ell_{\rm P}^2$, $\Delta I = 2\pi$ nats, every horizon	LVK ringdowns $15.6^{+20.5}_{-13.3}\,\ell_{\rm P}^2$ ($+0.5\sigma$)	an area quantum away from $8\pi\ell_{\rm P}^2$ at 5σ
24	Acceleration scale from Λ (§24.15)	$a_0 = (2/3\pi)\, c^2\sqrt{\Lambda/3} = 1.149 \times 10^{-10}\ \mathrm{m\,s^{-2}}$	RAR 1.20 ± 0.24; likelihood 1.04 ± 0.11	a_0 and Λ, measured independently, off the relation at 5σ
25	Quantum of the rotation (§24.15)	$\Omega_0 = jH_\infty$, $j = 1$	birefringence 0.88 ± 0.18 ($j = 2$ at 6.2σ); Ω_Λ with the resonance 0.9994 ± 0.0053	Ω_0/H_∞ settled 5σ from 1
26	Budget at the resonance (§24.16)	$\Omega_\Lambda/\Omega_c = 2.580$, $\Omega_m = 0.3160$, $H_0 = 67.35$, $\omega_b = 0.02248$	CMB ratio 2.611 ± 0.024 (-1.3σ); ω_b $+0.7\sigma$; DESI $+2.2\sigma$	the dark ratio, with the derived baryon ratio, settled 5σ from 2.580
27	Acceleration scale with redshift (§24.17)	$a_0(z) = a_0 E(z)$	MUSE-DARK [77]: rising, constant excluded	a constant a_0, or $a_0(z)/a_0$ off $E(z)$ at 5σ

#	Prediction	TRSD value	Current data	Falsified if
28	Cosmic dawn at the resonance (§24.10)	$t_\star = \alpha\, t_{\rm res} = 100.6$ Myr, $z = 29.6$, 46.4 MHz	JWST $z = 14.44$	21-cm coupling below 45.5 MHz
29	Cosmic dawn from the twist (§24.10)	$H_\infty t_\star = (\pi/3)\beta$, so the twist fixes the onset ratio, eq. (71)	$\beta = 0.277° \pm 0.057°$: $42.7^{+5.6}_{-6.0}$ MHz	onset and twist off the relation by 5σ together
30	Cluster baryon fraction (§21.7)	$f_b = 0.158$; $M_{\rm tot}/M_b = 6.35$	X-COP: cosmic value at R_{200} [85]; Bullet dark-to-gas convergence ≈ 7 [28]	f_b at R_{200} settled 5σ from 0.158 after the depletion correction
31	Dark self-interaction and offsets (§21.7)	$\sigma/m = 0$; no dark–galaxy offset in mergers	$\sigma/m < 0.63$ cm^2 g^{-1} (95%) [86]; Bullet peaks within 1σ of the galaxies	a self-interaction or a dark–galaxy offset established at 5σ
32	Splashback (§21.7)	the caustic edge of cold dark matter	detected in stacked clusters [88]	no splashback steepening where cold dark matter places it
33	The dilation beyond the acceleration (§21.7)	$\mathcal{S}$ set by capture and relaxation as well as by g	weak-lensing RAR: early and late types differ by $\geq 6\sigma$ at fixed stellar mass [89]; clusters off the galactic relation [85, 91]	one $\mathcal{S}(g)$ for every type and scale at 5σ
34	How far the dilation grows (§21.7, §15.1)	$r_t = (M_\chi/M_b)\,\ell$ if drawn from the bound store; otherwise $\mathcal{S} \simeq r/\ell$ to the field of the surroundings	weak lensing: flat with no clear decline to ≈ 1 Mpc around isolated galaxies [90]	a turn inside 1 Mpc around isolated galaxies (no store), or flat profiles beyond r_t (store)
35	No screening of the dilation (§21.7)	$\mu_{\rm Y} = 1/R_\infty$: no cut inside 5.4 Gpc	weak lensing flat to ≈ 1 Mpc [90]	an exponential cut of the lensing excess at a fixed scale inside the horizon
36	The dilation on the unbound web (§21.7)	horizon-set from $z = 0.63$, held above 68 Mpc at the fixed point $\sigma_{1D} = 494$ km s^{-1}: bulk flows rms 321 and 288 km s^{-1} within 150 and 200 h^{-1} Mpc; local-rate scatter 0.88%; no shift of the ladder's H_0	Cosmicflows-4 387 ± 28 and 419 ± 36 km s^{-1}, 78th and 90th percentiles (ΛCDM below 0.03%, 0.003%) [98]; shell dipole 98th [99]; network 6.1σ from 67.35; no local void in the supernovae [95]	bulk flows settled at the ΛCDM expectation at 5σ, or growth enhanced before $z = 0.7$ or below 68 Mpc
37	Light on the web (§21.7)	a conformal factor: light sees only the matter; late Sachs–Wolfe signal ×0.28–0.71 at $\ell = 2$–20; S_8 +0.16%; shear and CMB lensing unchanged	DES Y6 0.789 ± 0.012 [97], 2.8σ as for the background; $S_8^{\rm CMBL} = 0.825^{+0.015}_{-0.013}$ [100]; ISW detected at 4σ [101]; the light-coupled form excluded	an ISW–galaxy correlation at the background's amplitude for $z < 0.5$ tracers at $\ell \leq 20$ at 5σ, or lensing that follows the dilated pull above 68 Mpc

#	Prediction	TRSD value	Current data	Falsified if
38	The present in Planck cells (§24.18, H4)	$\ln(R_H/\ell_{\rm P}) = 1/\alpha + \ln 8\pi + \frac{3\pi}{2}\alpha$: $H_0 = 67.3765 \pm 0.0007$, $\Lambda\ell_{\rm P}^2 = 2.843 \times 10^{-122}$	CMB fit 67.35 ± 0.06 ($+0.4\sigma$); $\dot\alpha/\alpha = (1.0 \pm 1.1) \times 10^{-18}$ yr^{-1}	H_0 away from 67.38 at 5σ, or a drift of α
39	One gauge group (§30)	$(SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$ from the coil's $\mathbb{Z}_2 \times \mathbb{Z}_3$	charges of all known particles	a free fractionally charged colour singlet
40	No unification scale (§30)	no gauge-mediated proton decay	$\tau/B(p \to e^+\pi^0) > 2.4 \times 10^{34}$ yr [49]	gauge-mediated proton decay detected
41	Mergers: the excess stays on the galaxies (§15.3)	offset from the galaxies $\le f(\Delta t)D$: 0.5% of the gas–galaxy separation for the Bullet Cluster, $\le 6\%$ after 2 Gyr; each peak carries its own cluster's cosmic share; the swapped assignment is excluded ($T = 99$ on synthetic maps)	$\sigma/m < 0.63$ cm^2 g^{-1} [86]; MACS J0025 < 4 [105]	an offset above $0.1D$ in a clean merger, or a peak mass ratio off the pre-merger baryon ratio by more than the depletion scatter
42	No galaxy-free peak on the gas (§15.3)	above $f(\Delta t) \le 0.06$ of the excess; Abell 520 is the test	contested: found by [106, 107], not at that significance by [108, 109]	a $\ge 5\sigma$ galaxy-free gas-coincident peak holding $\gtrsim 10\%$ of the excess

B Variance analysis

Variance V = TRSD − observed, pull V/σ; tiers: on track $|V/\sigma| < 1$, monitor $1 \le |V/\sigma| < 2$, material $|V/\sigma| \ge 2$. Two further labels mark rows that are not independent tests: *calibration* (a calibrated value against the data it was chosen to approach) and *consistency* (a comparison with a fit sharing an input). The likelihood row combines the scaled statistical error with the nuisance-treatment systematic (§14.3).

#	Quantity	TRSD	Observed	pull	tier
1	a_0 (likelihood) [10^{-10} m s^{-2}]	1.148	1.037 ± 0.107	+1.04	monitor
2	a_0 (published RAR) [10^{-10} m s^{-2}]	1.148	1.20 ± 0.24	−0.22	on track
3	Torridity n (draft 1)	1.047	0.955 ± 0.040	+2.20	material
4	Mercury perihelion [mas/cy]	−0.004	-2.0 ± 3.0	+0.67	on track
5	Earth perihelion [mas/cy]	−0.028	0.19 ± 0.19	−1.15	monitor
6	Ω_K	0	0.0007 ± 0.0019	−0.37	on track
7	Ω_Λ/Ω_c	2.580	2.611 ± 0.024 (CMB spectra)	−1.28	monitor
8	Ω_b/Ω_c	0.1870	0.1864 ± 0.0022	+0.26	on track
9	Ω_m (Planck)	0.3160	0.3153 ± 0.0073	+0.10	on track
10	Ω_m (Pantheon+)	0.3160	0.334 ± 0.018	−1.00	monitor
11	Ω_m (DES-Y5)	0.3160	0.352 ± 0.017	−2.12	material
12	Ω_m (DESI DR2 BAO)	0.3160	0.2975 ± 0.0086	+2.15	material
13	Ω_m (DESI Lyα AP)	0.3160	0.325 ± 0.018	−0.50	on track
14	H_0 (Planck ΛCDM)	67.35	67.36 ± 0.54	−0.02	consistency
15	H_0 (SH0ES)	67.35	73.04 ± 1.04	−5.47	material
16	H_0 (DESI Lyα + BBN)	67.35	66.5 ± 1.3	+0.65	on track
17	Q_2 (§15.1) [10^{-27} s^{-2}]	0.0009	1.6 ± 1.8	−0.89	on track
18	ω_b (Planck)	0.02248	0.02237 ± 0.00015	+0.72	on track
19	β (ACT + Planck, §24.15) [deg]	0.3137	0.277 ± 0.057	+0.64	on track
20	H_0 (H0DN 2026)	67.35	73.50 ± 0.81	−7.59	material
21	H_0 (CCHP red giants)	67.35	70.39 ± 1.94	−1.57	monitor

All rows are at the resonance budget (§24.16). Ten rows are on track, five are monitored, five are material, and one is a consistency check. The Torridity row compares draft 1's $\pi/3$ with a fixed-exponent fit; $\pi/3$ is the running exponent at galactic accelerations (§17). The two material Ω_m rows are ΛCDM's own CMB–late-time split (Planck's fit lies $+2.1\sigma$ from DESI DR2 and -2.2σ from DES-Y5). The three material H_0 rows are the distance ladder: the framework has one expansion rate and predicts that ladders calibrated alike converge on 67.3 (§21.7); with the 0.88% scatter of the local rate they are -4.8σ and -6.1σ. The withdrawn 5/2 budget (I5) had two material variances of its own, ω_b at $+2.5\sigma$ and DESI's Ω_m at $+2.8\sigma$; both are gone. Data: [10, 33, 7, 21, 23, 24, 25, 38, 22].

Bridge 1, matter fraction. From Planck's ratios the three-component budget gives 0.31426; the derived baryon ratio moves it by +0.00010 (12%) and the resonance's vacuum ratio 2.5804 by +0.00071 (88%), to 0.31507; radiation and the neutrino masses give the budget's 0.3160, 0.1σ from Planck's 0.3153.

Bridge 2, Hubble constant. From SH0ES (73.04) the CMB acoustic calibration within ΛCDM moves H_0 by −5.68 (99.8%) and the resonance budget at fixed θ_* by −0.01, to 67.35: the ladder's offset is the offset between the two calibrations, and the budget adds nothing to it.

Galactic scale. The published RAR scale and the rotation-curve likelihood both place the prediction within 1σ; the nuisance treatment is the dominant uncertainty, and the result does not depend on the tail order. The dark-matter calibration of draft 1 (10^{-22} eV) is excluded by the Lyman-α forest and moved to $m_\delta > 2 \times 10^{-20}$ eV (§21.6).

C Numerical values

With $c, \hbar$ exact, $G = 6.674\,30 \times 10^{-11}$, 1 Mpc $= 3.085\,677\,58 \times 10^{22}$ m.

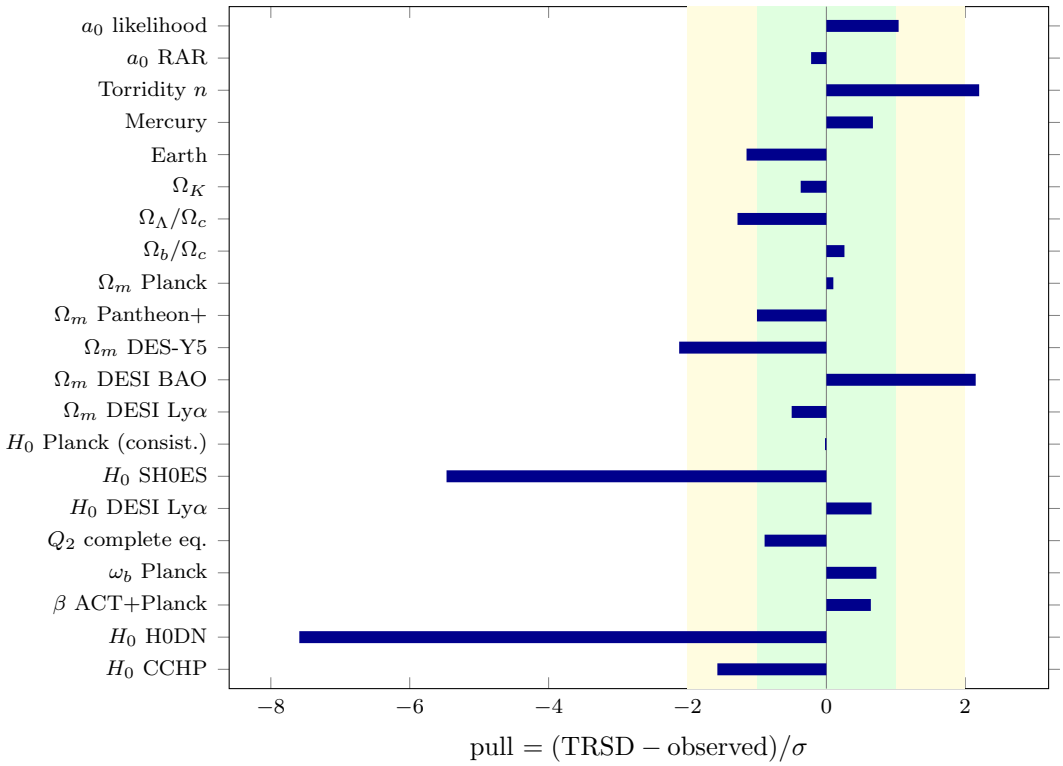


Figure 5: Pulls of the twenty-one comparisons at the resonance budget. Green: on track; yellow: monitor.

Quantity	$H_0 = 67.35$ (resonance budget)
R_H [m]	1.3735×10^{26}
a_c [m s^{-2}]	8.612×10^{-11}
a_0 [m s^{-2}]	1.1483×10^{-10}
N	4.538×10^{122}
$E_{\rm total}$ [J]	2.078×10^{69}
$M_{\rm total}$ [kg]	2.312×10^{52}
$M_{\rm Hubble}$ [kg]	9.248×10^{52}
$I_{\rm total}$ [nats]	2.269×10^{122}
E/I [J/nat]	9.159×10^{-54}
$m_U c^2$ [eV]	3.24×10^{-34}

Entries scale as $R_H, E, M \propto H_0^{-1}$, $N, I \propto H_0^{-2}$ and $a_c, a_0, E/I \propto H_0$ (at the legacy 73.04, $a_0 = 1.2454 \times 10^{-10}$). $\alpha_{\rm TRSD} = 0.175493$; $k_3 = 0.131620$; $f_\delta(\pi) = 3.8895 \times 10^{21}$ and $f_\delta(\pi/3) = 2.2456 \times 10^{21}$ $\sqrt{\rm N}$; $c^4/4G = 3.0256 \times 10^{43}$ N; $\mu_1(1) = 0.42056$, $\mu_{1.75}(1) = 0.45571$, $\mu_2(1) = 0.46234$, $\mu_3(1) = 0.47900$; withdrawn 5/2 budget $\Omega_m = 0.32194$, $\mathcal{W} = 3.68697$; $F/M_{\rm Hubble}a_0 = 2.849109$; $g_\chi = 1.24 \times 10^{-95}$ J m^3; SPARC fits as in §14. Spin (§24.15; resonance budget): $\omega = H_\infty = 1.8050 \times 10^{-18}$ s^{-1}, $R_\infty = 1.661 \times 10^{26}$ m, $\omega\ell_{\rm P}/c = 9.73 \times 10^{-62}$, $k_B T_\infty = \hbar\omega/2\pi$ (2.19×10^{-30} K), $\hbar\omega = 1.19 \times 10^{-33}$ eV, $L = \hbar N/4\pi = 3.611 \times 10^{121}\hbar$ today; area quantum $8\pi\ell_{\rm P}^2 = 6.565 \times 10^{-69}$ m^2; at the resonance $\theta_0 = 0.785699$ (0.785447 with radiation) and $\beta = 0.31370°$ (0.31359°). Resonance budget (§24.16): $\Omega_\Lambda = 0.683918$, $\Omega_m = 0.31600$, $\Omega_\Lambda/\Omega_c = 2.5804$, $H_0 = 67.35 \pm 0.06$, $\omega_b = 0.02248$, $\omega_c = 0.12022$, $S_8 = 0.833$, $a_0 = 1.148 \times 10^{-10}$ m s^{-2}, $t_0 = 13.79$ Gyr, $N_\infty = 6.635 \times 10^{122}$, $\eta = 6.16 \times 10^{-10}$; rotation $\Omega_0 = H_\infty$ ($j = 1$), quantum $\hbar H_\infty = 1.19 \times 10^{-33}$ eV. Cosmic dawn in cycles (§24.10): $\nu_\star/\nu_{21} = 1/30.81$ (46.10 MHz), 1/30.61 (46.41 MHz) with radiation; from $\beta = 0.277° \pm 0.057°$, $z_\star = 32.23$ and $42.74^{+5.64}_{-6.04}$ MHz. Dark sector as one geometry (§21.7): $n_* = 3/8\pi\ell_{\rm P}^2 R_\infty = 2.75 \times 10^{42}$ m^{-3} quanta of $\hbar H_\infty = 1.19 \times 10^{-33}$ eV; $\lambda_* = f_\delta^2(\pi)/2\delta_{\rm gen} = 7.22 \times 10^{42}$ N, $\lambda_0 = 2.80 \times 10^{42}$ N; $\Theta_0 = 0.279$, $w_{\rm dark,0} = -0.721$, equality at $z = 0.372$, $w_{\rm dark} = -\frac{1}{3}$ at $z = 0.728$; threshold $n_b = 0.65$ m^{-3} (2.58 times the mean); $M_{\rm tot}/M_b = 6.35$,

$f_b = 0.158$; the vacuum inside r_{200} holds 0.34% of the halo. The dilation: $\mathcal{S} = 3.67$, 1.58, 1.067, 1.019 at $g_{\rm bar} = 0.1$, 1, 10, $30\,a_0$; the Euclidean secant 2, 3.86, 11.47 at $\theta = 60°$, 75°, 85° ($g = 0.577$, 0.268, $0.0875\,a_0$), the dynamic law 2.67, 4.67, 12.37 there; around $10^{11}M_\odot$, $\ell = 11.0$ kpc and $\mathcal{S} = 2$, 4, 11 at 16, 39, 116 kpc, 91 at 1 Mpc; r_t with the cosmic share 59 kpc. The dilation field: $S - 1 = 2|\Phi|/c^2 \approx 10^{-6}$ in galaxies, 10^{-5}–10^{-4} in clusters; $R_{a_0} = 3.91\times10^{26}$ m; $\mu_{\rm Y} = 1/R_\infty = 6.0\times10^{-27}$ m^{-1}; $\ell = \sqrt{r_s R_{a_0}} = 7{,}190$ AU (Sun), 8.54 kpc ($6\times10^{10}M_\odot$). The web: rms field 0.0010–$0.0024\,a_0$ (12–500 Mpc), $0.0079\,a_0$ at a point, its baryons' share dilated 28.9-fold (at most 4.5 against the cosmic dilation); horizon-set from $z = 0.630$; $\sigma_{1D} = 494$ km s^{-1} (dilated flows 417, haloes 264); cut 68 Mpc; σ_8 +0.16%; late ISW $\times0.28$–0.71 ($\ell = 2$–20); bulk flows rms 321, 288 km s^{-1}; local-rate scatter 0.88%; KBC void 37%, $H_{\rm loc} = 74.2 \pm 0.6$ inside, an 8.3% step excluded; field-only $S_8 = 0.89$–1.42.

D The angular integrals

For $n = 1$, long division gives $u^3/(1+xu) = u^2/x - u/x^2 + 1/x^3 - 1/[x^3(1+xu)]$, and term by term $\int_0^1 = 1/3x - 1/2x^2 + 1/x^3 - \ln(1+x)/x^4$; then $\sigma_1 = 3x^2\int_0^1 u^3/(1+xu)\,du$ gives (23). For $n = 3$, $1 - \mu_3 = 3\int_0^1 u^2(1+xu/3)^{-3}du$; with $a = x/3$ and $w = 1 + au$, $\int_0^1 u^2(1+au)^{-3}du = a^{-3}\int_1^{1+a}(w-1)^2w^{-3}dw = a^{-3}[\ln w + 2/w - 1/2w^2]_1^{1+a}$, which gives (25).

E Verification and data

The companion script `trsd_v8_verify.py` reproduces every number (the rotation-curve likelihood with `--likelihood` and the shape profiles of §14.3 and §17 with `sparc_likelihood.py --shapes`, the running exponent with `--running`, through the module `sparc_likelihood.py` and the SPARC tables `SPARC_Lelli2016c.txt` and `MassModels_Lelli2016c.txt`): (1) the identities—the perfect-square potential, the relaxation flow and the reconciliation of Theorem 3, the couplings (8), the physical quartic and g_χ, $\mathcal{F}' = \mu$, the information–energy relation, $\rho_c c^2 = \pi^3 a_0^2/8G$, $F = c^4/4G$, m_U, $M_{\rm Hubble} = 4M_{\rm total}$; (2) the coil crossings; (3) the family μ_n with high-order series; (4) the SPARC fits; (5) the planetary tails; (6) the phantom profile and r_*; (7) the dark sector; (8) the horizon numbers; (9) the CMB pattern and cosmic variance; (10) the budget and the CAMB fit. The RAR file is the 2696-point SPARC set of [10] ($g_{\rm bar}$, $g_{\rm obs}$ and errors in 10^{-10} m s^{-2}, galaxy index), as distributed with the RARinterpret repository; the script expects it at `data/RAR.dat`. Fits minimize the sum of squared $\log_{10}$ residuals; statistical errors come from a galaxy bootstrap with 200 draws and seed 20260925.

F Archived constructions

The following draft 1 constructions are withdrawn from the framework, each for the reason given; they are recorded so that nothing is lost.

1. *The Lissajous chain for the threefold count.* Two equal-frequency perpendicular oscillators with a 90° phase offset trace a circle, not a trefoil. The count follows from the crossing window $2 \le \omega < 3$ and Postulate I (§2.3). Release 1.4.0 restored at the opening of Section 3 the sentence "The threefold count is the Lissajous figure of the two counter-rotating arcs: two equal-frequency perpendicular motions a quarter turn apart trace on the sphere a closed figure with three nodes at $2\pi/3$, where the counter-rotation leaves no net tangential motion and the curvature concentrates." Release 1.6.0 corrects it: on the sphere, two semicircles counter-turning at one rate about perpendicular axes cross on a great circle if started together and on an arc if started a quarter turn apart, with no node,

and a point carried by both traces a figure-eight with at most one node; the $2\pi/3$ spacing is the beat of the fold against the conjugate lobe (§2.4).

2. *The Torridity canonical scale* $2^{-1/4}a_0$, argued from which cascade ratio has a terminating decimal expansion. Decimal termination depends on the base, and the coincidence $2^{-1/4}a_0 \simeq (\pi/3) \times 10^{-10}\ \mathrm{m\,s^{-2}}$ depends on the SI units.
3. *The quantized sequences* $(\pi/3)(k\pm1)/k\times10^{-10}\ \mathrm{m\,s^{-2}}$, the dwarf scale $(2\pi^{\pi/3}/3+\frac{1}{12})\times10^{-10}$ and the cluster scale $(4\pi/3) \times 10^{-10}$, with falsifiers F4–F6: each multiplies a pure number by a unit. A dwarf or cluster scale would have to enter as a derived multiple of a_0.
4. *The screening factor* $(1 + x)^{-\pi/2}$: it changes the deep slope to $\frac{3}{4} + \frac{\pi}{2}$ and misfits SPARC by 0.17 dex (§15).
5. *The two-direction test* of a_0–H_0 at 0.056% and 0.04%: one relation read twice, compared at a precision the data do not have (§14.2).
6. *G from the cell count to* 0.001%: N was computed from G (§24.2).
7. *The horizon resonance ring*: the phantom profile has no maximum (§24.7).
8. *Octupole modulus handedness*: impossible for a real sky (§25.3).
9. *The draft-1 self-coupling* $3\hbar^3/2m_\delta^2 c$ and *the vacuum constant* $V_0 = 2$ as dark energy (§21).
10. *The eikonal replacement* $\Box \to d/ds$; the same flow survives as relaxation (§7).
11. *The three-arc law as the adopted completion* (draft 2): the rotation-curve likelihood places $n = 3$ at $\Delta\chi^2/S^2 = 18.6$ from the optimum (§14.3).
12. *The quantized values* of draft 1's exponent quantization: $2\pi/3$ for disk galaxies (a fixed $2\pi/3$ lies at $\Delta\chi^2/S^2 = 397$, and disks read $m = 1.03$–1.28). The regime dependence itself is kept: it is the running exponent of §17.
13. *The two-stage reading* (draft 3): $n = 2$ as the two lobes of the conjugate pair. n is a tail order, not a count, and $n = 2$ lies 2.3σ outside the galaxy window (§14.4).

G Changes

Release 1.16.2

Patch release: specifies the local Auxoviscausal strain determinant and its sign-consistent evolution; gives the local Bullet Cluster strain-contrast calculation and data requirements; and adds the covariant matter/radiation/thermodynamic conservation and perturbation closure. No new field, force, or fitted parameter is introduced. A numerical Bullet Cluster map and a full cosmological perturbation integration remain explicit computational tasks.

Versioning. Releases are numbered MAJOR.MINOR.PATCH. MAJOR changes an axiom or the physical reading of a sector, so that some result changes meaning; MINOR adds derived results or predictions without changing existing ones; PATCH corrects errors, updates numbers to new data, or changes presentation. Version 1.0.0 is the first release; the drafts that preceded it are recorded below, unabridged.

Release 1.15.1

Patch release: derives the conditional Auxoviscausal junction rule from the spatial-volume determinant of the existing conformal metric. The divergent physical strain is not matched directly; the finite remainder $\widehat{\mathcal{A}}_{\mathrm{aux}} = \frac{1}{3}\ln(\hat{V}/V_*)$ matches if the conformal spatial metric and

identified flow tube remain regular across the boundary. In homogeneous FLRW this remainder carries no independent record of the previous aeon's total $\ln a$. No new field or independent transmission coefficient is added.

Release 1.14.0

Minor release: derives Auxoviscausality as the covariant retained logarithmic proper-volume strain of the existing phase-flow congruence. The volume determinant gives $D\mathcal{A}_{\rm aux} = \vartheta/3$; integration along a phase-flow worldline gives the causal memory integral; Raychaudhuri and the Einstein source give the local Ricci-load response, including the shear correction. The FLRW limit reduces to $\mathcal{A}_{\rm aux} = \ln(a/a_0)$ and links its rate to the horizon information density. No new field or coupling is added. The geometric memory law is closed; microscopic yield-threshold plasticity and dissipative entropy production are explicitly not claimed.

Release 1.13.0

Minor release: Section 32 derives the asymmetric gas/phase response at collision, the relaxation-centroid velocity from $f(t)$, the mass-conserving local transfer flux, proper-volume recoil from the congruence expansion scalar, its Einstein/Raychaudhuri source, the dilation-derived recoil $v_{\rm rec} = -R\dot{\Phi}/c^2$ and its retarded-current integral, the domain-average expansion relation, and the horizon identities $H \propto \iota_H$ and $V_H \propto \iota_H^{-3}$. No external streaming velocity or new force is added. The profile-derived flux must satisfy the existing geodesic phase equation; the sign of local volume change follows the metric solution rather than being imposed.

Release 1.12.0

Minor release: the mergers of §21.7 are worked in a new §15.3 (GRUT release 1.19.0, §47.9); nothing earlier changes. The phase equation has no baryonic source and the flow is geodesic; in the unsettled state the cross term between two clusters is superposition, and the nonlinear cross term of the support law enters through the settled share f alone; the relaxation clock restarts at core passage, $f = 0.0048$ at the Bullet Cluster and at most 0.031 (1 Gyr) and 0.062 (2 Gyr); each peak carries its own cluster's share. On synthetic maps a collision separates an excess on the galaxies from a law computed on the baryons present ($T = 97.7$) and excludes a swapped assignment ($T = 99.1$), and does not separate TRSD from particle dark matter ($T = -0.08$). Not adopted: the cross assignment of the condensate, and the Abell 520 dark core as a prediction (it is the test of row 42). Register rows 41–42; eight references added; the GRUT reference updated to its release 1.19.0. A plain-English edition accompanies this release.

Release 1.11.0

Minor release: §30 completes the unification. The coil's features write one action, the axiom with the rotation held at its rate, the Yang–Mills terms of $(SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$, three generations of fermions and one Higgs doublet (GRUT §54.5); its content fixes the running, $b_3 = -7$, $b_2 = -\frac{19}{6}$, $b_Y = \frac{41}{6}$, so the triangle's coupling closes in strong coupling at the nucleon and the phase and the pair are one electroweak coupling (GRUT §54.6); the horizon and the nucleon are transmutations from one Planck cell of the electric and the strong coupling, and with H4 the strong datum is $K = 19.7566 \pm 0.0002$. Correction: electromagnetism is the phase of the three crossings, acting on the fermions; the phase θ of $\phi_\pm$ is the global phase of the rotation and carries no gauge field.

Release 1.10.2

Patch release: §30 pointed to GRUT 1.17.2; it is replaced by the completed §30 of release 1.11.0.

Release 1.10.1

Minor release. §24.18 (new): the present in Planck cells, $\ln(R_H/\ell_{\rm P}) = 1/\alpha + \ln 8\pi + \frac{3\pi}{2}\alpha$ (H4 of GRUT §48.12), read as a selection of the epoch, not a law for α (as a law it would make α drift 10^5 times faster than atomic clocks allow); it gives $H_0 = 67.3765$ with no cosmological input, 0.4σ from the CMB fit, and $\Lambda\ell_{\rm P}^2$ as $e^{-2/\alpha}$. §30 (new): the four interactions from the coil, the gauge group $(SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$ from its $\mathbb{Z}_2 \times \mathbb{Z}_3$, the hierarchy $Gm_p^2/\hbar c = K^2(16\pi)^{-2/3}e^{-2/3\alpha-\pi\alpha}$, and no unification scale, so no gauge-mediated proton decay. Register rows 38–40.

Release 1.10.0

A consolidation release; no axiom, reading or result changes. *Consistency:* every current number is quoted at the resonance budget of §24.16, refitted reproducibly to Planck's measured acoustic angle $100\theta_* = 1.04110 \pm 0.00031$: $H_0 = 67.35 \pm 0.06$, $\omega_b = 0.02248$, $\omega_c = 0.12022$ (release 1.9.0 quoted 67.33, 0.02250, 0.12031, which were not one consistent point); the numerical-values appendix has one column, the variance analysis is redone at the resonance budget, and the register rows that still carried the withdrawn 5/2 budget (matter fraction, Hubble constant, freeze, finite aeon, first starlight, baryon density) are updated. *The Hubble constant* is stated as the framework's prediction: one expansion rate, 67.35, which the dilation cannot move (clocks and rulers shift by $|\Phi|/c^2 \lesssim 10^{-5}$) and a local void cannot carry (the supernova Hubble diagram), so distance ladders calibrated alike converge on 67.3 (§21.7, register row 7). *Presentation:* a one-page abstract; release narration removed from the running text; "new tags removed. A plain-English edition accompanies this release.

Release 1.9.0

The dark sector as one geometry, formalizing a proposal made for this release. No axiom, reading or earlier result changes—the vacuum stays the horizon's capacity and the cosmic share the rotation held at its rate, now shown to be the same quanta in two states—so the release is a minor one.

1. §21.7 (new): dark energy is spacetime unexcited, the spin's quanta held by the final horizon, $n_*\hbar H_\infty = \rho_\Lambda c^2$ with $n_* = 3/8\pi\ell_{\rm P}^2 R_\infty$, the capacity of §24.11 counted in quanta; what is called dark matter is spacetime excited, the same quanta carried by the rotation; one stress tensor, $\hbar H_\infty(n\,u_\mu u_\nu - n_* g_{\mu\nu})$, whose vacuum term is the generator-defect tension over the square of the final radius, $2\lambda_* = f_\delta^2(\pi)/\delta_{\rm gen}$. $w = -1 + \Theta$ with Θ the excited fraction, equality at $z = 0.372$; the threshold is the vacuum's own density, at the cosmic ratio a baryon density of 0.65 m^{-3}. The states do not convert, the vacuum is 0.34% of a halo, and the background expansion is unchanged; the inventory is spacetime, 0.949, and baryons, 0.050.
2. §21.7: the dilation. The mass lensing infers is the baryonic mass times $\mathcal{S} = 1/\mu = 1/\cos_n\theta$, $\tan\theta = a_0/g$, exact with the dynamic exponent and $g^2 = g_{\rm bar}(g^n + a_0^n)^{1/n}$; at $n = 2$ it is $1/\cos\theta$, the proposal's form, whose 2, 3.86, 11.5 fall at $g = 0.58$, 0.27, $0.087\,a_0$; $\mathcal{S} \to 1$ at the centre and r/ℓ toward the edge, 4 at 39 kpc and 11 at 116 kpc around $10^{11}M_\odot$; set by the horizon, $a_0 \propto \sqrt{\Lambda}$ at the present; varying with capture and relaxation (register

row 33); its excess carried by the excited geometry as the phantom density; how far it grows open (register row 34). The sky's slant $r/\sqrt{r^2-b^2}$ keeps the total; the closure's $M_{\rm Hubble}/M_{\rm total}=4$ is Cauchy's ratio; the Bullet Cluster's excess passed through with the galaxies, carried by the collisionless excitation. The heights of the proposal's first table and its single mean $\langle\mathcal{S}\rangle=4$ corrected.

3. §21.7: the proposal's second form. The dilation field $S=e^{-\psi}\simeq 1-2\Phi/c^2$ is the refractive index of the optical metric, and $\mathcal{S}=\partial_r \ln S/\partial_r \ln S_b$; the field equation is (28) in $\ln S$, its stiffness set by $R_{a_0}|\nabla \ln S|=g/a_0$, and the screening scale is the horizon's, $1/R_\infty$ (register row 35); the density switch is the capture fraction, $n=3/2$, $\alpha_0=1$, $\rho_c=3\sigma^2/4\pi GR^2$, gating the excess; the background's consistency condition is the closure law, and the middle scales are geometric means, $\ell_P^2=\frac{1}{2}r_s\lambda_C$, $\ell^2=r_sR_{a_0}$, $g^2=g_{\rm bar}a_0$. Corrected: the excess assigned to the value of S rather than its gradient, a coupling rising with density and multiplying the baryons, and the enclosed-mass form of $\ln S$ inside a source.

4. §21.7: the dilation on the unbound web. The background cannot move; the rate measured inside a region of contrast δ growing at f is $H(1-f\delta/3)$. The field cannot separate scales, and a dilation set by it alone is excluded by S_8 (6.4σ against DES Y6). The dilation keeps a clock's causes: it reaches the unbound web only where the horizon sets the clocks, mean density below $2\rho_\Lambda$, from the onset of acceleration ($z=0.630$), and holds where the well outweighs all of the medium's motion, above 68 Mpc at the fixed point $\sigma_{1D}=494\ \mathrm{km\,s^{-1}}$, with no input. Being the horizon's, it is a conformal factor: light crosses it undilated, and the late Sachs–Wolfe signal falls ($\times$0.28–0.71 at $\ell=2$–20; $\times$4,200 had light seen it). S_8, shear and CMB lensing keep the background's values, the Cosmicflows-4 bulk flows fall at the 78th and 90th percentiles, and the ladder's H_0 is not reached: a local void would put an 8.3% step in the supernova Hubble diagram, which shows none to 0.60%. The earlier forms of this release, which held the test run's tensions unchanged and then read the ladder's H_0 as the void's rate, corrected.

5. Terminology: the proposal's word "stretch" is replaced by its physics term, the dilation: $S\simeq 1-2\Phi/c^2$ is spacetime's dilation as light measures it, half from time and half from space, and $\mathcal{S}$ its gradient against the baryons' (§21.7). The cells of §24.9, which do not stretch, keep their word.

6. Title, abstract, contributions, §0.6, the §21 reading note, the §23 table, §27, conclusions, register rows 30–37 and numerical values updated; twenty-one references added, and the GRUT reference updated to its release 1.16.0.

Release 1.8.0

1. §24.10: cosmic dawn counted in cycles. The 21-cm onset is the ratio of received to emitted cycles, in closed form at the resonance, 1/30.61 (46.41 MHz); the birefringence counts turns of the same phase weighted by α, so α cancels, $H_\infty t_\star=(2\pi/\kappa)\beta$, and the measured twist places the onset at $42.7^{+5.6}_{-6.0}$ MHz with no clock and no Hubble constant.

2. Abstract, contributions, conclusions, register row 29 and numerical values updated; the GRUT reference updated to its release 1.15.0.

Release 1.7.0

1. §24.15: I7 derived up to an integer. The rotation's energy is its Hamiltonian, $E=\Omega_0 Q$, with Q conjugate to the phase and quantized in $\hbar$, so its quanta carry $\hbar\Omega_0$; crossing the cosmological horizon, whose entropy steps by 2π, they require $\Omega_0=jH_\infty$; the birefringence ($j=2$ out at 6.2σ) and Ω_Λ through the resonance fix $j=1$.

2. §24.16 (new): the budget at the resonance. The dust's amount is an integration constant that decides when the resonance falls, not the budget there; $\Omega_\Lambda/\Omega_c = 2.580$ (-1.3σ), $H_0 = 67.35$, $\omega_b = 0.02248$ ($+0.9\sigma$), $\omega_c = 0.12022$, $\Omega_m = 0.3160$, computed with CAMB. I5 is withdrawn; the 5/2 budget is kept in §24.5. The value 2.568 of §24.13 corrected to 2.580: release 1.6.0 left the neutrinos out of Ω_m.
3. §24.17 (new): the acceleration scale tracks the horizon, $a_0(z) = a_0 E(z)$; the MUSE-DARK radial acceleration relation at $0.33 < z < 1.44$ rises with redshift and excludes a constant scale.
4. §21.2: the vacuum ring $|\Phi| = \sqrt{2}$ is the director circle of the generators' phases; with each generation's amplitude the rise of its generator's invariant, Koide's ratio is $(\Xi + 2)/3\Xi$, $\frac{2}{3}$ at the vacuum (GRUT §42.8).
5. §24.10: first starlight restated between physical epochs, $t_\star = \alpha t_{\rm res}$, at the resonance budget; the claimed $z \simeq 32$ candidate is a brown dwarf.
6. §0.3 ledger (the baryon ratio, derived since draft 9, no longer listed as calibrated), §0.4, §0.6, abstract, contributions, §24.5, §24.13, §24.14, register rows 25–28 and the variance appendix updated; seven references added, and the GRUT reference updated to its release 1.14.0.

Release 1.6.0

1. §24.15 (new): the spinning-arc principle derived for the final horizon. Its thermal form is the four-sphere of radius R_∞; static time is the angle of a turn; every line of sight is a C-arc from horizon to horizon through the observer, spinning about its ends, with the observer on the rim; smoothness at the horizon fixes $\omega = H_\infty$, the rim moves at c, $\Lambda = 3\omega^2/c^2$ and $k_B T_\infty = \hbar\omega/2\pi$; the zeroth law and the Bianchi identity keep the rate fixed. The charge of the turn is $M_{\rm Hubble} c R_H = \hbar N/4\pi$; $I_{\rm total} = 2\pi L/\hbar$; the information–energy relation is the energy of one turn over four cones, $E_{\rm total} = (\hbar\omega/8\pi) I_{\rm total}$; a mechanical spin would cost the entropy (Hod). The quantum: $8\pi\ell_{\rm P}^2$ for every horizon ($+0.5\sigma$ against the LIGO–Virgo ringdowns); $\hbar/2$ at the trembling radius. I7 (the phase turns with the spin, $X_0 = \Lambda/3$); $\theta_0 = \frac{2}{3}\,\mathrm{artanh}(3\sqrt{3}/2\pi)$; $\beta = (2\alpha/\pi)\,\mathrm{artanh}(3\sqrt{3}/2\pi) = 0.3137°$ ($+0.6\sigma$); $a_0 = (2/3\pi)\, c^2\sqrt{\Lambda/3}$.
2. §2.4 (new): the threefold beat. The coil's conjugate lobes keep $\mathbb{Z}_2$, which is why they cross π apart; their fold against a lobe keeps $\mathbb{Z}_3$, $|\phi_+^2 + \phi_-|^2 = \phi^4 + \phi^2 + 2\phi^3 \cos 3\theta$. The geometric origin at the opening of Section 3 is corrected: its equal-frequency Lissajous sentence, restored in release 1.4.0, stays withdrawn (Appendix F, item 1).
3. I7 added to §0.4; abstract, contributions, §24.14, §25.3, conclusions, register rows 22–24, variance row 19, the numerical values and the archive updated; the garbled sentence of the release 1.4.0 entry corrected; nine references added and the GRUT reference updated to its release 1.13.0.

Release 1.5.0

1. The opening principle (new), before the axiom: reality, actuality, occurrence and representation, and the present as the occurrence at which actuality falls into step with reality. Abstract and contributions updated.

Release 1.4.0

1. §24.14 (new): the spinning-arc principle, $\omega R = c$ with a fixed spin; the cosmic sphere ($\Lambda = 3\omega^2/c^2$, the present), the spin measured by birefringence (0.6σ), and the quantum sphere (the Compton radius). Abstract and contributions updated, and the geometric origin of draft 1 restored at the opening of Section 3: the spinning arcs, the threefold wobble that creates the three defects, and the Lissajous count; the wobble is the C_3 mode that meets the spin at the present.

Release 1.3.1

1. §24.13: the grounds of the rotation's rate, the final horizon's surface-gravity frequency, by its thermal circle and its lowest conformal mode; the premise left open stated.

Release 1.3.0

1. A front section states the work's original contributions.
2. §24.13 (new): the present as a resonance. The rotation turns at the final expansion rate H_∞, and the horizon's C_3 mode meets it when $\Omega_\Lambda = 27/4\pi^2 = 0.6839$; Planck 0.6847 ± 0.0073 (0.1σ); the dark ratio becomes 2.568. Recorded as a hypothesis. Abstract updated.

Release 1.2.3

1. With the measured dark ratio the budget reproduces Planck's densities (GRUT §25.10); noted at the budget.

Release 1.2.2

1. The calibrated full-spectrum fit (GRUT §25.9) disfavours $\Omega_\Lambda/\Omega_c = 5/2$ at 4.6σ, short of the 5σ threshold for removing I5; recorded at the budget.

Release 1.2.1

1. The rotation reading, formerly §24.12, opens the dark sector as §21.1, before the condensate it reinterprets; the condensate subsections follow from §21.2, and the closure law is §24.12. References to GRUT follow its release 1.2.1 numbering. Nothing was removed.

Release 1.2.0

1. §24.12 (new): the closure law, $\sum r_s = (\chi/2)R_H$ and $\Omega_{\text{total}} = \chi/2 = 1$, and the scale ladder as its symmetric cuts.

Release 1.1.0

1. §25.3 (new): the birefringence angle and the acceleration scale related through the one rotation, $\beta = \frac{9}{2}\alpha\, a_0 t_0/c$.

2. Referee pass: the abstract, introduction and §15.1 state the dark sector in the rotation reading; §21–22 carry a reading note; the frame-dragging statements of §21.1 replaced by the bound medium, consistent with the zero windings.

Release 1.0.0

1. Versioning reset; "vn" in the text now reads "draft n".
2. §10: I3 derived from the Unruh response of a gapless mode extended around the cycle.

Pre-release drafts

G.0000000000000000000 From draft 21 to draft 22

1. §25.3: $\kappa = 6$ derived as the anomaly coefficient of the cone triples; $\beta = 0.312°$, $+0.6\sigma$.
2. Register rows 10–12 restated in the rotation reading: the radial mode is not the dark component; dwarf profiles from the law.

G.000000000000000000 From draft 20 to draft 21

1. §21.1: the rotation is internal, with zero vorticity, meeting the CMB vorticity bound; windings sum to zero; global frame dragging excluded.

G.00000000000000000 From draft 19 to draft 20

1. Open items closed: the solar-system quadrupole; the covariant completion and its seven requirements; the vacuum-mode cutoff; the birefringence sign and size in the rotation reading.

G.0000000000000000 From draft 18 to draft 19

1. §21.1: the law acts where the medium is bound; wide binaries Newtonian; the total-field alternative excluded by Cassini; the cosmic share as the rotation held at its rate by a constraint.

G.000000000000000 From draft 17 to draft 18

1. §21.1: the crossover set by the total field, with a 10% wide-binary boost; the cosmic share as the modulation of the rotation, diluting as dust with zero sound speed.

G.00000000000000 From draft 16 to draft 17

1. §21.1: the coupling length derived from the two-dimensional reduction, $L_{\mathrm{red}}^2 = L_\pi R_{a_0}$; the Tully–Fisher zero point; the wide-binary test; magnetic fields from ordinary dynamos.

G.0000000000000 From draft 15 to draft 16

1. §21.1: the two parts of the coupling (metric, $-\Phi/c^2$; geometric, $(a_0/g_b)^{1/2}$) and the baryonic defect cone, deficit π with slant $L_\pi = 8GM/c^2$ and coupling length $\ell = \sqrt{GM_b/a_0}$.

G.000000000000 From draft 14 to draft 15

1. §21.1 rewritten as the rotation of the universe carried by the winding phase, with the baryons tuning its local rate and the frame dragging of the phase momentum density as the observed extra gravity.

G.00000000000 From draft 13 to draft 14

1. §21.1 (new): the local baryonic coupling of the C_3 phase mode and the rotation of the universe (GRUT §23), with the slowly rotating universe of [52].

G.0000000000 From draft 12 to draft 13

1. Renumbering to the derived budget ($\Omega_m = 0.32194$, $H_0 = 67.12$): horizon quantities, $a_0 = 1.1444 \times 10^{-10}$, the SPARC residual and likelihood tables, the RAR figure, the register, the variance table and both bridges.
2. New material variance: the physical baryon density, $\omega_b = 0.02275$, $+2.5\sigma$ from Planck, driven by $\Omega_\Lambda/\Omega_c = 5/2$ (variance row 18, register); the likelihood row of a_0 moves to monitor ($+1.00\sigma$).

G.000000000 From draft 11 to draft 12

1. §24.8–24.11 (new): the three kinds of bound and the freeze, $N/N_\infty = \Omega_\Lambda(t)$; the maximum-power point and the finite aeon; the cells do not stretch and α sets when light escapes; the first-starlight hypothesis and its test; the dark energy as the horizon's capacity energy, $\rho_\Lambda = \frac{3}{4}\rho_P/N_\infty$, capped for every field by the horizon condition.
2. Register rows; abstract.

G.00000000 From draft 10 to draft 11

1. §15.2: the relaxation time, $t_{\rm rel} = \tau_1(t_f)/3$, $N_{\rm rel} = 25.3$, and the relaxation state as a function of $D = t/t_f$, $f = 1 - e^{-(D-1)/x_H}$.

G.0000000 From draft 9 to draft 10

1. §15.2 (new): the relaxation state—support, infall and resistance—and the cluster–galaxy difference as a difference in dynamical age; trap probability and scatter.

G.000000 From draft 8 to draft 9

1. §17: the dynamic exponent—the closure-weighted sum of threshold laws over the cascade, with orders N_a/N_s—and the gravity law with the capture fraction; nothing is fitted.
2. §18: the closure fraction is the exposure order, the factor $2^{-2(j-1)}$ the scale.
3. §24.5: the baryon ratio derived from the exponent at the horizon's surface gravity, $z_H = \pi^2/2\sqrt{3}$: 0.18697 ($+0.3\sigma$), $\Omega_m = 0.32194$, $H_0 = 67.12$; the ratio leaves the calibrations.

G.00000 From draft 7 to draft 8

1. §15: the Solar-System quadrupole is the Galactic tide alone; the baryon-only value is kept as the reason the condensate must be the source, and the failed variance row is removed.
2. §15.1: the bound density derived as the polarization of the surviving condensate, and the capture fraction of the field.
3. Verification: `trsd_v8_verify.py`.

G.0000 From draft 6 to draft 7

1. §15.1 (new): the complete equation, with the condensate as a source; Cassini passes; variance row 18.
2. Verification: `trsd_v7_verify.py`.

G.000 From draft 5 to draft 6

1. §15: the Solar-System quadrupole computed across the window (GRUT §49.10), excluded by Cassini at 17σ; variance row 17.
2. Verification: `trsd_v6_verify.py`.

G.00 From draft 4 to draft 5

1. §17: Proposition 10, the running exponent, with its limits, its values system by system and Figure 3; v1's $\pi/3$ identified as the running exponent at galactic accelerations, and v1's regime dependence kept; register row 5 and the variance note reworded.
2. Verification: `trsd_v5_verify.py`, with `--running`.

G.0 From draft 3 to draft 4

1. §11: the family in closed form for real n and Proposition 8: the transition and the Newtonian tail are separate in the threshold family and tied in the Torridity form.
2. §14.3–14.4: both forms profiled in the likelihood; the tail order is constrained to $1.62 \leq n \leq 1.91$, not selected; the empirical function treated as a benchmark chosen on the same data; the scale shown to be independent of n.
3. §15: the perihelion shift corrected to be retrograde, with the factor $(k+2)/2$ for a power-law anomaly; the planetary thresholds recomputed (1.62 and 1.53).
4. §17: v1's Torridity exponents tested in the likelihood, including $2\pi/3$; Figure 4; the exponent quantization archived.
5. §23, §26–28, register, variance, archive: updated to the window.

G.1 From draft 2 to draft 3

1. §14.3: the full rotation-curve likelihood, with 490 per-galaxy nuisance parameters, its scale, shape ranking and nuisance systematic; §14.4 and §15: the exposure index selected by data, $n = 2$, with μ_2 in closed form; Figure 2 shows μ_2.

2. §11: the hypotheses of the universality theorem stated, and Proposition 7: the linear hazard is required by the Tully–Fisher slope.
3. §21.6: the Lyman-α bound shown to apply to the radial mode.
4. §24.5, register, variance: DESI 2026 Lyman-α full shape.
5. §25.2: the status of $C_3/C_2 = 5/2$ as a source-model property.
6. §26.1–26.2: the classification of every number and the dependency of every prediction.
7. §27: acceptance criteria for a covariant completion.

G.2 From draft 1 to draft 2

1. Added: status tags and novelty statement (§0); the crossing proposition (§2.3); the potential scale ε_V and canonical normalization with Theorem 2 (§3–4); the reconciliation of the flow (§7); the exposure lemma, the family μ_n and the universality theorem (§10–11); the SPARC confrontation (§14); the closure-mass theorem (§16.3); the vanishing of V_0 and the physical self-coupling (§21); the dual of the Hubble mass (§24.6); the variance analysis (Appendix B); the verification script.
2. Corrected numbers: $\alpha_{\mathrm{TRSD}} = 0.175493$ (was 0.175435) and $a_0 = 1.2454 \times 10^{-10}$ at 73.04; Mercury at $x = 4.6 \times 10^8$; r_* values; $m_U c^2 = 3.5 \times 10^{-34}$ eV; information in nats; the tachyon at $\Xi = 0$; $\int R\sqrt{g} = 2\delta$; the sign error in the v1 normalized action; $f_\delta^2 = F_{\mathrm{P}}/8$.
3. Corrected references: SPARC is Lelli et al. 2016 (AJ 152, 157); SH0ES is Riess et al. 2022; linear Breit–Wheeler was observed by STAR at RHIC; emergent gravity gives $cH_0/6$.
4. The downstream appendix of v1 is merged into §19 and §21, with the cone volume used consistently.

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